CHAPTER 2

Mosaic

2.1 Hilbert's Nullstellensatz

Let A be an associative, not necccesarily commutative, $\mathbb{C}$-algebra with unit. For $a \in A$ define

$$\operatorname{Spec} a = \{\lambda \in \mathbb{C} \mid a - \lambda \text{ is } \textit{not} \text{ invertible}\}.$$

MOTIVATION: (i) Let $A = \mathbb{C}[0,1]$ be the algebra of continuous $\mathbb{C}$-valued functions on the segment $0 \leq t \leq 1$. A function $t \mapsto a(t)$ is invertible if and only if it does not vanish on $[0,1]$. Thus, for any $a \in A$, the set $\operatorname{Spec} a$ is equal to the set of values of a.

(ii) Let $A = M_n(\mathbb{C})$ be the algebra of $n \times n$-matrices with complex entries. For any $\lambda \in \mathbb{C}$ and $a \in M_n(\mathbb{C})$, the matrix $a - \lambda$ is invertible if and only if λ is not a root of the characteristic polynomial $\det(a - t \cdot \mathrm{Id})$. Thus $\operatorname{Spec} a$ is the set of roots of the characteristic polynomial of a.

Theorem 2.1.1. *[Nullstellensatz] Assume that A has no more than countable dimension over $\mathbb{C}$. Then*

(a) *Weak version: If A is a division algebra, then $A = \mathbb{C}$.*
(b) *Strong version: For all $a \in A$ we have $\operatorname{Spec} a \neq \emptyset$; furthermore, $a \in A$ is nilpotent if and only if $\operatorname{Spec} a = \{0\}$.*

Proof. Part (a) implies part (b) as follows. Suppose that A is a division algebra strictly containing $\mathbb{C}$ and $a \in A \backslash \mathbb{C}$. Then since $a - \lambda \neq 0$ it is invertible for all $\lambda \in \mathbb{C}$. Hence $\operatorname{Spec} a = \emptyset$ which contradicts (b).

Part (b): Let $a \in A$ and $\lambda \notin \operatorname{Spec} a$. Then $(a - \lambda)$ is invertible. Therefore if $\operatorname{Spec} a$ is empty then $(a - \lambda)$ is invertible for every $\lambda \in \mathbb{C}$. Thus $\{(a - \lambda)^{-1}, \lambda \in \mathbb{C}\}$ is an uncountable family of elements of A, for $\mathbb{C}$ has uncountable cardinality. Therefore, elements of this family cannot be linearly independent over $\mathbb{C}$, since A has countable dimension over $\mathbb{C}$. Hence, we may find finitely many λ_i and $\mu_i \in \mathbb{C}$, not all of which are zero,

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such that

$$\sum \mu_i (a - \lambda_i)^{-1} = 0$$

with $\lambda_i, \mu_i \in \mathbb{C}$. Therefore clearing the denominators we have a polynomial $P(t) \in \mathbb{C}[t]$ with $P(a) = 0$. The field $\mathbb{C}$ being algebraically closed, we can write the polynomial P in the factorized form $P(t) = (t - \alpha_1) \cdots (t - \alpha_n)$. It follows that

$$(a - \alpha_1) \cdot \ldots \cdot (a - \alpha_n) = 0 \tag{2.1.2}$$

which is a contradiction, since all $(a - \alpha_i)$ cannot be invertible or else we have the implication $1 = 0$.

Assume now that a is nilpotent, so $a^n = 0$ for some positive integer n. We have for $\lambda \neq 0$

$$(a - \lambda)^{-1} = -\lambda^{-1}(1 - \lambda^{-1}a)^{-1} = \lambda^{-1} \sum_{i=0}^{n} (\lambda^{-1}a)^i$$

by the geometric progression formula. Thus $a - \lambda$ is invertible, and $\lambda \notin \operatorname{Spec} a$.

Assume finally $\operatorname{Spec} a = \{0\}$. Since $\mathbb{C}\backslash\{0\}$ still has uncountable cardinality, the same argument as at the beginning of the proof shows that there exists a polynomial P such that $P(a) = 0$. This implies (2.1.2). Collecting all α_i such that $\alpha_i = 0$ we obtain

$$a^n (a - \alpha_{i_1}) \cdot \ldots \cdot (a - \alpha_{i_m}) = 0, \qquad \alpha_{i_j} \neq 0$$

and each $a - \alpha_{i_j}$ is invertible which implies $a^n = 0$ where $n > 0$. ∎

Lemma 2.1.3. *[Schur's Lemma] Let M be a simple A-module.*

(a) *Weak form:* $\operatorname{End}_A M$ *is a division algebra.*
(b) *Strong form: If A has countable dimension over $\mathbb{C}$, then* $\operatorname{End}_A M = \mathbb{C}$.

Proof. (a). If $0 \neq f \in \operatorname{End}_A M$, then $\operatorname{Ker} f$ and $\operatorname{Im} f$ are A-stable submodules of M. This yields $\operatorname{Ker} f = (0)$ and $\operatorname{Im} f = M$ by simplicity of M. Hence f is invertible and (a) follows.

We now prove (b).

Step 1. We claim that M is of countable (or finite) dimension over $\mathbb{C}$. Let $m \in M$. We have a map

$$A \to M \qquad a \mapsto a \cdot m,$$

and $A \cdot m$ is M or 0 by simplicity of M. Pick $m \in M$ such that $A \cdot m = M$ (this is possible or A annihilates each $m \in M$). Then A maps onto M so that $\dim_{\mathbb{C}} M \leq \dim_{\mathbb{C}} A$.

Step 2. We claim $\mathrm{End}_A M$ has no more than countable dimension over $\mathbb{C}$. Choose $m \in M$ such that $M = A \cdot m$ as above. We claim the map $\mathrm{End}_A M \to M$, $f \mapsto f(m)$ is an injection. Indeed, $M = A \cdot m$ and $f(am) = af(m)$ which implies that f is determined by $f(m)$. Therefore $\dim \mathrm{End}_A M \leq \dim M$ and the claim follows.

Now steps 1 and 2 together with (a) show that $\mathrm{End}_A M$ is a division algebra of countable dimension over $\mathbb{C}$, hence equal to $\mathbb{C}$ by the weak form of Nullstellensatz (2.1.1). ∎

Let $\mathfrak{g}$ be a complex finite dimensional Lie algebra and $\mathcal{U}\mathfrak{g}$ its universal enveloping algebra. Let $Z\mathfrak{g}$ be the center of $\mathcal{U}\mathfrak{g}$. Recall that any $\mathfrak{g}$-module has a canonical $\mathcal{U}\mathfrak{g}$-, hence a $Z\mathfrak{g}$-module structure.

Corollary 2.1.4. [Q3] *The center $Z\mathfrak{g}$ acts by scalars on any simple $\mathfrak{g}$-module M.*

Proof. It is clear from the definition that $\mathcal{U}\mathfrak{g}$ has countable dimension over $\mathbb{C}$, and that $Z\mathfrak{g}$ maps into $\mathrm{End}_{\mathcal{U}\mathfrak{g}} M$. But Lemma 2.1.3(b) implies that $\mathrm{End}_{\mathcal{U}\mathfrak{g}} M = \mathbb{C}$. ∎

Given a finitely generated *commutative* $\mathbb{C}$-algebra A, let $\mathrm{Specm}\, A$ denote the set of all maximal ideals of A. The following version of Nullstellensatz is at the origin of the relationship between commutative algebra and algebraic geometry (see e.g., [AtMa], [Ha], and the next section).

Theorem 2.1.5. *Any maximal ideal of A is the kernel of an algebra homomorphism $A \to \mathbb{C}$.*

Proof. Let $I \subset A$ be a maximal ideal. Then A/I is a field, cf., [AtMa]. Moreover, A being a finitely generated $\mathbb{C}$-algebra, implies that it has no more than countable dimension over $\mathbb{C}$. Hence the same is true for A/I. Thus, the weak version of Theorem 2.1.1 yields $A/I = \mathbb{C}$. It follows that the natural projection $A \to A/I$ may be identified with an algebra homomorphism $A \to \mathbb{C}$ whose kernel clearly equals I. ∎

Due to the above theorem, the set $\mathrm{Specm}\, A$ may be identified with the set of all homomorphisms $\chi : A \to \mathbb{C}$.

2.2 Affine Algebraic Varieties

Throughout this section the ground field is the field $\mathbb{C}$ of complex numbers, and an "algebra" means a commutative $\mathbb{C}$-algebra with unit. Given a $\mathbb{C}$-algebra A, we will write $\mathrm{Specm}\, A$ for the set of maximal ideals of A.

Definition 2.2.1. A (reduced) affine subvariety $X \subset \mathbb{C}^n$ is the subset consisting of all roots of a finite collection of polynomial equations

$$X = \{(x_1, \ldots, x_n) \in \mathbb{C}^n \mid f_1(x_1, \ldots, x_n) = 0, \ldots, f_m(x_1, \ldots, x_n) = 0\}.$$

It is clear that the above set depends only on the ideal

$$I = (f_1, \dots, f_m) \subset \mathbb{C}[x_1, \dots, x_n],$$

and not on the actual polynomials f_i. Therefore, given an ideal $I \subset \mathbb{C}[x_1, \dots, x_n]$, we write

$$\mathcal{V}(I) = \{(x_1, \dots, x_n) \in \mathbb{C}^n \mid f(x_1, \dots, x_n) = 0 \text{ for all } f \in I\}.$$

Observe that, for any $x \in \mathcal{V}(I)$, the evaluation map $f \mapsto f(x)$ vanishes on I, hence induces an algebra homomorphism $\mathbb{C}[x_1, \dots, x_n]/I \to \mathbb{C}$. Let $\mathfrak{m}_x \in \operatorname{Specm} \mathbb{C}[x_1, \dots, x_n]/I$ be the kernel of this homomorphism, a maximal ideal in $\mathbb{C}[x_1, \dots, x_n]/I$.

Proposition 2.2.2. *For any ideal $I \subset \mathbb{C}[x_1, \dots, x_n]$, the assignment $x \mapsto \mathfrak{m}_x$ yields a bijection $\mathcal{V}(I) \simeq \operatorname{Specm} \mathbb{C}[x_1, \dots, x_n]/I$.*

Proof. The injectivity of the map $x \mapsto \mathfrak{m}_x$ is clear. To prove surjectivity we use Theorem 2.1.5 which says that every maximal ideal of the finitely generated $\mathbb{C}$-algebra $\mathbb{C}[x_1, \dots, x_n]/I$ is the kernel $\operatorname{Ker} \chi$ of an algebra homomorphism $\chi : \mathbb{C}[x_1, \dots, x_n]/I \to \mathbb{C}$. For each $i = 1, \dots, n$, let $a_i \in \mathbb{C}$ denote the image of x_i under the composition.

$$\mathbb{C}[x_1, \dots, x_n] \twoheadrightarrow \mathbb{C}[x_1, \dots, x_n]/I \xrightarrow{\chi} \mathbb{C}.$$

Put $a = (a_1, \dots, a_n) \in \mathbb{C}^n$. By multiplicativity we have $\chi(f) = f(a)$ for all polynomials f and moreover $f(a) = \chi(f) = 0$ whenever $f \in I$. Thus $a \in \mathcal{V}(I)$ and $\operatorname{Ker} \chi = \mathfrak{m}_a$. ∎

To an affine subvariety $X \subset \mathbb{C}^n$ we associate its *defining ideal* $I(X)$ and its *coordinate ring*, $\mathcal{O}(X)$, as follows:

$$I(X) = \{f \in \mathbb{C}[x_1, \dots, x_n] \mid f(x) = 0, \forall x \in X\},$$
$$\text{and} \quad \mathcal{O}(X) = \mathbb{C}[x_1, \dots, x_n]/I(X).$$

EXAMPLES

(a) $I(\varnothing) = \mathbb{C}[x_1, \dots, x_n]$ and $I(\mathbb{C}^n) = 0$.

(b) If $X = \{(a_1, \dots, a_n)\}$ is a single point, then $X \subset \mathbb{C}^n$ is an affine subvariety and $I(X)$ is just the maximal ideal $(x_1 - a_1, \dots, x_n - a_n) \subset \mathbb{C}[x_1, \dots, x_n]$.

(c) If $I = (f) \subset \mathbb{C}[x_1, \dots, x_n]$ is a principal ideal, then the affine subvariety $\mathcal{V}(I)$ is called a *hypersurface*.

Let $X \subset \mathbb{C}^n$ and $Y \subset \mathbb{C}^m$ be affine subvarieties.

Definition 2.2.3. A morphism $f : X \to Y$ is a *polynomial* map $f : \mathbb{C}^n \to \mathbb{C}^m$ such that $f(X) \subset Y$. Two morphisms $f_1, f_2 : X \to Y$ are said to be equal (write $f_1 = f_2$) if $f_1(x) = f_2(x)$ for all $x \in X$.

Affine subvarieties $X \subset \mathbb{C}^n$ and $Y \subset \mathbb{C}^m$ are called *isomorphic* if there are morphisms $f : X \to Y$ and $g : Y \to X$ such that $g \circ f = id_X$ and $f \circ g = id_Y$. For example, any two linear subspaces of $\mathbb{C}^n$ of the same dimension are isomorphic to each other as affine subvarieties of $\mathbb{C}^n$. An isomorphism class of affine subvarieties is called a (reduced) *affine algebraic variety.*

Observe that isomorphic affine subvarieties have isomorphic coordinate rings so that there is a well-defined ring $\mathcal{O}(X)$ associated to an affine algebraic variety X which is independent of an embedding $X \hookrightarrow \mathbb{C}^n$.

It is therefore natural to ask what kind of rings arise as coordinate rings of affine varieties. To that end, recall that for any commutative ring A and an ideal $I \subset A$, one defines the *nil-radical* of I as the set

$$\sqrt{I} = \{a \in A \mid a^k \in I \text{ for some } k = k(a) \gg 0\}.$$

This is again an ideal, for if $a^n \in I$ and $b^m \in I$, then $(a+b)^{n+m} \in I$ by the binomial formula. Observe that

$$I \subset \sqrt{I}, \quad \text{and} \quad \mathcal{V}(\sqrt{I}) = \mathcal{V}(I).$$

The following classical version of Hilbert's Nullstellensatz provides the basic relation between affine subvarieties of $\mathbb{C}^n$ and ideals in $\mathbb{C}[x_1, \ldots, x_n]$.

Theorem 2.2.4. *(Nullstellensatz) (i) For any ideal $I \subset \mathbb{C}[x_1, \ldots, x_n]$ we have*

$$I(\mathcal{V}(I)) = \sqrt{I}$$

(ii) The assignment $X \mapsto \mathcal{O}(X)$ sets up a contravariant equivalence between the category of affine algebraic varieties and the category of finitely generated $\mathbb{C}$-algebras without nilpotents.

Proof. (i) We clearly have $\sqrt{I} \subset I(\mathcal{V}(\sqrt{I})) = I(\mathcal{V}(I))$, whence proving the equality amounts to showing that the canonical surjection

$$\pi : \mathbb{C}[x_1, \ldots, x_n]/\sqrt{I} \twoheadrightarrow \mathbb{C}[x_1, \ldots, x_n]/I(\mathcal{V}(I))$$

is a bijection. Let $f \in \mathbb{C}[x_1, \ldots, x_n]/\sqrt{I}$ and assume that $\pi(f) = 0$, that is, $f \in I(\mathcal{V}(I))$. By Proposition 2.2.2, there is a bijection

$$\operatorname{Specm}(\mathbb{C}[x_1, \ldots, x_n]/\sqrt{I}) = \mathcal{V}(\sqrt{I}) = \mathcal{V}(I), \qquad \mathfrak{m}_x \leftrightarrow x,$$

Therefore f belongs to all the maximal ideals of $\mathbb{C}[x_1, \ldots, x_n]/\sqrt{I}$, since f vanishes on every point of $\mathcal{V}(I)$. Assume $\lambda \in \mathbb{C}$ is such that $\lambda - f$ is not invertible. Then there exists a maximal ideal $\mathfrak{m}$ in $\mathbb{C}[x_1, \ldots, x_n]/\sqrt{I}$ containing $\lambda - f$. Since $f \in \mathfrak{m}$ it follows that $\lambda = (\lambda - f) + f \in \mathfrak{m}$. This implies $\lambda = 0$. Thus we have shown that $\operatorname{Spec} f = \{0\}$, cf., the beginning of Section 2.1 for the definition of Spec. The strong form of the Nullstellensatz

2.1.1(b) implies now that f is nilpotent. But the ring $\mathbb{C}[x_1,\dots,x_n]/\sqrt{I}$ clearly has no non-zero nilpotents by definition of the nil-radical. Thus $f=0$, and part (i) follows.

We now prove surjectivity of the assignment of part (ii). Let A be a finitely generated $\mathbb{C}$-algebra without nilpotents. Making a choice of generators $a_1,\dots,a_n$ of A is the same as giving a surjective algebra homomorphism $\mathbb{C}[x_1,\dots,x_n] \twoheadrightarrow A$ sending $x_i \mapsto a_i$. This yields a presentation $A \simeq \mathbb{C}[x_1,\dots,x_n]/I$ with a certain ideal I such that $\sqrt{I}=I$, since A has no nilpotents. Let $X=\mathcal{V}(I)$ be the algebraic variety corresponding to I. We have

$$A \simeq \mathbb{C}[x_1,\dots,x_n]/I = \mathbb{C}[x_1,\dots,x_n]/\sqrt{I} = \mathbb{C}[x_1,\dots,x_n]/I(\mathcal{V}(I)),$$

where the last equality is due to part (i). Thus $A \simeq \mathcal{O}(X)$, and the surjectivity claim follows.

Finally, we prove that the assignment of part (ii) is faithful. Assume that $X \subset \mathbb{C}^n$ and $Y \subset \mathbb{C}^m$ are affine subvarieties such that $\mathcal{O}(X) \simeq \mathcal{O}(Y)$, that is there are mutually inverse morphisms

$$\mathbb{C}[x_1,\dots,x_n]/I(X) \leftrightarrows \mathbb{C}[y_1,\dots,y_m]/I(Y).$$

Let $b_j, j=1,\dots,m$, denote the image of y_j under the composition

$$\mathbb{C}[y_1,\dots,y_m] \to \mathbb{C}[y_1,\dots,y_m]/I(Y) \to \mathbb{C}[x_1,\dots,x_n]/I(X).$$

Each element b_j can be expressed in terms of the generators $x_1,\dots,x_n$ as $b_j = f_j(x_1,\dots,x_n)$ for a certain polynomial f_j. The collection $f_1,\dots,f_m$ gives a polynomial map $f : \mathbb{C}^n \to \mathbb{C}^m$ that induces an isomorphism $\mathbb{C}[y_1,\dots,y_m]/I(Y) \xrightarrow{\sim} \mathbb{C}[x_1,\dots,x_n]/I(X)$, hence takes X into Y. By symmetry, one gets by means of a similar argument the inverse morphism $g : Y \to X$. Thus X and Y are isomorphic. ∎

An affine algebraic variety X is said to be *irreducible* if it *cannot* be written as the union of two proper non-empty affine subvarieties. Algebraically the irreducibility of a variety X is expressed by the property that the ideal $I(X)$ is a *prime* ideal in $\mathbb{C}[x_1,\dots,x_n]$, i.e., an ideal that is not the intersection of any two strictly bigger ideals. Equivalently $I \subset \mathbb{C}[x_1,\dots,x_n]$ is a prime ideal if and only if the quotient $\mathbb{C}[x_1,\dots,x_n]/I$ has no zero divisors [Mum3]. In other words we have, cf. [Mum3].

Proposition 2.2.5. *An affine algebraic variety X is irreducible if and only if its coordinate ring $\mathcal{O}(X)$ has no zero-divisors.*

Given a finitely generated $\mathbb{C}$-algebra A without zero-divisors, let $Q(A)$ be the corresponding field of fractions and $\tilde{A}$ the integral closure of A in $Q(A)$, cf. [Mum3].

Definition 2.2.6. The ring $\tilde{A}$ is called the normalization of A; it is the maximal subring among the subrings $A \subset B \subset Q(A)$ such that B is a finitely generated A-module. If $\tilde{A} = A$ the ring A is said to be normal.

If X is an irreducible affine algebraic variety then the affine variety $\tilde{X} := \operatorname{Specm} \tilde{A}$ corresponding to the normalization of the ring $A = \mathcal{O}(X)$ is called the normalization of X. The embedding of the ring A into its normalization induces a canonical morphism of affine varieties: $\tilde{X} \twoheadrightarrow X$. If $\tilde{X} = X$ the variety X is called normal. The coordinate ring of a *smooth* affine variety is known to be normal.

The affine algebraic varieties have a natural topology in which all morphisms are continuous. More precisely, if we take the weakest topology in which all the preimages by morphisms of points of $\mathbb{C}$ are closed we get the so called *Zariski topology*. The Zariski closed sets of $\mathbb{C}^n$ are by definition all sets of the form $\mathcal{V}(I)$. The Zariski topology on an affine subvariety $X \subset \mathbb{C}^n$ is the induced topology from the Zariski topology of $\mathbb{C}^n$. Due to Theorem 2.2.4, this gives a well-defined topology for the affine algebraic varieties in which all algebraic morphisms are continuous.

The Zariski topology is very weak. For example the Zariski closed subsets of $\mathbb{C}$ are precisely the finite subsets. In particular any bijection $f : \mathbb{C} \to \mathbb{C}$ is continuous in the Zariski topology. This shows that not every map continuous in the Zariski topology is a morphism of affine varieties.

Nevertheless, the Zariski topology is sometimes a convenient tool for geometric reinterpretation of certain algebraic notions. For instance, for an affine variety X, the property of being normal is equivalent to the following (Hartgos property): for any Zariski open set $U \subset X$ and for any morphism $f : U \to \mathbb{C}$ the function f extends by continuity to the whole X if and only if it extends as a morphism of affine algebraic varieties [Ha].

Let now $X \subset \mathbb{C}^n$ be an irreducible affine subvariety. Since the ring $\mathcal{O}(X)$ does not have zero divisors we can form its field of fractions $\mathcal{Q}(X)$. For any point $x \in X$ consider the ring

$$\mathcal{O}_x = \{\tfrac{f}{g} \mid f, g \in \mathcal{O}(X),\ g(x) \neq 0\} \subset \mathcal{Q}(X).$$

Now for a Zariski open set $U \subset X$ set $\mathcal{O}_X(U) = \cap_{x \in U} \mathcal{O}_x$. By construction $\mathcal{O}_X(U)$ is a subring of $\mathcal{Q}(X)$ and if $U \subset V$ is an inclusion of Zariski open sets we have $\mathcal{O}_X(V) \subset \mathcal{O}_X(U)$. This defines a sheaf $\mathcal{O}_X$ on X called the structure sheaf of X.

Let Γ denote the "global sections" functor. We complete our review of the relationship between algebra and geometry with the following important result due to Serre, cf. [Se2]:

Theorem 2.2.7. *For any affine algebraic variety X we have*
(i) $\Gamma(X, \mathcal{O}_X) = \mathcal{O}(X)$.

(ii) The assignment $\mathcal{M} \mapsto M = \Gamma(X, \mathcal{M})$ *gives an equivalence between the category of coherent* $\mathcal{O}_X$*-sheaves and the category of finitely generated* $\mathcal{O}(X)$*-modules. The inverse equivalence is given by the functor* $M \mapsto \mathcal{O}_X \otimes_{\mathcal{O}(X)} M$.

The first claim of the theorem amounts to the equality $\mathcal{O}(X) = \cap_{x \in X} \mathcal{O}_x$. This can be proved easily, see [AtMa]. The second part was proved by Serre [Se2].

2.2.8. Remarks. (1) Part (i) of the theorem insures consistency of the notation $\mathcal{O}(X)$, that is the space of global sections of the sheaf $\mathcal{O}_X$ is indeed equal to the space of regular functions on X.

(2) It follows from the second part of the theorem above that the global sections functor Γ is an exact functor on the abelian category of coherent sheaves on an affine algebraic variety.

In fact, the following stronger result (also due to Serre) holds. Let $\mathcal{F}$ be a coherent sheaf on a complex affine algebraic variety X equipped with the Zariski topology. Then,

$$H^i(X, \mathcal{F}) = 0 \quad \text{for any} \quad i > 0.$$

Theorem 2.2.4 provides a nice geometric interpretation of commutative $\mathbb{C}$-algebras without nilpotent elements. A generalization of this result to commutative $\mathbb{C}$-algebras that may have nilpotent elements requires the more sophisticated notion of a *scheme*. We will not attempt to review the theory of schemes here and refer the reader to e.g., [Mum3]. Since we will use this notion on several occasions, we just mention that there is a contravariant equivalence between the category of finitely generated commutative $\mathbb{C}$-algebras and the category of affine schemes of finite type over $\mathbb{C}$.

Definition 2.2.9. (cf.[Ku], [BeLu]) A finitely generated commutative $\mathbb{C}$-algebra A will be called *Cohen-Macaulay* if it contains a subalgebra of the form $\mathcal{O}(V)$ such that A is a free $\mathcal{O}(V)$-module of finite rank, and V is a *smooth* affine algebraic variety.

Remark 2.2.10. The property of being locally Cohen-Macaulay has the following geometric meaning. Write X^{scheme} for the (not necessarily reduced) affine scheme (see [Mum3]) corresponding to the algebra A. Then $A = \mathcal{O}(X^{scheme})$ is locally Cohen-Macaulay if and only if there is a smooth affine variety V and a finite morphism (cf.[Mum3]) $\pi : X^{scheme} \to V$ such that $\mathcal{O}(X^{scheme})$ is a free $\mathcal{O}(V)$-module, equivalently, by Theorem 2.2.7(ii), if the direct image sheaf $\pi_* \mathcal{O}_{X^{scheme}}$ is free over $\mathcal{O}_V$.

For the applications considered in this book it will suffice to take V to be a vector space, that is to find a polynomial subalgebra $\mathbb{C}[y_1, \dots, y_m] \subset A$ such that A is a free $\mathbb{C}[y_1, \dots, y_m]$-module of finite rank.

Now let $X = \mathcal{V}(I) \subset \mathbb{C}^n$ be the affine variety defined by an ideal

$$I = (f_1, \dots, f_p) \subset \mathbb{C}[x_1, \dots, x_n].$$

Here is one of the main results of this section.

Theorem 2.2.11. [BeLu], [Ku] *Assume that*

(a) *The algebra* $\mathbb{C}[x_1, \dots, x_n]/I$ *is Cohen-Macaulay.*

(b) *The differentials* $df_1, \dots, df_p$ *of the polynomials* f_i, $i = 1, \dots, p$, *are linearly independent at each point of a Zariski-open dense subset* $X^\circ \subset X$.

Then

(i) $I = \sqrt{I}$ *so that* $\mathcal{O}(X) = \mathbb{C}[x_1, \dots, x_n]/I$. *Furthermore,*

(ii) if X *is irreducible and* $\dim(X \setminus X^\circ) \leq \dim X - 2$, *then* $\mathcal{O}(X^\circ) = \mathcal{O}(X)$, *and the ring* $\mathcal{O}(X) = \mathbb{C}[x_1, \dots, x_n]/I$ *is normal, cf.* 2.2.6.

Proof. Let X^{scheme} be the scheme corresponding to the algebra $\mathbb{C}[x_1, \dots, x_n]/I$, and $\pi : X^{scheme} \to V$ the finite morphism of affine schemes (as in 2.2.10) corresponding to the algebra embedding $\mathbb{C}[V] \hookrightarrow \mathbb{C}[x_1, \dots, x_n]/I$. Proving (i) amounts to showing that the natural sheaf morphism $\mathcal{O}_{X^{scheme}} \to \mathcal{O}_X$ arising from the natural projection $\mathbb{C}[x_1, \dots, x_n]/I \twoheadrightarrow \mathbb{C}[X]/\sqrt{I}$ is an isomorphism. It suffices to show that the induced map

$$u : \pi_* \mathcal{O}_{X^{scheme}} \to \pi_* \mathcal{O}_X$$

is an isomorphism, since π is a finite morphism.

Write $X^{\mathrm{BAD}} := X \setminus X^\circ$. The linear independence of the differentials $df_1, \dots, df_p$ on X° (condition (b)) implies that the map u becomes an isomorphism when restricted to $V^\circ = V \setminus \pi(X^{\mathrm{BAD}})$, for $\pi^{-1}(V^\circ) \cap X^{scheme} \subset X^\circ$ is a smooth reduced subvariety of $\pi^{-1}(V^\circ)$. Clearly V° is a Zariski-open dense subset of V. Thus, the map u is a surjective morphism from a free sheaf and, moreover, u is an isomorphism on a Zariski open subset. Hence the kernel of such a morphism is a subsheaf of a free sheaf supported on a proper closed subvariety of V. But a free sheaf has no subsheaves supported on proper closed subvarieties. Thus, u is an isomorphism, $X = X^{scheme}$, and part (i) follows.

To prove part (ii) observe first that we have $\dim X = \dim V$, since π is finite. Hence, $\dim(X \setminus X^\circ) \leq \dim X - 2$ implies $\dim(V \setminus V^\circ) \leq \dim V - 2$. Therefore, any regular function on $V \setminus V^\circ$ can be extended to a polynomial function over all of V. Thus the same holds for sections of a free sheaf on V. Applying this to the sheaf $\pi_* \mathcal{O}_X$ yields $\mathcal{O}(X^\circ) = \mathcal{O}(X)$.

It remains to prove that the ring $\mathcal{O}(X)$ is normal. Let $\tilde{X}$ be the normalization of X. We have the canonical projection $\nu : \tilde{X} \to X$, a finite morphism which is an isomorphism over X°, since the latter is smooth. The natural restriction maps and the projection ν give rise to the following commutative diagram.

$$\begin{array}{ccc} \mathcal{O}(X) & \xrightarrow{r} & \mathcal{O}(X^\circ) \\ \downarrow{\scriptstyle \nu^*} & & \| \\ \mathcal{O}(\tilde{X}) & \xrightarrow{r} & \mathcal{O}(\nu^{-1}(X^\circ)) \end{array}$$

The vertical isomorphism $\mathcal{O}(X^\circ) = \mathcal{O}(\nu^{-1}(X^\circ))$ is due to the fact that X° is smooth. The diagram implies that the map $\nu^* : \mathcal{O}(X) \hookrightarrow \mathcal{O}(\tilde{X})$ is a bijection. ∎

Let V be a finite dimensional vector space. We endow $\mathbb{C}[V]$ with the standard grading $\mathbb{C}[V] = \oplus_{i \geq 0} \mathbb{C}^i[V]$, where $\mathbb{C}^i[V]$ stands for the space of the degree i homogeneous polynomials on V. Given a vector space embedding $E \hookrightarrow V$ we get a canonical restriction homomorphism of graded algebras $\mathrm{res}_E : \mathbb{C}[V] \to \mathbb{C}[E]$. Further, the projection $V \to V/E$ of vector spaces induces the "pullback" homomorphism of graded algebras $\mathbb{C}[V/E] \hookrightarrow \mathbb{C}[V]$.

Let A be a graded subalgebra of $\mathbb{C}[V]$. We consider the natural algebra homomorphism $\phi : \mathbb{C}[V/E] \otimes_{\mathbb{C}} A \to \mathbb{C}[V]$, given by the pullback of the first factor followed by multiplication with A.

The following result of Bernstein-Lunts provides a useful criterion for $\mathbb{C}[V]$ to be a free A-module.

Proposition 2.2.12. [BeLu] *Assume that there is a vector subspace $E \subset V$ with the following properties:*

(1) *The composition $A \hookrightarrow \mathbb{C}[V] \xrightarrow{res_E} \mathbb{C}[E]$ is injective. Write A_E for the image of the above composition.*
(2) *The algebra $\mathbb{C}[E]$ is a free graded A_E-module of rank r.*

Then the map $\phi : \mathbb{C}[V/E] \otimes_{\mathbb{C}} A \to \mathbb{C}[V]$ is injective. Moreover, $\mathbb{C}[V]$ viewed as a graded $\mathbb{C}[V/E] \otimes_{\mathbb{C}} A$-module is free of rank r.

The proof of the proposition is based on the following simple linear algebra considerations. The projection $V \to V/E$ makes V an affine bundle over V/E, that is, a principal bundle for the additive group of the vector space E. This puts a natural increasing filtration on $\mathbb{C}[V]$, cf. $n°2.3.10$:

(2.2.13) $F_p\mathbb{C}[V] := \{P \in \mathbb{C}[V] \mid P \text{ has degree} \leq p \text{ along the fibers}\}$.

Write $\mathrm{gr}^F \mathbb{C}[V] = \oplus_p \mathbb{C}[V](p)$ for the associated graded ring corresponding to the filtration $F_\bullet$. Clearly, we have

$$F_0\mathbb{C}[V] = \mathbb{C}[V](0) = \mathbb{C}[V/E]\,.$$

Thus, each graded component, $\mathbb{C}[V](p)$, is an (infinite dimensional) free $\mathbb{C}[V/E]$-module. More precisely, there is a canonical isomorphism of free $\mathbb{C}[V/E]$-modules

$$\mathbb{C}[V](p) \simeq \mathbb{C}[V/E] \otimes_{\mathbb{C}} \mathbb{C}^p[E] \quad , \quad p = 0, 1, \ldots \tag{2.2.14}$$

where $\mathbb{C}^p[E]$ stands for the space of the degree p homogeneous polynomials on E. Let $\sigma_p : F_p\mathbb{C}[V] \to \mathbb{C}[V](p) = \mathbb{C}[V/E] \otimes_{\mathbb{C}} \mathbb{C}^p[E]$ be the canonical projection (= "principal symbol map").

Lemma 2.2.15. *Let $p \geq 0$ and $f \in F_p\mathbb{C}[V]$ be a homogeneous degree p polynomial (relative to the ordinary grading) such that $\mathrm{res}_E(f) \neq 0$ in $\mathbb{C}[E]$. Then $\sigma_p(f)$ equals the image of the element $1 \otimes \mathrm{res}_E(f) \in \mathbb{C}[V/E] \otimes_{\mathbb{C}} \mathbb{C}^p[E]$ under the isomorphism* 2.2.14. *In particular, $\sigma_p(f) \neq 0$ in $\mathbb{C}[V](p)$.*

Proof. Choose a vector subspace W in V complementary to E, so that $V = E \oplus W$. Hence, a graded algebra isomorphism $\mathbb{C}[V] = \mathbb{C}[E] \otimes \mathbb{C}[W]$. Using this isomorphism we can write $F_p\mathbb{C}[V] = \sum_{i \leq p} \mathbb{C}^i[E] \otimes \mathbb{C}[W]$. Thus, the homogeneous polynomial $f \in F_p\mathbb{C}[V]$ has the form

$$f = e_p \otimes 1 + \sum_{i<p} e_i \otimes w_{p-i}\,, \quad e_i \in \mathbb{C}^i[E]\,,\ w_{p-i} \in \mathbb{C}^{p-i}[W]\,. \tag{2.2.16}$$

We see that $\mathrm{res}_E(f) = e_p$, while $\sigma_p(f) = e_p \otimes 1$. The lemma follows. ∎

Proof of Proposition 2.2.12. On $\mathbb{C}[V]$ consider the increasing filtration $F_\bullet\mathbb{C}[V]$ defined by formula (2.2.13) above, and let $F_\bullet A = A \cap F_\bullet\mathbb{C}[V]$ be the induced filtration on A. Assign $\mathbb{C}[V/E]$ grade degree zero and put $\mathbb{C}[V/E] \otimes A$ on the tensor product filtration. Then the map $\phi : \mathbb{C}[V/E] \otimes A \to \mathbb{C}[V]$ is clearly filtration preserving. Proving the injectivity of this map amounts to showing injectivity of the associated graded map $\mathrm{gr}\,\phi : \mathbb{C}[V/E] \otimes \mathrm{gr}^F A \to \mathrm{gr}^F \mathbb{C}[V]$, due to Proposition 2.3.20 of the next section. The target space of the latter map is isomorphic naturally to the graded vector space $\mathbb{C}[V/E] \otimes_{\mathbb{C}} \mathbb{C}^\bullet[E]$, due to equation 2.2.14. To prove injectivity we will now describe the resulting map

$$\mathrm{gr}\,\phi : \mathbb{C}[V/E] \otimes \mathrm{gr}^F A \to \mathbb{C}[V/E] \otimes_{\mathbb{C}} \mathbb{C}^\bullet[E] \tag{2.2.17}$$

more explicitly as follows.

Recall that $A = \oplus_i A^i$ is a graded subalgebra in $\mathbb{C}[V] = \oplus_i \mathbb{C}^i[V]$ (the standard grading).

Define the second filtration on $\mathbb{C}[V]$ by $G_p\mathbb{C}[V] = \oplus_{i \leq p} \mathbb{C}^i[V]$, and write $G_pA := A \cap G_p\mathbb{C}[V]$ for the induced filtration. Clearly, for any p, we have

$G_pA \subset F_pA$. This inclusion gives the associated graded map $\mathrm{gr}^G A \to \mathrm{gr}^F A$. Proof of the following claim will be postponed until the end of this section.

Claim 2.2.18. The map $\mathrm{gr}^G A \to \mathrm{gr}^F A$ is an algebra isomorphism.

Now given any p and any element $\overline{a} \in \mathrm{gr}_p^F A$, we can find using the claim, an element $a \in A^p$ such that $\overline{a} = \sigma_p(a)$. Due to Lemma 2.2.15, the map $\mathrm{gr}\ \phi$ from (2.2.17) is given by

$$\mathrm{gr}\ \phi : f \otimes \sigma_p(a) \mapsto f \otimes \mathrm{res}_E(a),\ \forall f \in \mathbb{C}[V/E]\,. \tag{2.2.19}$$

The formula shows that the map $\mathrm{gr}\,\phi$ is essentially the tensor product of $\mathrm{id}_{\mathbb{C}[V/E]}$ with the restriction morphism $A \to \mathbb{C}[E]$. The latter is injective by assumption (1), and the injectivity claim follows.

Finally, we prove by a similar argument that $\mathbb{C}[V]$ is a free graded $\mathbb{C}[V/E]\otimes A$-module. To that end, view $\mathbb{C}[V]$ as a filtered $\mathbb{C}[V/E]\otimes A$-module by means of the filtration $F_\bullet$. By Proposition 2.3.20 it suffices to show that $\mathrm{gr}^F \mathbb{C}[V]$ is a free $\mathbb{C}[V/E] \otimes \mathrm{gr}^F A$-module of rank r. But equation 2.2.14 and formula 2.2.19 yield a $\mathrm{gr}^F A$-module isomorphism

$$\mathrm{gr}^F \mathbb{C}[V] \simeq \mathbb{C}[V/E] \otimes_{\mathbb{C}} \mathbb{C}^\bullet[E],$$

where the $\mathrm{gr}^F A$-action on $\mathbb{C}[E]$ arises from the algebra homomorphism

$$\mathrm{gr}^F A \hookrightarrow \mathbb{C}[E]\ ,\ \sigma_p(a) \mapsto \mathrm{res}_E(a).$$

This homomorphism is injective and makes $\mathbb{C}[E]$ a free $\mathrm{gr}^F A$-module of rank r, by assumption. The proposition follows.

2.2.20. POINCARÉ SERIES. The formalism explained below serves as a replacement of "dimension arguments" in the standard linear algebra when a finite dimensional vector space is replaced by an infinite dimensional one.

Let $E = \oplus_{i\geq 0} E_i$ be a (possibly infinite dimensional) graded vector space with finite dimensional graded components E_i. Write $E^* := \oplus_{i\geq 0}(E^i)^*$ for the *graded dual* of E. For a graded subspace $F \subset E$, write $F^\perp \subset E^*$ for the annihilator of F. Thus there is a natural isomorphism of graded spaces

$$F^\perp \simeq (E/F)^*. \tag{2.2.21}$$

The tensor product $E \otimes E'$ of two graded vector spaces has a natural grading $E \otimes E' = \oplus_{k\geq 0}(E \otimes E')_k$ given by $(E \otimes E')_k = \oplus_{i+j=k} E_i \otimes E'_j$.

To a graded vector space E, as above, we associate its Poincaré series, $P(E)$, a formal power series in t given by

$$P(E) = \sum_{i=0}^{\infty} t^i \cdot \dim E_i. \tag{2.2.22}$$

Verification of the power series identities contained in the following lemma is straightforward and is left to the reader

Lemma 2.2.23. (a) $P(E \otimes E') = P(E) \cdot P(E')$,
(b) $P(E/F) = P(E) - P(F)$, *and*
(c) $P(E^*) = P(E)$.

Repeated application of property (b) yields the following

Corollary 2.2.24. *Let $F_\bullet E$ be a (possibly infinite) filtration on E by graded subspaces $F_i E$. Then the grading on E induces a natural grading on $\mathrm{gr}^F E$, the associated graded space, and $P(\mathrm{gr}^F E) = P(E)$.*

The following lemma is also clear.

Lemma 2.2.25. *Let $f : E \to E'$ be a grading preserving linear map. If f is either injective or surjective, and moreover, if $P(E) = P(E')$, then f is an isomorphism.*

Proof of Claim 2.2.18*:* Write $A^p := A \cap \mathbb{C}^p[V]$. Due the canonical isomorphism $G_pA/G_{p-1}A \simeq A^p$, the map r gives a chain of maps:

$$(2.2.26)\quad A^p = G_pA/G_{p-1}A \xrightarrow{r} (F_pA \cap G_pA)/F_{p-1}A) \overset{(2.2.14)}{\hookrightarrow} \\ \hookrightarrow (\mathbb{C}[V/E] \otimes \mathbb{C}^p[E]) \cap G_pA = 1 \otimes \mathbb{C}^p[E] \simeq \mathbb{C}^p[E]\,.$$

Summing up over all p, the composition above yields an embedding of the subspace $r(\mathrm{gr}^G A) \subset \mathrm{gr}^F A$ into $\mathbb{C}[E]$. It follows from Lemma 2.2.15 that this composition is nothing but the restriction map $\mathrm{res}_E : A \to \mathbb{C}[E]$. Hence, this composition is injective by assumptions of Proposition 2.2.12. Therefore, the map $r : \mathrm{gr}^G A \to r(\mathrm{gr}^G A)$ is injective, hence bijective.

It suffices to show that $r(\mathrm{gr}^G A) = \mathrm{gr}^F A$. To that end, consider the Poincaré series $P(A)$ with respect to the standard grading. Observe further that the filtration $F_\bullet$ is a filtration by graded subspaces relative to the standard grading. Hence, Corollary 2.2.24 yields $P(A) = P(\mathrm{gr}^F A)$. On the other hand, the isomorphism $\mathrm{gr}^G A \simeq r(\mathrm{gr}^G A)$ established in the previous paragraph implies that $P(\mathrm{gr}^G A) = P(r(\mathrm{gr}^G A))$. Hence, we obtain $P(r(\mathrm{gr}^G A)) = P(A) = P(\mathrm{gr}^F A)$. Since $r(\mathrm{gr}^G A)$ is part of $\mathrm{gr}^F A$, it follows from the latter equality and Lemma 2.2.25 that $r(\mathrm{gr}^G A) = \mathrm{gr}^F A$. ∎

2.3 The Deformation Construction

Let A be a ring with unit and t a central element which is not a zero-divisor such that

$$\bigcap_{i \geq 1} t^i A = \{0\}.$$

A basic example of this is $A = \mathbb{C}[t]$.

Let A_t denote the localization of A at the multiplicative set $S = \{t^k \mid k = 1, 2, \dots\}$. Let M be a finitely generated A_t–module.

Definition 2.3.1. A lattice in M is a finitely generated A-submodule $L \subset M$ such that $A_t \cdot L = M$, or equivalently $\cup_{k \geq 0} t^{-k} L = M$.

Proposition 2.3.2. *For any two lattices L and L' there are integers $k, l \geq 0$ such that*

$$t^k \cdot L \subset L' \subset t^{-l} \cdot L.$$

Proof. Let $u_1, \ldots, u_r$ be generators of the A-module L'. Since L is a lattice there is an integer l such that $u_1, \ldots, u_r \in t^{-l} \cdot L$. Consequently $L' \subset t^{-l} \cdot L$. The second inclusion follows by symmetry. ∎

Assume from now on that A is Noetherian. We will use the following elementary result whose proof is left to the reader.

Lemma 2.3.3. *Let $0 \to M' \to M \to M'' \to 0$ be a short exact sequence of A_t-modules, and let $L \subset M$ be an A-lattice. Then*

(i) $L' := L \cap M'$ and $L'' := L/L \cap M' \subset M''$ are lattices in M' and M'' respectively.

(ii) The sequence $0 \to L' \to L \to L'' \to 0$ is exact.

Associated to any ring A is the *Grothendieck semigroup*, $K^+(A)$, of finitely-generated A-modules. This is the free abelian semi-group generated by finitely-generated A-modules, modulo the subgroup generated by all relations given by short exact sequences of A-modules:

$$0 \to M' \to M \to M'' \to 0.$$

Given an A-module M write $[M]$ for the class of M in $K^+(A)$. Thus, if $0 \to M' \to M \to M'' \to 0$ is an exact sequence of A-modules then in $K^+(A)$ one has $[M] = [M'] + [M'']$.

Lemma 2.3.4. *In the setup of Definition 2.3.1, for any two lattices L and L', in $K^+(A/t \cdot A)$ we have*

$$[L/t \cdot L] = [L'/t \cdot L'].$$

Proof. Call L and L' *adjacent* if $t \cdot L' \subset t \cdot L \subset L' \subset L$. In this case the exact sequence $0 \to L'/t \cdot L \to L/t \cdot L \to L/L' \to 0$ implies

$$[L/t \cdot L] = [L/L'] + [L'/t \cdot L],$$

and similarly $0 \to t \cdot L/t \cdot L' \to L'/t \cdot L' \to L'/t \cdot L \to 0$ implies

$$[L'/t \cdot L'] = [L'/t \cdot L] + [t \cdot L/t \cdot L'].$$

The statements now follow since $t \cdot L/t \cdot L' \simeq L/L'$.

In the general case of arbitrary lattices L and L' set $L_j = L + t^j \cdot L'$, $j \in \mathbb{Z}$. Clearly $L_j = L$ for large j and $L_j = t^j \cdot L'$ for very negative j. It remains to note that for each j the lattices L_j and L_{j+1} are adjacent. ∎

2.3.5. Assume that B is a ring with a separating $\mathbb{Z}$-filtration, i.e.,

$$\ldots \subset B_{-1} \subset B_0 \subset B_1 \subset \ldots, \quad \cup B_i = B, \quad \cap B_i = \{0\}, \quad 1 \in B_0$$

where $B_i \cdot B_j \subset B_{i+j}$, $\forall i, j \in \mathbb{Z}$. In particular, B_0 is a subalgebra.

Form the graded ring $\hat{B} = \bigoplus_{i \in \mathbb{Z}} B_i$. It will be convenient to view $\hat{B}$ as a subset of the ring of Laurent polynomials $B[t, t^{-1}]$ over B by means of the natural bijection $B_j \leftrightarrow B_j \cdot t^j \subset B[t, t^{-1}]$. Thus we have

$$\hat{B} \simeq \sum_{i \in \mathbb{Z}} B_i t^i \subset B[t, t^{-1}] \tag{2.3.6}$$

that is, $\hat{B}$ is the set of all *finite* expressions $\sum b_i t^i$, $b_i \in B_i$.

Proposition 2.3.7. *We have*

(a) $\hat{B}$ *is a subring of* $B[t, t^{-1}]$,
(b) $\hat{B}$ *is a graded ring with the* $\mathbb{Z}$*-grading being given by the decomposition* $\hat{B} = \oplus B_i$, *i.e.,* $(\hat{B})_i = B_i t^i$.
(c) $t \in \hat{B}$, *is central and is not a zero-divisor,*
(d) $\cup t^{-k} \hat{B} = B[t, t^{-1}]$.

Proof. The proofs of (a), (b) and (c) are immediate. To prove (d), pick up $b_i \in B_i$. Then we can write $b_i t^i = b_i t^j \cdot (t^{j-i})$, and (d) follows because $B = \cup B_i$. ∎

Corollary 2.3.8. *(i) There is a natural isomorphism:* $\hat{B}/t \cdot \hat{B} \simeq \operatorname{gr} B$, *where* $\operatorname{gr} B = \oplus B_i / B_{i-1}$ *is the associated graded ring of* B.

(ii) The ring $B[t, t^{-1}]$ *is isomorphic to the localization of* $\hat{B}$ *at* t, *i.e.,* $B[t, t^{-1}] = (\hat{B})_t$.

Proof. Observe that multiplication by t shifts degree in $\hat{B}$ by 1, and can be identified with the natural embedding $\oplus_i B_{i-1} \hookrightarrow \oplus_i B_i$. Therefore we get

$$\hat{B}/t \cdot \hat{B} = \oplus_i (B_i / B_{i-1}) = \operatorname{gr} B$$

and part (i) follows. Part (ii) follows from Proposition 2.3.7 (d). ∎

2.3.9. GEOMETRIC INTERPRETATION. Let X be an affine algebraic variety over the complex numbers and let $B = \mathcal{O}(X)$ be the ring of regular functions on X. Suppose that there is a given $\mathbb{Z}$-filtration on B. Then we have the associated ring $\hat{B}$ which corresponds to a certain affine algebraic variety $\hat{X}$. The natural ring embedding $\mathbb{C}[t] \hookrightarrow \hat{B}$ gives rise to the surjective morphism of algebraic varieties: $f : \hat{X} \twoheadrightarrow \mathbb{C}$. Since t is not a zero-divisor,

$\hat{B}$ is a torsion-free $\mathbb{C}[t]$-module and therefore flat. Thus we have produced a flat family, $\hat{X}$, of algebraic varieties parametrized by $\mathbb{C}$.

To study this family observe first that part (i) of Corollary 2.3.8 yields

$$f^{-1}(0) = \operatorname{Specm} \hat{B}/t \cdot \hat{B} = \operatorname{Specm} \operatorname{gr} B$$

Similarly, using part (ii) of the corollary we obtain

$$f^{-1}(\mathbb{C}^*) = \operatorname{Specm} (\hat{B})_t = \operatorname{Specm} B[t, t^{-1}] = \mathcal{O}(X \times \mathbb{C}^*)$$

Thus, there is the following canonical diagram:

$$\begin{array}{ccccc} \operatorname{Specm}(\operatorname{gr} B) & \hookrightarrow & \hat{X} & \hookleftarrow & X \times \mathbb{C}^* \\ \downarrow & & \downarrow f & & \downarrow \text{2nd projection} \\ \{0\} & \hookrightarrow & \mathbb{C} & \hookleftarrow & \mathbb{C}^* \end{array}$$

2.3.10. PROJECTIVE COMPLETION OF AN AFFINE SPACE. (cf., [Fu]) Let V be a vector space. A principal homogeneous space, E, of the additive group of V is said to be an *affine linear space over* V. A choice of a base point $e \in E$ gives an isomorphism of varieties $T_e : V \xrightarrow{\sim} E$ defined by $v \mapsto v + e$ (traditionally '+' here stands for the v-action on e). Thus an affine linear space may be thought of as a "vector space without preferred origin."

A function f on E is said to be polynomial of degree $\leq n$ if, for some $e \in E$, the composition

$$V \xrightarrow{T_e}_{\sim} E \xrightarrow{f} \mathbb{C}$$

is a polynomial on V of degree $\leq n$. This does not depend on the choice of $e \in E$, since if P is a polynomial on V of degree $\leq n$ then the function $x \mapsto P(x + v)$ is again a polynomial on V of degree $\leq n$ for any fixed $v \in V$. Let $\mathbb{C}[E]$ denote the algebra of all polynomial functions on E. The subspaces $\mathbb{C}_n[E]$ of polynomials of degree $\leq n$ form an increasing filtration on $\mathbb{C}[E]$ making it a filtered algebra. Observe that this algebra has no natural grading, since the notion of a homogeneous polynomial on E is *not* well-defined.

The elements of $\mathbb{C}_1[E]$ form a vector space of *affine* functions with the distinguished subspace $\mathbb{C} = \mathbb{C}_0[E] \hookrightarrow \mathbb{C}_1[E]$ of constant functions.

Given $e \in E$, to any $f \in \mathbb{C}_1[E]$, associate a function f^e on V given by the formula $f^e(v) = f(v + e) - f(e)$. The assignment $f \mapsto f^e$ vanishes on the subspace $\mathbb{C}_0[E]$ and yields a canonical vector space isomorphism

$$\mathbb{C}_1[E]/\mathbb{C}_0[E] \simeq V^*, \tag{2.3.11}$$

which does not depend on the choice of $e \in E$. By multiplicativity, one extends this isomorphism of vector spaces to a natural graded algebra

isomorphism

$$\operatorname{gr} \mathbb{C}[E] \simeq \mathbb{C}[V], \tag{2.3.12}$$

where "gr" stands for the "associated graded ring" with respect to the above defined filtration on $\mathbb{C}[E]$, and $\mathbb{C}[V] = S^\bullet(V^*)$, is the polynomial algebra on V.

We now apply the general construction of Section 2.3.5 to the filtered algebra $\mathbb{C}[E]$ and form the algebra $\widehat{\mathbb{C}[E]}$. We have by definition

$$\widehat{\mathbb{C}[E]} \simeq \oplus \mathbb{C}_i[E] \simeq S^\bullet(\mathbb{C}_1[E]) \simeq \mathbb{C}[\hat{E}],$$

where $\hat{E} := \mathbb{C}_1[E]^*$ is a vector space of dimension $\dim E + 1$.

The general diagram of Section 2.3.9 reduces in the special case under consideration to a diagram

$$\begin{array}{ccccc} V & \hookrightarrow & \hat{E} & \hookleftarrow & E \times \mathbb{C}^* \\ \downarrow & & \downarrow & & \downarrow \\ \{0\} & \hookrightarrow & \mathbb{C} & \hookleftarrow & C^*, \end{array} \tag{2.3.13}$$

where we used (2.3.12) to identify $\operatorname{Specm}(\operatorname{gr} \mathbb{C}[E])$ with V. The maps in the diagram can be described explicitly as follows. The embedding $V \hookrightarrow \hat{E}$ is the adjoint to the projection $\mathbb{C}_1[E] \twoheadrightarrow V^*$ arising from (2.3.11). The projection $E \to \mathbb{C}$ is the adjoint of the canonical embedding $\mathbb{C} = \mathbb{C}_0[E] \hookrightarrow \mathbb{C}_1[E]$. Finally the embedding $E \times \mathbb{C}^* \hookrightarrow \hat{E}$ assigns to $(e, t) \in E \times \mathbb{C}^*$ a linear function $ev_{e,t}$ on $\mathbb{C}_1[E]$ given by the formula $ev_{e,t}(f) = t \cdot f(e)$, $f \in \mathbb{C}_1[E]$.

Set $\mathbb{P}E := \mathbb{P}(\hat{E})$, the projective space associated to the vector space $\hat{E}$. Thus $\mathbb{P}E$ is the space of $\mathbb{C}^*$-orbits in $\hat{E} \setminus \{0\}$ and diagram (2.3.13) yields the canonical decomposition

$$\mathbb{P}E \simeq E \sqcup \mathbb{P}(V) \tag{2.3.14}$$

with E being a Zariski-open cell in $\mathbb{P}(E)$ and $\mathbb{P}(V)$ the hyperplane at infinity. We call $\mathbb{P}E$ the *projective completion* of the affine space E. Note the different meaning of the symbol $\mathbb{P}$ in the two notations $\mathbb{P}E$ and $\mathbb{P}(V)$: the latter one is the projectivization of a vector space so that $\dim \mathbb{P}(V) = \dim V - 1$. The symbol $\mathbb{P}E$ *does not* stand since the projectivization of E, for the projectivization of an affine space is not defined, in particular, $\dim \mathbb{P}E = \dim E$.

2.3.15. Deformation to the Normal Bundle. (cf., [BFM1], [Ger], [DV]) Set $B = \mathcal{O}(X)$ for a smooth affine algebraic variety X. Let $Y \subset X$ be a smooth, closed subvariety, and $\mathcal{I}_Y \subset \mathcal{O}(X)$ the defining ideal of Y.

Set

$$B_i = \begin{cases} \mathcal{I}_Y^{-i} & i < 0, \\ \mathcal{O}(X) & \forall i \geq 0. \end{cases}$$

We now use natural graded algebra isomorphisms

$$\operatorname{gr} B = \bigoplus_{i \leq 0} \mathcal{I}_Y^{-i}/\mathcal{I}_Y^{-i+1} = \bigoplus_{i \leq 0} S^{-i}(\mathcal{I}_Y/\mathcal{I}_Y^2) = \mathcal{O}(T_Y X),$$

where $T_Y X$ is the normal bundle of X over Y. With this setup the diagram in section 2.3.9 becomes the following *deformation diagram*:

$$\begin{array}{ccccc} T_Y X & \hookrightarrow & \mathcal{X}_Y & \hookleftarrow & X \times \mathbb{C}^* \\ \downarrow & & \downarrow & & \downarrow \text{2nd projection} \\ \{0\} & \hookrightarrow & \mathbb{C} & \hookleftarrow & \mathbb{C}^* \end{array}$$

We see that $\mathcal{X}_Y$ is a flat family of varieties over $\mathbb{C}$ whose special fiber, i.e., the fiber over 0 is $T_Y X$. On the other hand the *generic* fibers are all isomorphic to X. Moreover, $\mathcal{X}_Y$ can be shown to be smooth.

Note that the above construction is completely natural so that it can be glued to yield the same construction globally on arbitrary smooth, but not necessarily affine algebraic varieties.

2.3.16. GOOD FILTRATIONS. Suppose that M is a finitely generated B-module. Assume that we are given a separating $\mathbb{Z}$-filtration on M (that is, a filtration $\ldots M_{-1} \subset M_0 \subset M_1 \ldots$, such that $\cap M_i = \{0\}$) which is compatible with the $\mathbb{Z}$-filtration on B in the sense that

$$B_i \cdot M_j \subset M_{i+j}, \quad \forall i, j \in \mathbb{Z}$$

The associated graded module $\operatorname{gr} M := \oplus M_i/M_{i-1}$ has a natural $\operatorname{gr} B$-module structure. Set $\hat{M} = \oplus_i M_i$, and view it as the submodule $\hat{M} = \sum t^i \cdot M_i \subset M[t, t^{-1}]$. Clearly $\hat{M}$ is a graded $\hat{B}$-module and we have $\hat{M}/t \cdot \hat{M} = \operatorname{gr} M$.

Lemma 2.3.17. *For a filtered B-module M the following conditions are equivalent:*

(a) *There are $m_1, \ldots, m_r \in M$ such that*

$$M_i = B_{i+k_1} \cdot m_1 + \cdots + B_{i+k_r} \cdot m_r, \quad k_1, \ldots, k_r \in \mathbb{Z}$$

(b) *$\hat{M}$ is a $\hat{B}$-lattice in $M[t, t^{-1}]$.*

Proposition 2.3.18. *If the filtration on B is bounded below, i.e., $B_i = 0$ for $i < 0$, then*

$$\operatorname{gr} B \text{ is Noetherian} \implies B \text{ is Noetherian.}$$

Moreover, the conditions of Lemma 2.3.17 *are equivalent to the condition*

$$\operatorname{gr} M \text{ is finitely generated over } \operatorname{gr} B.$$

Proof. Let $m_1, \dots, m_k \in \hat{M}$ be such that their images $\bar{m}_1, \dots, \bar{m}_k$ in $\operatorname{gr} M = \oplus M_i/M_{i-1}$ generate $\operatorname{gr} M$ over $\operatorname{gr} B$. For $m \in M_i$ we have

$$\bar{m} = \bar{b}_1 \bar{m}_1 + \cdots + \bar{b}_k \bar{m}_k, \qquad \bar{b}_i \in \operatorname{gr} B, \bar{m}_i \in \operatorname{gr} M.$$

Hence, $m - \sum b_i m_i \in M_{i-1}$. Iterating this procedure, we obtain eventually that $m - \sum b_i m_i \in M_k$, $k \ll 0$.

Note that since the filtration on B is bounded below and $\operatorname{gr} M$ is finitely generated over $\operatorname{gr} B$ we have $\operatorname{gr}_k M = 0$ for very negative k. Hence, $M_k = 0$ for $k \ll 0$, for the filtration on M is separating. Thus the procedure above terminates so that m can be written as a finite $\hat{B}$-linear combination of $m_1, \dots, m_k$. ∎

A filtration $\{M_\bullet\}$ on a B-module M is called *good* if it satisfies either of the two equivalent conditions of Lemma 2.3.17. Good filtrations are abundant in practice: for instance if M is finitely generated and we choose generators $m_1, \dots, m_k \in M$ then setting

$$M_n = B_n \cdot m_1 + \cdots + B_n \cdot m_k$$

gives rise to a good filtration on M. Of course this depends on the choice of generators and therefore is far from canonical. In spite of that, Lemma 2.3.4 implies the following important corollary.

Corollary 2.3.19. *The class* $[\operatorname{gr} M] \in K^+(\operatorname{gr} B)$ *does not depend on the choice of good filtration.*

The general proposition below has been already used in the previous section and will be frequently used throughout the book. Its proof is straightforward and is left to the reader.

Proposition 2.3.20. *Let M and N be filtered B-modules such that M_i and N_i vanish for all $i \ll 0$. Then we have:*

(i) If $\operatorname{gr} M$ *is a free* $\operatorname{gr} B$*-module of rank r, then M is a free B-module of the same rank.*

(ii) Let $\phi : M \to N$ be a filtration preserving B-module morphism such that the associated graded morphism $\operatorname{gr} \phi : \operatorname{gr} M \to \operatorname{gr} N$ *is an isomorphism. Then ϕ is an isomorphism.*

2.3.21. SPECIALIZATION IN K-THEORY. Let X be a complex affine algebraic variety. Assume we have a flat map $t : X \twoheadrightarrow \mathbb{C}$ which will be viewed as a complex valued regular function on X. Giving t algebraically means giving an algebra homomorphism $\mathbb{C}[t] \hookrightarrow \mathcal{O}(X)$ such that $\mathcal{O}(X)$ is t-torsion-free. Set $X^* = t^{-1}(\mathbb{C}^*)$ and $X_0 = t^{-1}(\{0\})$. Then $\mathcal{O}(X^*) = \mathcal{O}(X)_t$ is the localization of $\mathcal{O}(X)$ at t, and $\mathcal{O}(X_0) = \mathcal{O}(X)/t \cdot \mathcal{O}(X)$. Thus we have the diagram

$$\begin{array}{ccccc} X_0 & \hookrightarrow & X & \hookleftarrow & X^* \\ \downarrow & & \downarrow & & \downarrow \\ \{0\} & \hookrightarrow & \mathbb{C} & \hookleftarrow & \mathbb{C}^* \end{array}$$

We will define a morphism

$$\lim_{t\to 0} : K^+(X^*) \longrightarrow K^+(X_0),$$

from the Grothendieck semigroup of finitely generated $\mathcal{O}(X)_t$-modules to the Grothendieck semigroup of finitely generated $\mathcal{O}(X)/t \cdot \mathcal{O}(X)$-modules.

For $M \in K^+(X^*)$ choose *any* lattice $L \subset M$. Then by definition M is a finitely generated $\mathcal{O}(X)_t$-module, while L is a finitely generated $\mathcal{O}(X)$-module. Therefore $L/t \cdot L$ is an $\mathcal{O}(X)/t \cdot \mathcal{O}(X)$-module whose class in the Grothendieck semigroup is independent of the lattice chosen by Lemma 2.3.4. We set

$$\lim_{t\to 0}[M] = [L/t \cdot L].$$

Lemma 2.3.3 implies that the map $\lim_{t\to 0}$ so defined is a semi-group homomorphism. Hence, it can be extended, by additivity, from K^+ to K, the full Grothendieck group.

Example 2.3.22. Assume, in the above setup that $\mathcal{F}$ is a coherent sheaf on X which is flat over $\mathbb{C}$. Then

$$\lim_{t\to 0}(\mathcal{F}_{|X^*}) = \text{restriction of } \mathcal{F} \text{ to } X_0.$$

To see this, write $X = \operatorname{Specm} A$. Then $X^* = \operatorname{Specm} A_t$, and $X_0 = \operatorname{Specm} A/tA$. Let M be the finitely generated A_t-module corresponding to $\mathcal{F}$. Then the restriction to X_0 of $\mathcal{F}$ corresponds to $M \otimes_A A/tA$. Now, $\mathcal{F}$ being flat is the same as M being t-torsion free. This means in particular that M, regarded as an A-module, is a lattice in itself. Thus

$$\lim_{t\to 0}[M] = [M/tM] = M \otimes_A A/tA.$$

2.4 $\mathbb{C}^*$-actions on a projective variety

Let X be a smooth complex projective variety with an algebraic $\mathbb{C}^*$-action $\mathbb{C}^* \times X \to X$, $(z, x) \mapsto z \cdot x$. Embed the torus $\mathbb{C}^*$ into the Riemann sphere $\mathbb{CP}^1$ so that $\mathbb{CP}^1 \setminus \mathbb{C}^* = \{0\} \cup \{\infty\}$. The following result is a special case of the theorem of Borel [Bo1] on the existence of fixed points of solvable Lie group actions.

Lemma 2.4.1. *For any point $x \in X$ the map $z \mapsto z \cdot x$ has a limit as $z \in \mathbb{C}^*$ approaches $0 \in \mathbb{CP}^1$, resp. ∞. Furthermore, the limit points $\lim_{z\to 0} z \cdot x$ (resp. $\lim_{z\to\infty} z \cdot x$) are fixed points of the $\mathbb{C}^*$-action.*

Proof. Fix $x \in X$. Let $\mathrm{Graph} = \{(z, z \cdot x) \in \mathbb{C}^* \times X, z \in \mathbb{C}^*\}$ be the graph of the $\mathbb{C}^*$-action on x and $Y = \overline{\mathrm{Graph}} \subset \mathbb{CP}^1 \times X$ be its closure in $\mathbb{CP}^1 \times X$. Clearly, Y is a 1-dimensional algebraic variety. Furthermore, the first projection $\mathbb{CP}^1 \times X \to \mathbb{CP}^1$ gives a morphism $p : Y \to \mathbb{CP}^1$ which is an isomorphism over $\mathbb{C}^* = \mathbb{CP}^1 \setminus \{0, \infty\}$. We claim that p is an isomorphism. The claim being proved, the existence of the limits follows from explicit formulae:

$$\lim_{z\to 0} z \cdot x = p^{-1}(0), \qquad \lim_{z\to\infty} z \cdot x = p^{-1}(\infty).$$

To prove the claim consider the normalization (cf. §2.2) $\pi : \tilde{Y} \to Y$ of the curve Y. By [Mum2], $\tilde{Y}$ is a smooth compact complex curve, and the map $p \circ \pi : \tilde{Y} \to \mathbb{CP}^1$ is a birational isomorphism, i.e., induces an isomorphism of the fields of rational functions (possibly with poles) on $\tilde{Y}$ and $\mathbb{CP}^1$ respectively. Hence, $\tilde{Y} \simeq \mathbb{CP}^1$, for there are no birational isomorphisms between curves of different genera. Moreover, any birational automorphism of $\mathbb{CP}^1$ is an isomorphism (this is a special case of the Lüroth theorem, cf. [Ful], [Hum]). It follows that the map $p \circ \pi$, and hence p, are actually isomorphisms.

Observe next that the closure of the orbit $\mathbb{C}^* \cdot x \subset X$ equals $p(Y)$, the second projection of Y. Since $Y \simeq \mathbb{CP}^1$, the closure of $\mathbb{C}^* \cdot x$ is obtained by adding to the orbit at most two points, the images of 0 and ∞. These points form a $\mathbb{C}^*$-stable set. Finally, any $\mathbb{C}^*$-orbit is connected so that each of the points must be a fixed point. ∎

Corollary 2.4.2. [Bo3] *The fixed point set of an algebraic $\mathbb{C}^*$-action on a projective variety is always non-empty.*

Let $\mathbf{W}$ denote the fixed point set of the $\mathbb{C}^*$-action on X, which we will assume to be *finite*. For each $w \in \mathbf{W}$ we define the *attracting set*

$$X_w = \{x \in X \mid \lim_{z\to 0} z \cdot x = w\}.$$

Clearly $w \in X_w$. Since $\mathbb{C}^*$ fixes w there is a natural $\mathbb{C}^*$-action on T_wX, the tangent space of X at w. We have the weight space decomposition into positive and negative eigenspaces:

$$T_wX = T_w^+X \oplus T_w^-X, \qquad T_w^+ = \bigoplus_{n>0\ n\in\mathbb{Z}} T_wX(n), \qquad T_w^- = \bigoplus_{n<0\ n\in\mathbb{Z}} T_wX(n),$$

where the group $\mathbb{Z}$ is identified with $\mathrm{Hom}_{alg}(\mathbb{C}^*, \mathbb{C}^*)$ by means of the isomorphism sending $n \in \mathbb{Z}$ to the homomorphism $z \mapsto z^n$, and where $T_wX(n) = \{x \in T_wX \mid z \cdot x = z^n x, \forall z \in \mathbb{C}^*\}$. Observe that $n = 0$ is not an eigenvalue, since w is an isolated fixed point of the $\mathbb{C}^*$-action. We now state without proof

Theorem 2.4.3. *(Bialynicki-Birula Decomposition* [BiaBi]*)*

(a) *The attracting sets form a decomposition $X = \sqcup_{w\in W} X_w$ into smooth locally closed algebraic subvarieties;*
(b) *There are natural isomorphisms of algebraic varieties $X_w \simeq T_w(X_w) = T_w^+X$ which commute with the $\mathbb{C}^*$-action.*

There is a generalization of this result where the fixed point set $\mathbf{W}$ is not supposed to be discrete in which case the pieces X_w of the Bialynicki-Birula decomposition are parametrized by connected components of $\mathbf{W}$.

In the remainder of this section we discuss the relationship between the Bialynicki-Birula decomposition and Morse theory for Kähler manifolds. Our exposition will be rather sketchy, [At], [GH],[GS2] and [We] for more details.

2.4.4. Linear Algebra of a hermitian Vector Space. Given a real vector space V_R let $(V_R)_{\mathbb{C}} = \mathbb{C} \otimes_{\mathbb{R}} V_R$ denote its complexification and $1 \otimes u + i \otimes v \mapsto \overline{1 \otimes u + i \otimes v} = 1 \otimes u - i \otimes v$ the "complex conjugation" endomorphism on $(V_R)_{\mathbb{C}}$. Assume now that V is a complex vector and write $V_{\mathbb{R}}$ for the real vector space arising from V by forgetting the complex structure. We have an $\mathbb{R}$-linear endomorphism $I : V_{\mathbb{R}} \to V_{\mathbb{R}}$ given by the multiplication by $\sqrt{-1}$ and we extend it, by $\mathbb{C}$-linearity, to a complex endomorphism $I_{\mathbb{C}} : (V_{\mathbb{R}})_{\mathbb{C}} \to (V_{\mathbb{R}})_{\mathbb{C}}$. There is a canonical (complex) direct sum decomposition: $(V_{\mathbb{R}})_{\mathbb{C}} = V' \oplus V''$, where

$$V' = \{z \in (V_{\mathbb{R}})_{\mathbb{C}}\ ,\ I_{\mathbb{C}}(z) = i \cdot z\} \quad , \quad V'' = \{z \in (V_{\mathbb{R}})_{\mathbb{C}}\ ,\ I_{\mathbb{C}}(z) = -i \cdot z\}.$$

Further, the assignment $v \mapsto 1 \otimes v - i \otimes I(v)$, resp. $v \mapsto 1 \otimes v + i \otimes I(v)$, gives a canonical isomorphism $V \simeq V'$, resp. $V \simeq V''$, of complex vector spaces. Note that $v \in V' \Leftrightarrow \bar{v} \in V''$. We write V^{hol} for V' and $\overline{V^{hol}}$ for V'' in the future. Observe also that the operator $I_{\mathbb{C}}$, hence the complex structure on V, is completely determined by the above direct sum decomposition

$$(V_{\mathbb{R}})_{\mathbb{C}} = V^{hol} \oplus \overline{V^{hol}}. \tag{2.4.5}$$

A few remarks on hermitian metrics on a complex vector space V are now in order. First, let $V = \mathbb{C}$ be the complex line with coordinate function $z = x + i \cdot y$. The standard hermitian form on $\mathbb{C}$ is given, for $z_1 = x_1 + i \cdot y_1$ and $z_2 = x_2 + i \cdot y_2 \in \mathbb{C}$ by

$$\langle z_1, z_2 \rangle = z_1 \cdot \bar{z_2} = (x_1 \cdot x_2 + y_1 \cdot y_2) + i \cdot (x_1 \cdot y_2 - x_2 \cdot y_1).$$

More generally, given a finite dimensional complex vector space V, let $\Omega : V_{\mathbb{R}} \times V_{\mathbb{R}} \to \mathbb{C}$ be an $\mathbb{R}$-bilinear form on the underlying real vector space. The form Ω is called a *hermitian metric* on V if the following holds for any $u, v \in V$:

$$\Omega(v,u) = \overline{\Omega(u,v)}, \qquad \Omega(i \cdot u, v) = i \cdot \Omega(u,v) = -\Omega(u, i \cdot v),$$

$$u \neq 0 \;\Rightarrow\; \Omega(u,u) > 0.$$

Given a hermitian metric Ω, write $\Omega(u,v) = (u,v) + i\omega(u,v)$, where $(\cdot,\cdot)$ stands for the real and ω for the imaginary part of Ω. The above conditions imply that $(\cdot,\cdot)$ is a symmetric positive-definite bilinear form on the real vector space $V_{\mathbb{R}}$ and ω is a non-degenerate skew-symmetric form on $V_{\mathbb{R}}$. The forms Ω, ω and $(\cdot,\cdot)$ are related by means of the following identities:

$$(u,v) = \operatorname{Re}\Omega(u,v), \quad \omega(u,v) = \operatorname{Im}\Omega(u,v), \quad \omega(u,v) = (u, i \cdot v).$$

Thus, the real vector space $V_{\mathbb{R}}$ comes equipped with three additional structures:

(a) Euclidean form $(\cdot,\cdot)$ and orientation (coming from the complex structure),
(b) Symplectic form ω,
(c) Complex structure operator $I = \sqrt{-1} : V_{\mathbb{R}} \to V_{\mathbb{R}}$.

One can check also that the orientation in (a) induced by the complex structure is the same as the one induced by the symplectic structure (b). Conversely, given a real $2n$-dimensional vector space V_R with three structures $(\cdot,\cdot)$, ω, I, as above, we obtain a complex n-dimensional vector space V with a hermitian metric, provided the following compatibility conditions hold:

(2.4.6)
$$\omega(u,v) = (u, I \cdot v), \quad (I \cdot u, I \cdot v) = (u,v), \quad \omega(I \cdot u, I \cdot v) = \omega(u,v).$$

Observe that any two of the three structures determine the third one completely due to the compatibility conditions.

Given a complex vector space V with hermitian form $\Omega = (\cdot,\cdot) + i \cdot \omega$, extend the 2-form ω to a complex symplectic 2-form $\omega_{\mathbb{C}} : (V_{\mathbb{R}})_{\mathbb{C}} \times (V_{\mathbb{R}})_{\mathbb{C}} \to \mathbb{C}$. The form $\omega_{\mathbb{C}}$ is compatible with complex conjugation in the sense that $\omega_{\mathbb{C}}(\bar{u}, \bar{v}) = \overline{\omega_{\mathbb{C}}(u,v)}$. Further, recall decomposition 2.4.5 $(V_{\mathbb{R}})_{\mathbb{C}} = V^{hol} \oplus \overline{V^{hol}}$.

Claim 2.4.7. V^{hol} and $\overline{V^{hol}}$ are complex lagrangian subspaces relative to the symplectic 2-form $\omega_{\mathbb{C}}$.

Proof. We prove the claim for V^{hol}; the other one follows by complex conjugation. Recall that $V^{hol} \simeq V$ so that any element of V^{hol} is of the form $1 \otimes v - i \otimes I(v)$, $v \in V$. Thus, we calculate

$$\omega_{\mathbb{C}}(1 \otimes v - i \otimes I(v), 1 \otimes u - i \otimes I(u)) = \\ \omega(v,u) - i \cdot \omega(v, I(u)) - i \cdot \omega(I(v), u) - \omega(I(v), I(u)).$$

Both the real and the imaginary parts of this expression vanish due to the identities (2.4.6). Hence, V^{hol} is an isotropic subspace. The dimension equality $\dim_{\mathbb{C}} V^{hol} = \dim_{\mathbb{C}} V = 1/2 \dim_{\mathbb{C}} (V_{\mathbb{R}})_{\mathbb{C}}$ completes the proof. ∎

A similar computation shows that $\sqrt{-1} \cdot \omega_{\mathbb{C}}(v, \bar{v}) > 0$ for any $v \in (V_{\mathbb{R}})_{\mathbb{C}}$, $v \neq 0$. This, together with the fact that the complex structure on V is completely determined by the subspace $V^{hol} \subset (V_{\mathbb{R}})_{\mathbb{C}}$, yields the following result (see [We], [GS1] for details).

Proposition 2.4.8. *Let (V_R, ω) be a real symplectic vector space. Giving a hermitian structure on V_R with imaginary part equal to ω amounts to giving a complex lagrangian subspace $V^{hol} \subset (V_R)_{\mathbb{C}}$ such that there is a complex direct sum decomposition*

$$(V_R)_{\mathbb{C}} = V^{hol} \oplus \overline{V^{hol}}, \quad \text{and } \sqrt{-1} \cdot \omega_{\mathbb{C}}(v, \bar{v}) > 0, \quad \forall v \in (V_R)_{\mathbb{C}}, v \neq 0.$$

Given a hermitian vector space V of complex dimension n, let

$$SO_{2n}(\mathbb{R}), \; Sp_{2n}(\mathbb{R}), \; GL_n(\mathbb{C})$$

denote the groups of $\mathbb{R}$-linear automorphisms of $V_{\mathbb{R}}$ preserving the structures (a), (b) and (c) of (2.4.6) respectively, and let SU_n be the group preserving all three structures, the special unitary group. Since any two of the structures determine the third one, the group SU_n turns out to be equal to the intersection of any pair of the groups above:

$$SO_{2n}(\mathbb{R}) \cap Sp_{2n}(\mathbb{R}) = Sp_{2n}(\mathbb{R}) \cap GL_n(\mathbb{C}) = GL_n(\mathbb{C}) \cap SO_{2n}(\mathbb{R}) = SU_n.$$

2.4.9. KÄHLER MANIFOLDS. Given a C^∞-manifold X_R, let TX_R denote the tangent bundle on X and $(TX_R)_{\mathbb{C}}$ its complexification, a complex C^∞-vector bundle on X_R. If X is a complex manifold, write $X_{\mathbb{R}}$ for the underlying real C^∞-manifold. We have a vector bundle analogue of decomposition (2.4.5)

$$(TX_{\mathbb{R}})_{\mathbb{C}} = TX^{hol} \oplus \overline{TX^{hol}}. \tag{2.4.10}$$

A hermitian form on the complex manifold X is a C^∞-map $\Omega : x \mapsto \Omega_x$ assigning to a point $x \in X$ a hermitian form $\Omega_x : T_x X_{\mathbb{R}} \times T_x X_{\mathbb{R}} \to \mathbb{C}$ on

the tangent space at x (that is, Ω is a C^∞-section of the vector bundle $(TX^{hol} \otimes \overline{TX^{hol}})^*$). The real part, $\mathrm{Re}\,\Omega$, gives a Riemannian metric on the real manifold $X_{\mathbb{R}}$ and the imaginary part, $\mathrm{Im}\,\Omega$, gives a differential 2-form ω on $X_{\mathbb{R}}$. Here is a "non-linear" extension of Proposition 2.4.8 from the setup of linear algebra to that of differential geometry.

Theorem 2.4.11. *(cf.,*[GH], [GS2]*) Let X_R be a real C^∞-manifold. Then the following three sets of structure on X_R are equivalent:*

(a) The structure of a complex manifold X with a hermitian form Ω having the following additional property: for any point $x \in X$, there are holomorphic local coordinates $z_1, \ldots, z_n$ with the origin at x such that

$$\Omega(\tfrac{\partial}{\partial z_i}, \tfrac{\partial}{\partial z_j}) = \delta_{ij} + O(|z|^2).$$

(b) The structure of a complex manifold with a hermitian form Ω such that the real 2-form $Im\Omega$ is closed.

(c) The real symplectic structure ω on X_R and a complex vector subbundle $TX^{hol} \subset (TX_R)_{\mathbb{C}}$ such that:

(1) TX^{hol} has lagrangian fibers with respect to the complex 2-form $\omega_{\mathbb{C}}$;

(2) TX^{hol} is an integrable subbundle of $(TX_R)_{\mathbb{C}}$ in the sense of Frobenius Theorem 1.5.4 *extended to the complexification by complex linearity;*

(3) there is a vector bundle direct sum decomposition

$$(TX_R)_{\mathbb{C}} = TX^{hol} \oplus \overline{TX^{hol}};$$

(4) Positivity: $\sqrt{-1} \cdot \omega_{\mathbb{C}}(v, \bar{v}) > 0$ for any $v \in TX^{hol}$, $v \neq 0$.

The hermitian form Ω in (a), (b) and the symplectic form ω in (c) are related by the equation $\omega = \mathrm{Im}\ \Omega$.

The real manifold X is called *Kähler* if it is equipped with any of the equivalent structures of the theorem above. For example, the standard hermitian form $\Omega(u, v) = \sum_{i=0}^{n} u_i \cdot \bar{v}_i$ on $\mathbb{C}^n$ makes $X = \mathbb{C}^n$ a Kähler manifold.

2.4.12. FUBINI-STUDI METRIC. We will study in more detail the special case $X = \mathbb{CP}^n$, to be denoted $\mathbb{P}^n$ for short. There is a distinguished Kähler metric on $\mathbb{P}^n$ which we now define.

Let Ω be the standard hermitian form on $\mathbb{C}^{n+1}$. Let $S^{2n+1} \subset \mathbb{C}^{n+1}$ be the unit sphere and let $S^1 = \{z \in \mathbb{C}^* \mid |z| = 1\}$ be the unit circle. Recall that the group $\mathbb{C}^*$ acts freely on $\mathbb{C}^{n+1} \setminus \{0\}$ and by definition $\mathbb{P}^n := (\mathbb{C}^{n+1} \setminus \{0\})/\mathbb{C}^*$. This orbit space of a complex group inherits the natural structure of a complex manifold. Observe further that the action of the subgroup $S^1 \subset \mathbb{C}^*$ on $\mathbb{C}^{n+1}$ preserves the sphere S^{2n+1}. Moreover, the embedding $S^{2n+1} \hookrightarrow \mathbb{C}^{n+1}$ yields an isomorphism of *real* C^∞-manifolds:

$$S^{2n+1}/S^1 = (\mathbb{C}^{n+1} \setminus \{0\})/\mathbb{C}^* = \mathbb{P}^n. \tag{2.4.13}$$

The real part of the hermitian form on $\mathbb{C}^{n+1}$ induces by restriction a Riemannian metric on S^{2n+1}. The group S^1 clearly acts on S^{2n+1} by isometries, so that the metric on S^{2n+1} gives rise to a Riemannian metric on the quotient S^{2n+1}/S^1. When transported to $\mathbb{P}^n$ by means of the isomorphism 2.4.13, this metric turns out to be compatible with the natural complex structure on $\mathbb{P}^n$ and, moreover, turns out to be Kähler.

In more detail, consider the tautological complex line bundle L on $\mathbb{P}^n$, i.e., the line bundle associated with the principal $\mathbb{C}^*$-bundle $\mathbb{C}^{n+1} \setminus \{0\} \to (\mathbb{C}^{n+1} \setminus \{0\})/\mathbb{C}^* = \mathbb{P}^n$. In other words, the fiber of L at a point $[x] \in \mathbb{P}^n$ is the line in $\mathbb{C}^{n+1}$ spanned by x. Thus we may view L as a subbundle of the trivial bundle $\mathbb{C}_{\mathbb{P}}^{n+1}$ on $\mathbb{P}^n$ with fiber $\mathbb{C}^{n+1}$. It is clear that the (holomorphic) tangent bundle on $\mathbb{P}^n$ can be expressed in terms of L as follows:

$$T^{hol}\mathbb{P}^n = T((\mathbb{C}^{n+1} \setminus \{0\})/\mathbb{C}^*) = \operatorname{Hom}(L, \mathbb{C}_{\mathbb{P}}^{n+1}/L)$$

The standard hermitian 2-form, $\Omega(u,v)$, on $\mathbb{C}^{n+1}$ gives rise to a hermitian metric on the trivial bundle $\mathbb{C}_{\mathbb{P}}^{n+1}$. The later one induces a hermitian metric on the subbundle L by restriction and on the quotient bundle $\mathbb{C}_{\mathbb{P}}^{n+1}/L$, by orthogonal projection. This way the tangent bundle $T^{hol}\mathbb{P}^n = \operatorname{Hom}(L, \mathbb{C}_{\mathbb{P}}^{n+1}/L)$ acquires a natural hermitian metric which we will denote again by Ω.

The standard $GL_{n+1}(\mathbb{C})$-action on $\mathbb{C}^{n+1}$ induces a natural $GL_{n+1}(\mathbb{C})$-action on $\mathbb{P}^n$ by projective linear transformations. By construction the group $\mathbb{U} \subset GL_{n+1}(\mathbb{C})$ of all unitary transformations preserves the metric on $T^{hol}\mathbb{P}^n$. It follows that the real 2-form $\omega = \operatorname{Im}\Omega$ must be $\mathbb{U}$-invariant and in particular closed. Hence Ω is a Kähler metric on $\mathbb{P}^n$. This Kähler metric is known as the Fubini-Studi metric.

Due to the fact that the group $\mathbb{U}$ acts on $\mathbb{P}^n$ preserving the symplectic structure ω, each element of the Lie algebra $\operatorname{Lie}\mathbb{U}$ gives rise to a symplectic vector field on $\mathbb{P}^n$.

Proposition 2.4.14. *The $\mathbb{U}$-action on $\mathbb{P}^n$ is Hamiltonian with respect to the Fubini-Studi symplectic form ω. More precisely, the assignment $a \mapsto H_a$ given by the formula*

$$H_a(v) = (a \cdot v, v), \quad v \in S^{2n+1}/S^1$$

is a Lie algebra homomorphism $\operatorname{Lie}\mathbb{U} \to C^\infty(\mathbb{P}^n)$ such that $i_a\omega = dH_a$.

Proof. Left to the reader. ∎

Remark 2.4.15. Observe that the expression $(a \cdot v, v)$ is well-defined, i.e., the function $v \mapsto (a \cdot v, v)$ on S^{2n+1} is S^1-invariant, hence, descends to S^{2n+1}/S^1.

Now let G be a complex reductive group with maximal compact subgroup $K \subset G$ and let X be a smooth quasi-projective G-variety. By the equivariant projective embedding theorem 5.1.25 of part 5 below, we can find an algebraic group homomorphism $\rho : G \to \mathrm{GL}_{n+1}(\mathbb{C})$ and a projective embedding $i : X \hookrightarrow \mathbb{P}^n$ so that $i(g \cdot x) = \rho(g) \cdot i(x), \forall g \in G, x \in X$ since the image of K is a compact subgroup of $\mathrm{GL}_n(\mathbb{C})$, there exists a K-invariant hermitian form on $\mathbb{C}^{n+1}$. Fix such a form and let Ω be the corresponding Fubini-Studi metric on $\mathbb{P}^n$. Then $i^*\Omega$, the pullback of Ω to X, is a Kähler metric on X. Furthermore, we have $\rho(K) \subset \mathbb{U}$, hence, the Kähler form $i^*\Omega$ is K-invariant. Thus, any G-equivariant projective embedding $X \hookrightarrow \mathbb{P}^n$ makes X a symplectic C^∞-manifold with symplectic 2-form $\mathrm{Im}\,(i^*\Omega)$, the imaginary part of the Kähler form. Moreover, Proposition 2.4.14 yields

Corollary 2.4.16. *The K-action on X is Hamiltonian with respect to the form* $\mathrm{Im}\,(i^*\Omega)$.

We return finally to the setup of the beginning of this section, i.e., to the special case where $G = \mathbb{C}^*$ and X is a smooth projective $\mathbb{C}^*$-variety with isolated fixed points. Let $S^1 \subset \mathbb{C}^*$ be the unit circle, the maximal compact subgroup of $\mathbb{C}^*$, and let ξ be the vector field on X generating the S^1-action. From Corollary 2.4.16 we obtain (setting $K = S^1$) the following result.

Lemma 2.4.17. *Let Ω be the S^1-equivariant Kähler form on X arising from a $\mathbb{C}^*$-equivariant projective embedding as above. Then there exists a function $H \in C^\infty(X)$ such that $i_\xi(\mathrm{Im}\,\Omega) = dH$.*

Remark 2.4.18. The above result is not quite trivial. Its alternative proof based on Hodge theory is given in [CL].

2.4.19. MORSE THEORY REVIEWED. We recall a few basic definitions. Let $H : M \to \mathbb{R}$ be a C^∞-function on a smooth manifold M. The point $x \in M$ is called a *critical point* of H if d_xH, the differential of H at x, vanishes.

Let x be a critical point of the function H. Associate to it a bilinear form on the space of vector fields on a small neighborhood of x by the following assignment: $(u, v) \mapsto u(v(H))_{|x}$. The expression on the right clearly depends not on u but only on its value at x. The equation $[u, v](H)_{|x} = 0$ shows that $u(v(H))_{|x} = v(u(H))_{|x}$ so that the dependence of the form on v is only by means of its value at x. Thus, the bilinear form on the vector fields descends to a symmetric bilinear form on the tangent space T_xM, called the Hessian of H and denoted d_x^2H. In local coordinates $\{x_i\}$ near the critical point, the matrix of the Hessian is given by the second partial derivatives:

$$d_x^2H = \|\frac{\partial^2 H}{\partial x_i \partial x_j}(x)\|.$$

Definition 2.4.20. (see [Mi]) A real C^∞-function H on a smooth manifold M is called a *Morse function* if the following conditions hold:

(a) The critical points of H are isolated;
(b) At each critical point x, the Hessian form $d_x^2 H$ is a non-degenerate bilinear form.

Now let M be a compact C^∞-manifold with a Riemannian metric $(\cdot,\cdot)$. Any C^∞-function H on M gives rise to the gradient vector field on M defined by the equation $(\mathrm{grad}H,\cdot) = dH(\cdot)$. There is a *gradient flow* corresponding to the gradient vector field, a C^∞-map $\Phi : M \times \mathbb{R} \to M$ satisfying the differential equation

$$\frac{d\Phi(x,t)}{dt} = \mathrm{grad}H_{|\Phi(x,t)} \quad , \quad \Phi(x,0) = x \ , \ x \in M.$$

Assume that H is a Morse function on the Riemannian manifold M. There are two symmetric non-degenerate bilinear forms on the tangent space T_xM at each critical point $x \in M$. The first form is the Riemannian metric $(\cdot,\cdot)_x$ and the second is the Hessian $d_x^2 H$. Hence, there is a self-adjoint linear operator $A_x : T_xM \to T_xM$ uniquely determined by the condition $d_x^2 H(u,v) = (A_x u, v)$, $\forall u, v \in T_xM$. Let T_x^+M (resp. T_x^-M) denote the span of the eigenspaces of A_x corresponding to positive (resp. negative) eigenvalues. We have $T_xM = T_x^+M \oplus T_x^-M$, and there are no zero-eigenspaces since A_x is non-degenerate. Here is a version of one of the main results of Morse theory. See [GM1],[Mi] and references therein for proofs and more details.

Theorem 2.4.21. *Let H be a Morse function on a compact Riemannian manifold M with the set $\mathbf{W}$ (necessarily finite) of critical points. Then we have*

(i) For any $x \in M$, the gradient flow $t \mapsto \Phi(x,t)$ has limits as $t \to \pm\infty$ and those limits are critical points of H.

(ii) For each critical point $w \in \mathbf{W}$ the attracting set

$$M_w = \{x \in M | \lim_{t \to -\infty} \Phi(x,t) = w\}.$$

is diffeomorphic to the vector space T_w^-M.

(iii) The attracting sets M_w, $w \in \mathbf{W}$, form a cell decomposition $M = \coprod_w M_w$.

We are now in a position to establish a link between the Bialynicki-Birula decomposition and Morse theory.

Let X be a smooth complex projective variety with an algebraic $\mathbb{C}^*$-action having finitely many fixed points. Let $S^1 \subset \mathbb{C}^*$ be the unit circle and let Ω be the S^1-invariant Kähler form on X arising from an equivariant projective embedding (cf., Proposition 5.1.25). Let $\mathbb{R}^{>0} \subset \mathbb{C}^*$ be the

multiplicative subgroup of the positive real numbers. The following result plays a key role in relating the *complex* geometry of the $\mathbb{C}^*$-action on X with the *real* symplectic geometry arising from the Kähler form.

Proposition 2.4.22. *(cf.* [At],[GS1],[Kirw]*)*

(a) *The function H, introduced in Lemma* 2.4.17*, is a Morse function on X whose critical points are precisely the fixed points of the $\mathbb{C}^*$-action.*

(b) *The orbits of the gradient flow (with respect to the Kähler metric) associated to the function H coincide with the orbits of the natural $\mathbb{R}^{>0}$-action on X obtained from the $\mathbb{C}^*$-action by restriction.*

Proof. (a) Observe that given the $\mathbb{C}^*$-action on X holomorphic we have that x is a $\mathbb{C}^*$-fixed point if and only if x is an S^1-fixed point $\Leftrightarrow \xi(x) = 0$.

Since the Kähler form ω is non-degenerate we thus obtain: x is a $\mathbb{C}^*$-fixed point $\Leftrightarrow \xi(x) = 0$ if and only if $(i_\xi\omega)(x) = 0 \Leftrightarrow \mathrm{d}H(x) = 0 \Leftrightarrow x$ is a critical point of H.

We refer the reader to [Mi] for the proof of the non-degeneracy of the Hessian at critical points.

(b) Let η be the vector field on X generating the $\mathbb{R}^{>0}$-action. Proving part (b) amounts to showing that grad $H = \eta$ or, equivalently, that

$$\mathrm{Re}\,\Omega(\mathrm{grad}\,H, u) = \mathrm{Re}\,\Omega(\eta, u) \tag{2.4.23}$$

for any vector field u on X. The LHS of (2.4.23) can be rewritten as $\mathrm{d}H(u)$, by definition of the gradient vector field. On the other hand, the RHS equals $\mathrm{Im}\,\Omega(\sqrt{-1}\cdot\eta, u)$. Observe further that the Lie algebra of the subgroup $S^1 \subset \mathbb{C}^*$, resp. $\mathbb{R}^{>0} \subset \mathbb{C}^*$, gets identified by means of the exponential map $\exp : \mathbb{C} \to \mathbb{C}^*$ with the subspace $\sqrt{-1}\cdot\mathbb{R} \subset \mathbb{C}$, resp. $\mathbb{R} \subset \mathbb{C}$. Therefore we have $\sqrt{-1}\cdot\eta = \xi$. Also $\mathrm{Im}\,\Omega$ is by definition the symplectic 2-form ω associated with the Kähler form Ω. Since $i_\xi\omega = \mathrm{dH}$ we find

$$\text{RHS of (2.4.23)} = \mathrm{Im}\,\Omega(\xi, u) = (i_\xi \mathrm{Im}\,\Omega)(u) = \mathrm{dH}(u) = \text{LHS of (2.4.23)}.$$

This completes the proof. ∎

Corollary 2.4.24. *The Bialynicki-Birula decomposition of X associated to the $\mathbb{C}^*$-action coincides with the cell decomposition of X associated to the Morse function H in Theorem* 2.4.21 *by means of Morse theory.*

Thus, the Bialynicki-Birula decomposition on a Kähler manifold can be deduced (at least as a C^∞-cell decomposition) from Morse theory.

2.5 Fixed Point Reduction

In this section we prove a general result relating the cohomology of a variety with that of a fixed point subvariety. This result will play an important role in our approach to the classification of irreducible representations of affine Hecke algebras given in Chapter 8, and is useful in many other matters as well.

Let L be a Lie group and X a "reasonable" (see beginning of 2.6) topological space, such as a possibly singular closed complex subvariety of a complex manifold or a finite-dimensional CW-complex, with a continuous L-action. Let T be compact torus (a product of several copies of the circle S^1) contained in the center of L. Let X^T denote the fixed point set of T. Since the L-action commutes with that of T, the subvariety X^T is L-stable. The following result relates the cohomology of X with that of X^T.

Proposition 2.5.1. *Assume that the group L has finitely many connected components. Then*

(i) The following equality holds in the Grothendieck group of finite-dimensional L-modules: $[H^*(X,\mathbb{C})] = [H^*(X^T,\mathbb{C})]$.

(ii) $H^{odd}(X,\mathbb{C}) = 0$ *implies* $H^{odd}(X^T,\mathbb{C}) = 0$.

Let $\bar{L}$ denote the group of components of L which is finite by assumption. An L-action on cohomology always factors through $\bar{L}$. Hence, the equation of part (i) holds in effect in the representation ring $R(\bar{L})$ of the finite group $\bar{L}$. This remark applies everywhere below: the group L may always be replaced, as long as the cohomology is concerned, by its finite quotient.

Proof. Arguing by induction on $\dim T$, one may assume without loss of generality that $T = S^1$. Let $\mathbb{C}^\infty$ be an infinite dimensional complex vector space with a hermitian metric. One may take $\mathbb{C}^\infty$ to be either the direct limit of the standard direct system $\mathbb{C} \hookrightarrow \mathbb{C}^2 \hookrightarrow \mathbb{C}^3 \hookrightarrow \dots$ of finite dimensional hermitian vector spaces (with the direct limit topology), or a complete separable complex Hilbert space. The unit circle $S^1 \subset \mathbb{C}^*$ acts freely on S^∞, say on the right, and we write $\mathbb{CP}^\infty$ for the orbit space (which is either the direct limit of a system of finite dimensional projective spaces or the projective space associated to a Hilbert space, depending on our choice of $\mathbb{C}^\infty$). Then the natural projection $S^\infty \xrightarrow{S^1} \mathbb{CP}^\infty$ is the standard model for a universal S^1-bundle. (A universal S^1-bundle is a principal S^1-bundle with contractible total spaces: we exploit that the sphere S^∞ is known to be *contractible*.)

For any topological space X with a left S^1-action, we define the Borel mixing construction (cf. [AtBo]) on X as

$$X_T := S^\infty \times_{S^1} X = \{(s,x) \in S^\infty \times X\}/(sg,x) \sim (s,gx), \quad g \in S^1\}.$$

The second projection $S^\infty \times X \to X$ induces a canonical fibration

$$\pi : X_T \xrightarrow{X} \mathbb{CP}^\infty \tag{2.5.2}$$

with fiber X. Thus, the cohomology $H^*(X_T, \mathbb{C})$ becomes an $H^*(\mathbb{CP}^\infty, \mathbb{C})$-module by means of π^*. Recall that, see [Sp],

$$H^*(\mathbb{CP}^\infty, \mathbb{C}) \simeq \mathbb{C}[u], \qquad deg\, u = 2 \tag{2.5.3}$$

is the polynomial algebra in one variable u of degree 2.

The proof of the proposition is based on the Leray spectral sequence for the fibration (2.5.2):

$$E_2^{p,q} = H^p(X) \otimes H^q(\mathbb{CP}^\infty) \Rightarrow H^{p+q}(X_T). \tag{2.5.4}$$

This is a multiplicative first quadrant spectral sequence with finitely many rows, since $H^p(X)$ vanishes for $p >> 0$. In particular, the spectral sequence converges. Further, the differentials in the spectral sequence are $H^*(\mathbb{CP}^\infty)$-linear maps that commute, in addition, with the natural action of the finite group $\bar{L}$.

Let $\mathbf{K}$ be the Grothendieck group of finitely generated graded $\mathbb{C}[u]$-modules equipped with $\bar{L}$-action preserving the grading. Any graded component of such a module is clearly a finite dimensional representation of $\bar{L}$. Next, write $E_k^\pm = \oplus_{r=p+q} E_k^{p,q}$, where the summation is taken over all *even* values of r in the "+" case and over all *odd* values of r in the "-" case, respectively; we use similar notation for the other graded spaces and identify $H^*(\mathbb{CP}^\infty)$ with $\mathbb{C}[u]$. This way both $H^\pm(X_T)$ and all the terms $E_k^\pm$ of the spectral sequence become graded $\bar{L}$-equivariant $\mathbb{C}[u]$-modules. Observe that these modules give rise to well-defined classes in the Grothendieck group $\mathbf{K}$.

We now recall the so-called Euler-Poincaré principle. It says, that for any bounded complex $\{C^i, d\}$ with cohomology groups $H^i = \mathrm{Ker}\,(C^i \xrightarrow{d} C^{i+1})/\mathrm{Im}\,(C^{i-1} \xrightarrow{d} C^i)$ in an appropriate Grothendieck group, one has an equation

$$C^+ - C^- = H^+ - H^-. \tag{2.5.5}$$

Further, the differential of the spectral sequence makes each term $E_k^\pm$ of the spectral sequence (2.5.4) a complex of $\mathbb{C}[u]$-modules with $\bar{L}$-action. The cohomology of this complex is by construction of the spectral sequence, just the next term $E_{k+1}^\pm$ of the spectral sequence. Hence, (2.5.5) yields the following equation in the Grothendieck group $\mathbf{K}$:

$$E_k^+ - E_k^- = E_{k+1}^+ - E_{k+1}^-.$$

The spectral sequence being convergent, for each n, the n-th graded component of the modules $E_2^\pm, E_3^\pm, \ldots$ stabilizes to that of the limit module

$E_\infty^\pm$. Therefore, iterating the above argument, we find that

$$E_2^+ - E_2^- = E_\infty^+ - E_\infty^- \quad \text{in} \quad \mathbf{K}.$$

On the other hand, the term E_∞ of the spectral sequence is known to be, see [BtTu], the associated graded (with respect to a certain filtration) to the RHS of (2.5.4). A similar argument then yields the equation

$$E_\infty^+ - E_\infty^- = H^+(X_T) - H^-(X_T).$$

Combining the last two equalities with isomorphism $E_2^\pm = H^\pm(X) \otimes \mathbb{C}[u]$ we obtain

$$H^+(X) \otimes \mathbb{C}[u] - H^-(X) \otimes \mathbb{C}[u] = H^+(X_T) - H^-(X_T). \tag{2.5.6}$$

Next, we apply the specialization map 2.3.21 in K-theory (for $\bar{L}$-equivariant $\mathbb{C}[u]$-modules). Since $H^\pm(X) \otimes \mathbb{C}[u, u^{-1}]$ is a free $\mathbb{C}[u, u^{-1}]$-module, we clearly have

$$\lim_{u \to 0} \left[H^\pm(X) \otimes \mathbb{C}[u, u^{-1}] \right] = H^\pm(X).$$

Using equation (2.5.6) we get

$$H^+(X) - H^-(X) = \lim_{u \to 0} \left[\mathbb{C}[u, u^{-1}] \otimes_{\mathbb{C}[u]} H^+(X_T) - \mathbb{C}[u, u^{-1}] \otimes_{\mathbb{C}[u]} H^-(X_T) \right]. \tag{2.5.7}$$

The RHS of this expression can be related to the cohomology of X^T, the fixed point set, by means of the localization theorem for equivariant cohomology, see [AtBo], (analogous to the Localization Theorem 5.10 in our book for equivariant K-theory). It says that there is a *parity preserving* (but not degree preserving) isomorphism of $\mathbb{C}[u, u^{-1}]$-modules:

$$\mathbb{C}[u, u^{-1}] \otimes_{\mathbb{C}[u]} H^\pm(X_T, \mathbb{C}) \simeq \mathbb{C}[u, u^{-1}] \otimes_{\mathbb{C}} H^\pm(X^T, \mathbb{C}). \tag{2.5.8}$$

This isomorphism combined with (2.5.7) yields the following equation in $\mathbf{K}$.

$$\begin{aligned} H^+(X) - H^-(X) &= \lim_{u \to 0} \left[\mathbb{C}[u, u^{-1}] \otimes_{\mathbb{C}[u]} H^+(X_T) - \mathbb{C}[u, u^{-1}] \otimes_{\mathbb{C}[u]} H^-(X_T) \right] \\ &= \lim_{u \to 0} \left[\mathbb{C}[u, u^{-1}] \otimes_{\mathbb{C}} H^+(X^T) - \mathbb{C}[u, u^{-1}] \otimes_{\mathbb{C}} H^-(X^T) \right] \\ &= H^+(X^T) - H^-(X^T). \end{aligned}$$

This completes the proof of part (i) of the proposition. To prove (ii) assume $H^{odd}(X) = 0$. Then all the terms in the spectral sequence involving non-even p or q vanish, due to (2.5.3). Thus, all the differentials in the spectral sequence are zero, and the spectral sequence collapses. It follows that the spectral sequence degenerates at the E_2-term, and we have

$$H^-(X) = 0 \implies E_2^- = E_\infty^- = 0 \implies H^-(X_T) = 0.$$

The localization isomorphism (2.5.8) now implies $H^-(X_T) = 0 \Longrightarrow H^-(X^T) = 0$, and part (ii) follows. ∎

Corollary 2.5.9. *If the variety X has no odd-dimensional rational homology, then in $R(\bar{L})$ we have an equation $H^{even}(X) = H^{even}(X^T)$.*

Remark 2.5.10. For a smooth projective variety X with an algebraic $\mathbb{C}^*$-action, Proposition 2.5.1 can be deduced from the Bialnicki-Birula decomposition. This argument gives a stronger result than part (ii): it shows that $H^{odd}(X, \mathbb{C}) = 0$ *if and only if* $H^{odd}(X^T, \mathbb{C}) = 0$. For such an argument see e.g. [KL4, sec. 4]. A similar result can also be obtained (using Morse theory methods) for an arbitrary Hamiltonian S^1-action on a general compact symplectic manifold.

2.6 Borel-Moore Homology

Borel-Moore homology will be the principal functor we use in this book for constructing representations of Weyl groups, enveloping algebras, and Hecke algebras. We review here the most essential properties of the Borel-Moore homology theory and refer the reader to the monographs [Bre] and [Iv] for a more detailed treatment of the subject.

We have to say a few words about the kind of spaces we will be dealing with, which will sometimes be called "reasonable." By a "space" (in the topological sense) we will mean a locally compact topological space X that has the homotopy type of a finite CW-complex, in particular, has finitely many connected components and has finitely generated homotopy and homology groups (with $\mathbb{Z}$-coefficients). Furthermore, our space X is assumed to admit a closed embedding into a countable at infinity C^∞-manifold M (in particular, X is paracompact). We assume also that there exists an open neighborhood $U \supset X$ in M such that X is a homotopy retract of U. Similarly, by a closed "subset" of a C^∞-manifold we always mean a subset X which has an open neighborhood $U \supset X$ such that X is a homotopy retract of U. In this case one can also find a smaller *closed* neighborhood $V \subset U$ such that X is a *proper* homotopy retract of V (recall that a continuous map $f : X \to Y$ is called *proper* if the inverse image of any compact set is compact). It is known, cf. [GM], [RoSa], that any complex or real algebraic variety satisfies the above conditions. These are mostly the spaces we will use in applications.

We now give a list of the various equivalent definitions of Borel-Moore homology of a space X, see [BoMo],[Bre]. Everywhere below all homology and cohomology are taken with complex coefficients, which may be replaced by any field of characteristic zero.

(1) Let $\hat{X} = X \cup \{\infty\}$ be the one-point compactification of X. Define $H_*^{BM}(X) = H_*(\hat{X}, \infty)$, where H_* is ordinary relative homology of the pair $(\hat{X}, \infty)$.

(2) Let $\overline{X}$ be an arbitrary compactification of X such that $(\overline{X}, \overline{X}\backslash X)$ is a CW-pair. Then, $H_*^{BM}(X) \simeq H_*(\overline{X}, \overline{X}\backslash X)$, see [Sp]. The fact that this definition agrees with (1) is proved in [Bre].

(3) Let $C_*^{BM}(X)$ be the chain complex of *infinite* singular chains $\sum_{i=0}^{\infty} a_i\sigma_i$, where σ_i is a singular simplex, $a_i \in \mathbb{C}$, and the sum is locally finite in the following sense: for any compact set $D \subset X$ there are only finitely many non-zero coefficients a_i such that $D \cap \operatorname{supp}\sigma_i \neq \emptyset$. The usual boundary map ∂ on singular chains is well-defined on $C_*^{BM}(X)$ because taking boundaries cannot destroy the finiteness condition. We then have

$$H_*^{BM}(X) = H_*(C_*^{BM}(X), \partial).$$

(4) Poincaré duality: let M be a smooth, oriented manifold, and $\dim_{\mathbb{R}} M = m$. Let X be a closed subset of M which has a closed neighborhood $U \subset M$ such that X is a proper deformation retract of U. Then there is a canonical isomorphism (cf. [Iv], [Bre]):

$$H_i^{BM}(X) \simeq H^{m-i}(M, M\backslash X), \tag{2.6.1}$$

where each side of the equality is understood to be with complex coefficients. In particular, setting $X = M$ we obtain, for any *smooth* not necessarily compact variety M, a canonical isomorphism (depending on the orientation of M)

$$H_i^{BM}(M) \simeq H^{m-i}(M). \tag{2.6.2}$$

We will often use the "Poincaré duality" definition to prove many of the basic theorems about Borel-Moore homology by appealing to the same theorems for singular cohomology. In these instances we will refer the reader to [Bre], [Sp] for the proofs in singular cohomology, despite the fact that Borel-Moore homology is not explicitly developed there.

There is one more definition of Borel-Moore homology with real coefficients based on the de Rham approach, which generalizes the ordinary de Rham complex on a smooth manifold. Its equivalence to the other definitions can be taken to be a "singular-space de Rham" theorem.

Let M be a smooth m-dimensional manifold, and $X \subset M$ a (possibly singular) closed subset. Write $\Omega_c^*(M)$ for the vector space of C^∞-forms on M with compact support equipped with the standard topology, see e.g. [Ru], of uniform convergence (of partial derivatives) on each compact set. A continuous linear function $\Phi : \Omega_c^{m-k} \to \mathbb{R}$ is called a *distribution* of degree k. The continuity condition can be spelled out explicitly as follows. Given Φ and any compact subset $K \subset M$ there exist: (1) integers $L > 0$, $N \geq$

$m-k$; (2) C^∞-vector fields $\xi_1,\dots,\xi_L$ and $\eta_1,\dots,\eta_N$ defined on an open neighborhood of K, and (3) a constant $C>0$ (all the data depend on Φ and K), such that

$$|\,\Phi(\omega)\,| \leq C\cdot \max_{x\in K} \sum_{0<i_1,\dots i_l\leq L; j_1,\dots j_{m-k}\leq N} |\,\xi_{i_1}\cdot\dots\cdot\xi_{i_l}\,\omega(\eta_{j_1},\dots,\eta_{j_{m-k}})\,|$$

for any $m-k$-form ω supported on K.

2.6.3. DISTRIBUTION DE RHAM COMPLEX. Let $D_i(M)$ denote the vector space of degree i distributions on M. The exterior differential on $\Omega_c^*(M)$ induces, by adjunction, a differential $d: D_k(M)\to D_{k-1}(M)$. The complex $(D_\bullet(M),d)$ is called the *distribution de Rham complex* of M. Write $D_\bullet(X)$ for the subcomplex of distributions supported on X.

The following singular version of the de Rham theorem is due to L. Shwartz and M. Kashiwara (see [Ka] and references therein).

Theorem 2.6.4. *Assume M is an oriented real analytic manifold and X is a closed analytic subset of M. Then there is a natural isomorphism*

$$H_*(D_\bullet(X),d)\simeq H_*^{BM}(X).$$

Remark 2.6.5. Let $m=\dim_{\mathbf{R}}M$. Then any C^∞-differential form $\phi\in\Omega^{m-k}(M)$ gives a degree k distribution on M by means of the formula $\omega\mapsto\int_M\phi\wedge\omega$. This way we obtain a natural embedding of complexes

$$\Omega^{m-\bullet}(M)\hookrightarrow D_\bullet(M). \tag{2.6.6}$$

Assume now that M is compact. Choose a Riemannian metric on M and let $\Delta=dd^*+d^*d$ denote the corresponding Laplace–Beltrami operator on distributions (resp. differential forms). Then the cohomology of the complexes $(D_\bullet(M),d)$ and $(\Omega^\bullet(M),d)$ are isomorphic, by Hodge theory (cf. [GH]), to the corresponding spaces of *harmonic* elements, i.e., such that $\Delta\phi=0$. By elliptic regularity (cf. [Ru], [GH]) any harmonic distribution is actually smooth. Hence the embedding of complexes 2.6.6 induces an isomorphism of cohomology. This proves Theorem 2.6.4 for smooth manifolds Λ. The singular case is much more complicated and was proved by Kashiwara using resolution of singularities.

2.6.7. NOTATION. From now on we will reserve the notation H_i for the Borel-Moore homology groups, since these will be used most frequently. We will write H_i^{ord} for the ordinary homology. Note that if X is compact then the ordinary homology of X and the Borel-Moore homology of X coincide, as follows for instance from definition (3) above.

We now study the functorial properties of Borel-Moore homology.

2.6.8. PROPER PUSHFORWARD. Borel-Moore homology is a covariant functor with respect to proper maps. It $f : X \to Y$ is a proper map, then we may define the direct image (or proper pushforward) map

$$f_* : H_*(X) \to H_*(Y)$$

by extending f to a map $\bar{f} : \bar{X} \to \bar{Y}$ where $\bar{X} = X \cup \{\infty\}$, resp. $\bar{Y} = Y \cup \{\infty\}$, and $\bar{f}(\infty) = \infty$ (observe that f being proper ensures that $\bar{f}$ is continuous).

2.6.9. LONG EXACT SEQUENCE OF BOREL-MOORE HOMOLOGY. Given an open subset $U \subset X$ there is a natural restriction morphism $H_*(X) \to H_*(U)$ induced by the composition of maps

$$H_*(X) = H_*^{ord}(\overline{X}, \overline{X} \setminus X) \to H_*^{ord}(\overline{X}, \overline{X} \setminus U) = H_*(U),$$

where $\overline{X}$ stands for a compactification of X, cf., definition (2) of Borel-Moore homology, and the map in the middle is induced by the natural morphism of pairs $(\overline{X}, \overline{X} \setminus X) \to (\overline{X}, \overline{X} \setminus U)$. For an alternative ad hoc definition of the restriction to an open subset, see [Iv].

Suppose next that F is a closed subset of X. Write $i : F \hookrightarrow X$ for the (closed) embedding, set $U = X \setminus F$, and consider the diagram

$$F \overset{i}{\hookrightarrow} X \overset{j}{\hookleftarrow} U$$

Since i is proper and j is an open embedding, the functors i_* and j^* are well-defined. Then there is a natural long exact sequence in Borel-Moore homology (see [Bre], [Sp] for more details):

$$\cdots \to H_p(F) \to H_p(X) \to H_p(U) \to H_{p-1}(F) \to \cdots \tag{2.6.10}$$

To construct this long exact sequence, choose an embedding of X as a closed subset of a smooth manifold M. Then the Poincaré duality isomorphism (2.6.1) gives

$$H^{m-p}(M, M \setminus X) \simeq H_p(X) \quad \text{and} \quad H^{m-p}(M, M \setminus F) \simeq H_p(F).$$

Further, the set U being locally closed in M, we may find an open subset $M' \subset M$ such that U is a closed subset of M'. Then, the excision axiom, see [Sp], combined with Poincaré duality yields

$$H^{m-p}(M, M \setminus U) \simeq H^{m-p}(M', M' \setminus U) \simeq H_p(U).$$

Thus, we see that terms of the standard relative cohomology long exact sequence, cf. [Sp]:

$$\to H^k(M, M \setminus F) \to H^k(M, M \setminus X) \to H^k(M, M \setminus U) \to H^{k+1}(M, M \setminus F) \to \tag{2.6.11}$$

get identified by means of the above isomorphisms with the corresponding terms of (2.6.10). This way we define (2.6.10) to be the exact sequence induced by the cohomology exact sequence above.

2.6.12. FUNDAMENTAL CLASS. Any smooth oriented manifold X has a well-defined *fundamental class* in Borel-Moore homology:

$$[X] \in H_m(X), \qquad m = \dim_{\mathbb{R}} X.$$

Note that there is no fundamental class in ordinary homology unless X is compact. As a particular example, write the long exact sequence of the ordinary homology associated with the pair $(S^n = \mathbb{R}^n \cup \{\infty\}, \mathbb{R}^n)$. Using the definitions we find

$$H_i(\mathbb{R}^n, \mathbb{C}) = \begin{cases} \mathbb{C} \cdot [\mathbb{R}^n] & \text{if} \quad i = n \\ 0 & \text{if} \quad i \neq n \end{cases} \tag{2.6.13}$$

The essential feature of Borel-Moore homology is the existence of a fundamental class, $[X]$, of any (not necessarily smooth or compact) complex algebraic variety X. If X is irreducible of real dimension m, then $[X]$ is the unique class in $H_m(X)$ that restricts to the fundamental class of the non-singular part of X. More precisely, write X^{reg} for the Zariski open dense subset consisting of the non-singular points of X. Being a smooth complex manifold, X^{reg} has a canonical orientation coming from the complex structure, and hence a fundamental class $[X^{reg}] \in H_m(X^{reg})$. The inequality $\dim(X \setminus X^{reg}) \leq m$ yields (e.g., by definition (1) of Borel-Moore homology)

$$H_k(X \setminus X^{reg}) = 0 \quad \text{for any} \quad k > m - 2.$$

The long exact sequence of Borel-Moore homology (see 2.6.9) shows that the restriction $H_m(X) \to H_m(X^{reg})$ is an isomorphism. We define $[X]$ to be the preimage of $[X^{reg}]$ under this isomorphism. If X is an arbitrary complex algebraic variety with irreducible components $X_1, X_2, \ldots, X_n$ then $[X]$ is set to be a non-homogeneous class equal to $\sum [X_i]$.

The top Borel-Moore homology of a complex algebraic variety is particularly easy to understand in light of the following proposition.

Proposition 2.6.14. *Let X be a complex variety of complex dimension n and let $X_1, \ldots, X_m$ be the n-dimensional irreducible components of X. Then the fundamental classes $[X_1], \ldots, [X_m]$ form a basis for the vector space $H_{top}(X) = H_{2n}(X)$.*

2.6.15. INTERSECTION PAIRING. Let M be a smooth, oriented manifold and $Z, \tilde{Z}$ two closed subsets (in the sense explained at the beginning of this

section) in M. We define a bilinear pairing

$$(2.6.16) \quad \cap : H_i(Z) \times H_j(\tilde{Z}) \to H_{i+j-m}(Z \cap \tilde{Z}), \quad m = \dim_{\mathbb{R}} M$$

which refines the standard intersection of cycles in a smooth variety. The only new feature is that instead of regarding cycles as homology classes in the ambient manifold M we take their supports into account. So, given two singular chains with supports in the subsets Z and $\tilde{Z}$ respectively, we would like to define their intersection to be a class in the homology of the set-theoretic intersection $Z \cap \tilde{Z}$. To that end we use the standard $\cup$-product in relative cohomology (cf. [Sp]):

$$\cup : H^{m-i}(M, M\backslash Z) \times H^{m-j}(M, M\backslash \tilde{Z}) \to H^{2m-j-i}\Big(M, (M\backslash Z) \cup (M\backslash \tilde{Z})\Big).$$

Applying Poincaré duality (2.6.1) to each term of this $\cup$-product, we get the intersection pairing (2.6.16).

The above introduced intersection pairing has especially clear geometric meaning in the case when M is a real analytic manifold and Z, $\tilde{Z}$ are closed analytic subsets in M. One can then use the definition of Borel-Moore homology as the homology of the complex formed by subanalytic chains, see e.g. [GM1] or [KS, §8.2]. It is known further, see [RoSa], that the set $Z \cap \tilde{Z}$ has an open neighborhood U in M such that $Z \cap \tilde{Z}$ is a proper homotopy retract of $\overline{U}$, the closure of U (this is a general property of analytic sets). Now given two subanalytic cycles $c \in H_*(Z)$ and $\tilde{c} \in H_*(\tilde{Z})$, one can give the following geometric construction of the class $c \cap \tilde{c} \in H_*(Z \cap \tilde{Z})$.

First choose V, an open neighborhood of Z in M such that Z is a proper homotopy retract of $\overline{V}$, and $\overline{V} \cap \tilde{Z} \subset U$. Second, since V is smooth, one can find a subanalytic cycle c' in $\overline{V}$ which is homologous to c in $\overline{V}$ and such that the set-theoretic intersection of c' with $\tilde{c}$ is contained in V and, moreover, c' intersects $\tilde{c}$ transversely at smooth points of both c' and $\tilde{c}$. Hence, the set-theoretic intersection $c' \cap \tilde{c}$ gives a well-defined subanalytic cycle in $H_*(\overline{V} \cap \tilde{Z})$, and therefore in $H_*(\overline{U})$. Finally, one defines $c \cap \tilde{c} \in H_*(Z \cap \tilde{Z})$ as the direct image of $c' \cap \tilde{c}$ under a proper contraction $\overline{U} \to Z \cap \tilde{Z}$ which exists by assumption. It is fairly straightforward to check that this way one obtains the same class as the one defined in (2.6.16) by means of the $\cup$-product in cohomology. It follows in particular that the result of the geometric construction above does not depend on the choices involved in the construction.

Occasionally, we will use an intersection pairing involving both Borel-Moore and ordinary homology. To define it, recall that for a closed subset Z of a smooth oriented manifold M (where Z is of the type explained at the beginning of this section), one has a natural Poincaré duality analogue

of (2.6.1) for the cohomology H_c^* with *compact* support

$$H_c^{m-i}(M, M\backslash Z) \simeq H_i^{ord}(Z).$$

Given two such subsets $Z, \tilde{Z}$, there is also a standard cup-product map

$$\cup : H_c^{m-i}(M, M\backslash Z) \times H^{m-j}(M, M\backslash \tilde{Z}) \to H_c^{2m-j-i}\Big(M, (M\backslash Z) \cup (M\backslash \tilde{Z})\Big).$$

Transporting it to homology by means of the Poincaré duality, one obtains a well-defined intersection pairing

$$(2.6.17) \cap : H_i^{ord}(Z) \times H_j(\tilde{Z}) \to H_{i+j-m}^{ord}(Z \cap \tilde{Z}), \qquad m = \dim_{\mathbb{R}} M.$$

In the special case $Z = \tilde{Z} = M$ and $i + j = m$ we have the following important result.

Proposition 2.6.18. *(Poincaré duality) Assume M is an oriented connected (not necessarily compact) smooth variety. Then, for any j, the map*

$$\cap : H_{m-j}^{ord}(M) \times H_j(M) \to H_0^{ord}(M) = \mathbb{C}$$

arising from (2.6.17) *is a non-degenerate pairing.*

Thus the intersection pairing of the ordinary and Borel-Moore homology of a smooth variety gives a natural isomorphism

$$H_j(M) \simeq H_{m-j}^{ord}(M)^*.$$

On the other hand, for any topological space M, there is a canonical isomorphism $H^{m-j}(M) \simeq H_{m-j}^{ord}(M)^*$. Composing the two isomorphisms one finds

$$H_j(M) \simeq H_{m-j}^{ord}(M)^* \simeq H^{m-j}(M),$$

which is a concrete form of the Poincaré duality isomorphism (2.6.2).

Warning: Since taking intersection of transverse cycles involves counting points *with signs* that depend on orientation, the first of the above isomorphisms (but not the second) changes sign if the orientation of M is changed.

2.6.19. KÜNNETH FORMULA. Let M_1 and M_2 be arbitrary CW-complexes. Then there is a natural isomorphism (see [Sp])

$$(2.6.20) \qquad \boxtimes : H_*(M_1) \otimes H_*(M_2) \to H_*(M_1 \times M_2).$$

This follows, by means of Definition (3) of the Borel-Moore homology, from the standard Künneth formula for the ordinary homology, see [Sp]:

$$H_*^{ord}(\overline{M_1}, \overline{M_1} \setminus M_1) \otimes H_*^{ord}(\overline{M_2}, \overline{M_2} \setminus M_2) \simeq$$
$$\simeq H_*^{ord}\Big(\overline{M_1 \times M_2}\,,\, \overline{M_1 \times M_2} \setminus (\overline{M_1} \times M_2 \cup M_1 \times \overline{M_2})\Big)\,,$$

where $\overline{M_i}$ stands for a compactification of M_i.

2.6.21. RESTRICTION WITH SUPPORTS. Let $i : N \hookrightarrow M$ be a closed embedding of oriented manifolds, and $d = \dim M - \dim N$. Given a closed, possibly singular, subset $Z \subset M$ we define the restriction *with support* in Z functor

$$i^* : H_k(Z) \to H_{k-d}(Z \cap N)\,, \quad c \mapsto c \cap [N]\,, \tag{2.6.22}$$

where $c \cap [N]$ stands for the intersection pairing in the ambient manifold M, see (2.6.16). When transported to cohomology by means of the Poincaré duality isomorphism (2.6.1) the map i^* gets identified with the standard restriction $H^*(M, M \setminus Z) \to H^*(N, N \setminus (N \cap Z))$.

It should be emphasized that the map i^* in (2.6.22) depends on the ambient variety M in an essential way, although it maps homology classes of Z to those of $Z \cap N$ and M is not explicitly present in the notation. Note in particular that the map shifts homology degree by $d = \dim M - \dim N$. Thus, if for example one replaces M by a larger smooth manifold $M' \supset M$ (without changing Z and N) then the restriction map with respect to the the ambient space M' will even become zero if $\dim M' > \dim Z + \dim N$.

2.6.23. DIAGONAL REDUCTION. Let $M_\Delta \subset M \times M$ be the diagonal where M continues to be an oriented manifold. Observe that for closed subsets $Z, \tilde{Z} \subset M$ we have a set-theoretic equality

$$(Z \times \tilde{Z}) \cap M_\Delta = Z_\Delta \cap \tilde{Z}_\Delta\,,$$

where the subscript "Δ" indicates that the varieties are placed into the diagonal M_Δ. There is a similar formula for homology classes. To obtain it, recall that the standard $\cup$-product in cohomology is defined as the pullback induced by the diagonal embedding $i_\Delta : M_\Delta \hookrightarrow M \times M$. Transporting this pullback to Borel-Moore homology by means of the Poincaré duality we obtain the following result

$$c \cap \tilde{c} = i_\Delta^*(c \boxtimes \tilde{c}) \quad , \quad c \in H_*(Z),\, \tilde{c} \in H_*(\tilde{Z}). \tag{2.6.24}$$

The RHS here may be rewritten also as $(c \boxtimes \tilde{c}) \cap [M_\Delta]$. An analogous formula in the context of algebraic cycles is proved in [Fu].

Remark 2.6.25. Observe that we may use the RHS of (2.6.24) to be an alternative definition of the intersection pairing (2.6.16) in terms of restriction with supports.

2.6.26. SMOOTH PULLBACK IN BOREL-MOORE HOMOLOGY. Let X be a locally compact space and $p : \tilde{X} \to X$ a locally trivial fibration with smooth oriented fiber F (note that X is not assumed smooth). We say that p is oriented if all transition functions of the fibration preserve the orientation of the fiber.

For an oriented fibration p, as above, with $\dim F = d$ one can define a natural pullback morphism

$$p^* : H_\bullet(X) \to H_{\bullet+d}(\tilde{X}). \tag{2.6.27}$$

We will not give a definition of the map p^* here, and only describe here some important properties of this map (taking its existence for granted). The most natural way to define this map is by means of a sheaf-theoretic approach to Borel-Moore homology, to be explained in Chapter 8. We will see in (8.3.32) that to define the smooth pullback above one needs to know that the Dualizing complex on X pulls back, up to shift by d, to the Dualizing complex on $\tilde{X}$, that is $p^*\mathbb{D}_X = \mathbb{D}_{\tilde{X}}[-d]$.

In the case of a trivial fibration $p : \tilde{X} = X \times F \to X$ the morphism (2.6.27) is defined by the assignment $c \mapsto c \boxtimes [F]$. In the general case of an arbitrary oriented fibration, the pullback morphism p^* has the property that it restricts to the map $c \mapsto c \boxtimes [F]$ on any open subset $U \subset X$ such that the fibration $p : p^{-1}(U) \to U$ is trivial. More precisely, whenever there is a cartesian square like the one on the left

$$\begin{array}{ccc} U \times F & \xrightarrow{\tilde{i}} & \tilde{X} \\ {\scriptstyle pr_1}\downarrow & & \downarrow{\scriptstyle p} \\ U & \xrightarrow{i} & X \end{array} \qquad \begin{array}{ccc} H_{\bullet+d}(U \times F) & \xleftarrow{\tilde{i}^*} & H_{\bullet+d}(\tilde{X}) \\ {\scriptstyle c \mapsto c\boxtimes[F]}\uparrow & & \uparrow{\scriptstyle p^*} \\ H_\bullet(U) & \xleftarrow{i^*} & H_\bullet(X) \end{array}$$

the induced square on the right commutes.

Further, if in the above situation $i : X \to \tilde{X}$ is a continuous section of p, one can define the Gysin pullback $i^* : H_\bullet(\tilde{X}) \to H_{\bullet-d}(X)$ (though no ambient smooth space is assumed to be given). Moreover, one has a Thom-type formula

$$i^* \circ p^* = \mathrm{Id}_X \tag{2.6.28}$$

Again, the definition of i^* will be postponed until Chapter 8. We will see there that for $i^* : H_\bullet(\tilde{X}) \to H_{\bullet-d}(X)$ to be defined one needs to know that the dualizing complex on $\tilde{X}$ restricts to the dualizing complex on $i(X)$, up to shift, i.e., we have $i^*\mathbb{D}_{\tilde{X}} = \mathbb{D}_X[d]$. At this point, the reader should note

that the above mentioned condition involves only the local structure of the embedding $i(X) \hookrightarrow \tilde{X}$ while the projection $p : \tilde{X} \to X$ is not explicitly present. Therefore, such a Gysin pullback i^* can be defined for any map i which is locally isomorphic to a section of a locally trivial oriented fibration.

We now explicitly define i^* in the case of a trivial fibration $p : \tilde{X} = X \times F \to X$ with a section $i : x \mapsto (x, f(x))$. In this case we have $H_*(\tilde{X}) = H_*(X) \otimes H_*(F)$, by the Kunneth formula, and the map $i^* : H_\bullet(\tilde{X}) \to H_{\bullet - d}(X)$ sends a class of the form $c \boxtimes [F]$ to c, and any class $c \boxtimes h$ with $\deg h < d$ to zero. Formula (2.6.28) then trivially holds. In the general case, the morphism i^* has the property that it restricts to the one above on any open subset $U \subset X$ such that the fibration $p : p^{-1}(U) \to U$ is trivial.

We claim that if there exists a way to define *natural* morphisms p^* and i^* that satisfy the above mentioned properties then this way is unique. To see this, assume U and V are two open subsets of X. We have the following natural commutative diagram whose horizontal rows are given by the Maeyer-Vietoris exact sequences

$$\begin{array}{ccccccc}
\cdots \longrightarrow & H_i(\tilde{U} \cup \tilde{V}) & \longrightarrow & H_i(\tilde{U}) \oplus H_i(\tilde{V}) & \longrightarrow & H_i(\tilde{U} \cap \tilde{V}) & \longrightarrow \cdots \\
& {\scriptstyle i^*}\downarrow\uparrow{\scriptstyle p^*} & & {\scriptstyle i^*}\downarrow\uparrow{\scriptstyle p^*} & & {\scriptstyle i^*}\downarrow\uparrow{\scriptstyle p^*} & \\
\cdots \longrightarrow & H_{i-d}(U \cup V) & \longrightarrow & H_{i-d}(U) \oplus H_{i-d}(V) & \longrightarrow & H_{i-d}(U \cap V) & \longrightarrow \cdots
\end{array}$$

We emphasize that vertical maps in the diagram exist due to the "naturality" assumption on p^* and i^* that we have made. Standard diagram chase shows that if we have already proved the uniqueness of p^* and i^*, and formula (2.6.28) for the restriction of $\tilde{X} \to X$ over U, V and $U \cap V$ respectively, then the same is true for $U \cup V$. The claim for the whole of X is now deduced using an appropriate open covering of X.

We will usually use the morphisms p^* and i^* in the situation where X is imbedded in a smooth oriented variety M, and the fibration $p : \tilde{X} \to X$ is the restriction to $X \subset M$ of a locally trivial oriented fibration $\overline{p} : \tilde{M} \to M$ with the same fiber F. In such a case the pullback morphism (2.6.27) is induced, by means of Poincaré duality, by the standard pullback morphism in cohomology

$$H^*(M, M \backslash X) \xrightarrow{\overline{p}^*} H^*(\tilde{M}, \tilde{M} \backslash \tilde{X}) .$$

A similar formula applies to i^*.

Now let $\overline{p} : \tilde{M} \to M$ be as above, and let $Z \subset M$ and $Z' \subset \tilde{M}$ be two closed subsets. Then the restriction of $\overline{p}$ gives an oriented fibration $p : \overline{p}^{-1}(Z) \to Z$. Assuming that the projection

$$\overline{p}^{-1}(Z) \cap Z' \to M \quad \text{is proper},$$

we write $Z \circ Z'$ for its image, a closed subset in M, cf. condition (2.7.8) in

the next section. Then, for any $c \in H_*(Z)$ and $c' \in H_*(Z')$, we have the following equation in $H_*(Z \circ Z')$:

(2.6.29) PROJECTION FORMULA: $\overline{p}_*(p^*c \cap c') = c \cap (\overline{p}_* c')$,

where $p^* : H_*(Z) \to H_*(\overline{p}^{-1}(Z))$. Proof of this formula is entirely analogous to the proof of its counterpart for algebraic cycles given, for example, in [Fu].

2.6.30. SPECIALIZATION IN BOREL-MOORE HOMOLOGY. (cf. [FM]) Let (S, o) be a smooth manifold with base point o. Put $S^* := S \setminus \{o\}$. Given a possibly singular space Z and a map $\pi : Z \to S$, we set $Z_o = \pi^{-1}(o) \cap Z$ and, for any subset $S' \subset S$, we write $Z(S') := \pi^{-1}(S')$. Assuming that the restriction $\pi : Z(S^*) \to S^*$ is a locally trivial fibration with possibly singular fibers (warning: the map $\pi : Z \to S$ is *not* assumed to be locally trivial near o) we define a *specialization* map

$$\lim_{s \to 0} : H_\bullet(Z(S^*)) \to H_{\bullet - d}(Z_o), \quad d = \dim_{\mathbb{R}} S$$

as follows, cf. [FM].

Let (B, o) be an open neighborhood of o in S diffeomorphic to $(\mathbb{R}^d, 0)$. Choose a decomposition $\mathbb{R}^d = \mathbb{R} \times \mathbb{R}^{d-1}$, and introduce the following notation

$$\mathbb{R}_{>0} = (0, \infty) \ , \ \mathbb{R}_{\geq 0} = [0, \infty) \ , \ \mathbb{R}^d_{>0} = \mathbb{R}_{>0} \times \mathbb{R}^{d-1} \subset \mathbb{R}^d.$$

Accordingly, we write $B_{>0} \subset B$ for the open half-space corresponding to $\mathbb{R}^d_{>0}$ under the isomorphism $B \simeq \mathbb{R}^d$. Further, we let $I_{>0}$, resp. I, be the positive, resp. non-negative, part of the one dimensional vector space in B corresponding to $\mathbb{R} \times \{0\} \subset \mathbb{R} \times \mathbb{R}^{d-1} = \mathbb{R}^d$. Thus, we have $I := I_{>0} \cup \{o\}$. Clearly we have canonical isomorphisms, cf. (2.6.13)

(2.6.31) $$H_d(B_{>0}) \simeq H_d(\mathbb{R}^d_{>0}) \simeq H_1(\mathbb{R}_{>0}) \simeq H_1(I_{>0}).$$

Using the local triviality of the fibration $Z(S^*) \to S^*$ we may assume, shrinking B if necessary, that the projection $\pi : Z(B_{>0}) \to B_{>0}$ is a trivial fibration with fiber F (we have used here that $B_{>0}$ is contractible; note that we cannot claim that $Z(B \setminus \{o\}) \to B \setminus \{o\}$ is a trivial fibration because $B \setminus \{o\}$ is not contractible). Then, the Künneth theorem combined with (2.6.31) yields a chain of isomorphisms

$$\begin{aligned} H_\bullet(Z(B_{>0})) &\xrightarrow{\sim} H_{\bullet - d}(F) \otimes H_d(B_{>0}) \\ &\xrightarrow{\sim} H_{\bullet - d}(F) \otimes H_1(I_{>0}) \xrightarrow{\sim} H_{\bullet - d + 1}(Z(I_{>0})), \end{aligned} \tag{2.6.32}$$

shifting gradation by $d - 1$. Write ψ for the composition of the above isomorphisms. We now define the specialization map $\lim_{s \to 0}$ to be the

following composition

$$(2.6.33)\ H_\bullet(Z(S^*)) \to H_\bullet(Z(B_{>0})) \xrightarrow{\psi} H_{\bullet-d+1}(Z(I_{>0})) \xrightarrow{\partial} H_{\bullet-d}(Z_o)\,.$$

Here the first map is induced by restriction to the open subset $Z(B_{>0}) \subset Z(S^*)$, the second map, ψ, was defined above, and the last map, ∂, is the connecting homomorphism in the long exact sequence of Borel-Moore homology, cf. (2.6.10):

(2.6.34)

$$\cdots \to H_j(Z_o) \to H_j(Z(I)) \to H_j(Z(I_{>0})) \xrightarrow{\partial} H_{j-1}(Z_o) \to \cdots\,.$$

Lemma 2.6.35. *The construction of the specialization map does not depend on the choices involved.*

Proof. Our construction was based on the choice of B, an isomorphism $B \simeq \mathbb{R}^d$ and on the decomposition $\mathbb{R}^d = \mathbb{R} \times \mathbb{R}^{d-1}$. Let us analyze the dependence on these choices. Independence of the choice of B is immediate from the transitivity of restriction to open subsets. We claim that the composition of the first two maps in (2.6.33) depends only on the choice of the segment $I_{\geq 0} \subset B$. Indeed, let U be a small tubular open neighborhood of $I_{>0}$ in B such that there is a smooth contraction $p : Z(U) \to Z(I_{>0})$. Then the composition of the first two maps in (2.6.33) equals the restriction from $Z(S^*)$ to $Z(U)$ followed by the Gysin pullback, see (2.6.28), induced by the embedding $Z(I_{>0}) \hookrightarrow Z(U)$, which is isomorphic to a section of the smooth contraction $p : Z(U) \to Z(I_{>0})$. It is clear that both maps in homology are independent of all the choices.

Finally, the last map, ∂, in (2.6.33) depends only on the choice of a semi-line $I_{>0}$. Such a map is clearly defined if $I_{>0}$ is replaced by any path given by the image of the positive semi-line under an immersion $\gamma : (-\epsilon, +\infty) \hookrightarrow S$ such that $\gamma(0) = o$.

Thus, we are essentially reduced to the following setup. There is a fixed open subset $B_{>0} \subset S$, and two different paths $\gamma^{(1)}$ and $\gamma^{(2)}$ as above, taking the positive semi-line into subsets $I^{(1)}_{>0}$ and $I^{(2)}_{>0}$, respectively. We must show that the two maps $\psi \circ \partial$ corresponding to the two paths are equal.

To simplify the setup assume $d = 2$, the case $d > 2$ being totally analogous. We can then find a smooth map $\Phi : \mathbb{R}^2 \to B$ with the following properties:

(a) $\Phi(\{0\} \times \mathbb{R}) = \{o\}$;
(b) Φ restricts to a diffeomorphism $\mathbb{R}^2_{>0} = R_{>0} \times \mathbb{R} \xrightarrow{\sim} B_{>0}$;
(c) $\Phi|_{\mathbb{R}_{>0} \times \{1\}} = \gamma^{(1)}$ and $\Phi|_{\mathbb{R}_{>0} \times \{2\}} = \gamma^{(2)}$.

We now pull back the map $Z \to B$ by means of Φ, i.e., form the cartesian

diagram:

$$\begin{array}{ccc} \tilde{Z} & \xrightarrow{\tilde{\Phi}} & Z \\ {\scriptstyle\tilde{\pi}}\downarrow & & \downarrow{\scriptstyle\pi} \\ \mathbb{R}^2_{\geq 0} & \xrightarrow{\Phi} & B \end{array}$$

The variety $\tilde{Z}$ is defined by this diagram as a fiber product, and we also put

$$\tilde{Z}_\circ := \tilde{\pi}^{-1}(\{0\} \times \mathbb{R}) \quad , \quad \tilde{Z}_{>0} := \tilde{\pi}^{-1}(\mathbb{R}^2_{>0}) = \tilde{Z} \setminus \tilde{Z}_\circ$$

We are going to use the variety $\tilde{Z}$ to construct a kind of homotopy between the specialization maps $\lim_{s\to 0}{}^{(1)}$ and $\lim_{s\to 0}{}^{(2)}$ corresponding to the two choices of semi-lines $I^{(1)}_{>0}$ and $I^{(2)}_{>0}$, respectively. To that end, consider the following long exact sequence analogous to (2.6.34)

$$(2.6.36) \quad \cdots \to H_j(\tilde{Z}_\circ) \to H_j(\tilde{Z}) \to H_j(\tilde{Z}_{>0}) \xrightarrow{\tilde{\delta}} H_{j-1}(\tilde{Z}_\circ) \to \cdots .$$

The connecting homomorphism in this exact sequence gives a map $\tilde{\delta} : H_j(\tilde{Z}_{>0}) \to H_{j-1}(\tilde{Z}_\circ)$. Observe now that we have, by property (a) of the map Φ, a canonical isomorphism $\tilde{Z}_\circ = \mathbb{R} \times Z_\circ$. Therefore, each of the two embeddings $\{i\} \hookrightarrow \mathbb{R}$, $i = 1, 2$ gives a Künneth isomorphism

$$(2.6.37) \quad \phi : H_j(Z_\circ) = H_j(Z_\circ \times \{i\}) \xrightarrow{\sim} H_{j+1}(Z_\circ \times \mathbb{R}) = H_{j+1}(\tilde{Z}_\circ) .$$

Further, by property (c) of the map Φ and triviality of the bundle $Z(B_{>0}) \to B_{>0}$, we have the following two canonical isomorphisms similar to the isomorphism ψ used in the construction of the specialization map (2.6.33):

$$\tilde{\psi}_i : H_\bullet(Z(I^{(i)}_{>0})) \xrightarrow{\sim} H_{\bullet+1}(\tilde{Z}_{>0}) \quad , \quad i = 1, 2 .$$

All the maps that we have introduced can be assembled in the diagram below. Moreover, our construction shows that the diagram commutes:

$$\begin{array}{ccccc} H_\bullet(Z(I^{(1)}_{>0})) & \xrightarrow{\tilde{\psi}_1} & H_{\bullet+1}(\tilde{Z}_{>0}) & \xrightarrow{\tilde{\psi}_2} & H_\bullet(Z(I^{(2)}_{>0})) \\ {\scriptstyle\psi_1\circ\partial}\downarrow & & \downarrow{\scriptstyle\tilde{\delta}} & & \downarrow{\scriptstyle\psi_2\circ\partial} \\ H_{\bullet-1}(Z_\circ \times \{1\}) & \xrightarrow{\tilde{\phi}_1} & H_\bullet(\tilde{Z}_\circ) & \xrightarrow{\tilde{\phi}_2} & H_{\bullet-1}(Z_\circ \times \{2\}) \end{array}$$

The diagram yields $\psi_1 \circ \partial = \psi_2 \circ \partial$. Therefore, $\lim^{(1)}_{s\to 0} = \lim^{(2)}_{s\to 0}$, and the lemma follows. ∎

Specialization enjoys the following transitivity property. Let $S_1 \subset S$ be a smooth k-codimensional submanifold such that $o \in S_1$. Write $\lim_{s\to 0}^{S_1}$ for the specialization map for the variety $Z(S_1) \to S_1$, as opposed to the original specialization map $\lim_{s\to 0}^{S}$ for $Z(S) \to S$. Let $\varepsilon^* : H_\bullet(Z(S^*)) \to H_{\bullet-k}(Z(S_1^*))$ be the Gysin pullback, see (2.6.28), induced by the embedding $\varepsilon : Z(S_1^*) \hookrightarrow Z(S^*)$. Locally on S_1^* the map ε may be viewed as a section of a smooth fibration $Z(S^*) \to Z(S_1^*)$, so that the Gysin map is well-defined. From the construction of specialization and the above lemma we deduce the following result.

Lemma 2.6.38. *The specialization is compatible with restricion from S to S_1, i.e., we have $\lim_{s\to 0}^{S_1} = \varepsilon^* \circ \lim_{s\to 0}^{S}$.*

Next, assume given a smooth oriented manifold M, a smooth locally trivial oriented fibration $\pi : M \to S$ over a smooth base S, and a point $o \in S$. Write $S^* = S \setminus \{o\}$ and, for any subset $M' \subset M$, use the notation $M'(S^*) := M' \cap \pi^{-1}(S^*)$. Assume that the restriction $\pi : M(S^*) \to S^*$ is a trivial fibration and fix its trivialization. Suppose further we have two closed subsets $Z, Z' \subset M$ such that the projections $Z(S^*) \to S^*$ and $Z'(S^*) \to S^*$ are both trivialized by the chosen trivialization of $M(S^*)$.

We write the subscript "o" to denote the special fiber over o. Note that we have $Z_o \cap Z'_o = (Z \cap Z')_o$ and, similarly, $Z(S^*) \cap Z'(S^*) = (Z \cap Z')(S^*)$. Recall $d = \dim S$.

Proposition 2.6.39. *Intersection pairing commutes with the specialization, i.e., in the above setup the following diagram commutes*

$$\begin{array}{ccc} H_\bullet(Z(S^*)) \otimes H_\bullet(Z'(S^*)) & \xrightarrow{\cap} & H_\bullet(Z(S^*) \cap Z'(S^*)) \\ \downarrow{\scriptstyle \lim_{s\to 0}} & & \downarrow{\scriptstyle \lim_{s\to 0}} \\ H_{\bullet-d}(Z_o) \otimes H_{\bullet-d}(Z'_o) & \xrightarrow{\cap} & H_{\bullet-d}(Z_o \cap Z'_o) \end{array}$$

Sketch of Proof. By diagonal reduction, see 2.6.23, the proposition is reduced to the assertion that the specialization commutes with restriction with supports. Thus, it suffices to consider the situation where we are given only one closed subset $Z \subset M$ as above, and we are also given a locally trivial oriented smooth subbundle N in M. We get a diagram

$$\begin{array}{ccccc} N & \xrightarrow{i} & M & \hookleftarrow & Z \\ & {\scriptstyle\pi}\searrow & \downarrow{\scriptstyle\pi} & \swarrow{\scriptstyle\pi} & \\ & & S & & \end{array}$$

Observe next that due to the construction of the specialization we may assume without loss of generality that $S = I$ is a closed semi-line isomorphic to $[0, \infty)$. Then the specialization map reduces essentially to the

connecting homomorphism in the long exact sequence of Borel-Moore homology of the pair (Z, Z_o). The claim now follows from the functoriality of long exact sequences, that is from the fact the embedding of pairs $i : (Z \cap N, Z_o \cap N) \hookrightarrow (Z, Z_o)$ induces a morphism of the long exact sequences, hence commutes with the connecting homomorphisms. ∎

2.6.40. COHOMOLOGY ACTION. The Borel-Moore homology, $H_*(Z)$, of any space Z has a natural $H^*(Z)$-module structure. To define it, choose a closed embedding $i : Z \hookrightarrow U$ into a C^∞-manifold U such that Z is a homotopy retract of U. Then the restriction to Z induces an isomorphism of cohomology $i^* : H^*(U) \xrightarrow{\sim} H^*(Z)$. Now, we have the standard cup-product on cohomology, see [Sp]: $\cap : H^i(U) \times H^j(U, U\backslash Z) \to H^{i+j}(U, U\backslash Z)$. Setting $k = \dim U - j$, this cup-product gets identified, by means of the Poincaré duality and the isomorphism i^* above, with an action map

$$H^i(Z) \otimes H_k(Z) \to H_{k-i}(Z) \quad , \quad a \otimes c \mapsto a \cdot c .$$

The construction does not depend on the choice of a closed embedding $i : Z \hookrightarrow U$, as will become clear from the sheaf-theoretic definition of the Borel-Moore homology given in Chapter 8. In particular, the construction applies for any closed subset Z of a C^∞-manifold M by taking U to be an appropriate tubular neighborhood of Z in M.

Assume now that Z and Z' are closed subsets of a C^∞-manifold M. Given a cohomology class $a \in H^*(Z)$, write $a_{|_{Z \cap Z'}}$ for its image in $H^*(Z \cap Z')$ under the natural restriction. We may lift a to a cohomology class of an appropriate neighborhood of Z. Then the associativity of the $\cup$-product in cohomology yields the following compatibility formula between the intersection pairing in Borel-Moore homology and the cohomology action (denoted by dot)

$$(2.6.41)\quad (a \cdot c) \cap c' = a_{|_{Z \cap Z'}} \cdot (c \cap c') \quad , \quad \forall c \in H_*(Z),\ c' \in H_*(Z') .$$

2.6.42. THOM ISOMORPHISM. Let $\pi : V \to X$ be a locally-trivial oriented C^∞-vector bundle. Recall that associated to V is its Euler class $e(V) \in H^r(X)$ where $r = rkV$, see e.g., [BtTu]. If V is a *complex* vector bundle with the orientation induced by the complex structure on the fibers, then $e(V)$ is known to be equal to the top Chern class of V.

Let $i : X \hookrightarrow V$ denote the zero-section. Then one has the following well-known result, see e.g. [BtTu], usually referred to as the Thom isomorphism in homology.

Proposition 2.6.43. *(i) The Gysin pullback morphisms i^* and π^*, cf.* (2.6.28)*, give rise to mutually inverse isomorphisms of Borel-Moore homology:*

$$H_*(X) \leftrightarrows H_{*+r}(V)$$

(ii) For any $c \in H_*(X)$ one has: $i^* i_*(c) = e(V) \cup c$.

Next let N be an oriented closed submanifold of an oriented C^∞-manifold M. Set $d = \dim_{\mathbb{R}} M - \dim_{\mathbb{R}} N$. The inclusion $i : N \hookrightarrow M$ induces the direct image morphism $i_* : H_*(N) \to H_*(M)$ and restriction with support in N morphism $i^* : H_*(M) \to H_{*-d}(N)$, respectively. The following result describes the composition $i^* i_* : H_*(N) \to H_{*-d}(N)$.

Corollary 2.6.44. *For any $c \in H_*(N)$ one has: $i^* i_*(c) = e(T_N M) \cup c$.*

Sketch of Proof. First recall that there exists an open tubular neighborhood $U \supset N$ in M and a diffeomorphism $T_N M \xrightarrow{\sim} U$, see e.g. [Mi], that takes the zero section $N \subset T_N M$ to $N \subset U$. Such a diffeomorphism may be constructed, for example, by choosing a Riemannian metric on M and taking the exponential geodesic mapping $\exp : T_N M \to M$. The exponential mapping is known to be a diffeomorphism of a small enough disk-subbundle in $T_N M$ with a tubular neighborhood of N in M. One then composes this diffeomorphism with a diffeomorphism between $T_N M$ and the disk-subbundle.

Now, using an appropriate excision axiom, one may replace the manifold M by U in the claim of the corollary. Then, the pair (M, N) gets replaced, due to the diffeomorphism above, by the pair $(T_N M, N)$. Thus, we are reduced to the situation of Proposition 2.6.43. Part (ii) of the proposition yields the assertion. ∎

Next, assume given $p : V \to Z$, an oriented vector bundle and $W \subset V$, an oriented vector subbundle. Write $j : W \hookrightarrow V$ for the embedding of the total spaces. Then we have the following formula

$$j_*[W] = p^* e(V/W) \cdot [V] \quad \text{in} \quad H_*(V)\,. \tag{2.6.45}$$

To prove this formula, it suffices to show, due to part (i) of Proposition 2.6.43, that $j^* j_*[W] = j^*(p^* e(V/W) \cdot [V])$. The LHS of this expression equals $p^* e(V/W) \cdot [W]$, by part (ii) of the proposition. The RHS can be rewritten as $p^* e(V/W) \cdot j^*[V] = p^* e(V/W) \cdot [W]$, since we have $j^*[V] = [W]$. Thus, we have proved that LHS = RHS, and formula (2.6.45) follows.

2.6.46. Access Intersection Formula. Let Z_1 and Z_2 be two closed oriented submanifolds of an oriented C^∞-manifold M. Assume that $Z = Z_1 \cap Z_2$, their set-theoretic intersection, is smooth. On Z define the vector bundle $T_{1,2} := T_z M / (T_z Z_1 + T_z Z_2)$.

The following formula for the intersection pairing of two fundamental classes is used quite frequently and is essentially the most general case among those where such an intersection can be directly calculated.

Proposition 2.6.47. *Assume in addition that the intersection of Z_1 and Z_2 "clean" in the sense that*

$$T_z Z_1 \cap T_z Z_2 = T_z Z \quad , \quad \forall z \in Z \,. \tag{2.6.48}$$

Then, we have $[Z_1] \cap [Z_2] = e(T_{1,2}) \cdot [Z]$, where the LHS involves the intersection pairing $\cap : H_*(Z_1) \times H_*(Z_2) \to H_*(Z)$ in the ambient space M, and the dot on the RHS stands for the $H^*(Z)$-action on Borel-Moore homology introduced at the beginning of this subsection.

Note that any transverse intersection is necessarily "clean," but not conversely.

2.6.49. *Sketch of Proof of Proposition* 2.6.47. We first show that "clean" intersection locally looks like intersection of vector subspaces in a vector space. As a first step, one replaces M by a small open neighborhood of Z. One then argues as in the proof of Corollary 2.6.44. Using condition (2.6.48) and the exponential map $\exp : T_z M \to M$ relative to a Riemannian metric on M, one constructs a local diffeomorphism between the quadruples (M, Z_1, Z_2, Z) and $(T_z M\,,\, T_z Z_1\,,\, T_z Z_2\,,\, Z = \text{zero-section})$.

Thus we are reduced to computing the intersection pairing of $[T_z Z_1]$ and $[T_z Z_2]$ in $T_z M$. To that end, view $T_z Z_1$ and $T_z Z_2$ as vector subbundles in the vector bundle $T_z M$, and form the quotient vector bundle $E := T_z M / T_z Z_2$. Let $p : T_z M \to E$ be the vector bundle projection and $\bar{Z}\,(\simeq Z)$ the zero-section of E. Clearly we have $[T_z Z_2] = p^*[\bar{Z}]$. Hence, using the projection formula (2.6.29), we obtain

$$p_*([T_z Z_1] \cap [T_z Z_2]) = p_*([T_z Z_1] \cap p^*[\bar{Z}]) = p_*[T_z Z_1] \cap [\bar{Z}] \,.$$

Put $\overline{T_z Z_1} := p_*(T_z Z_1)$, a vector subbundle in E. Observe that condition (2.6.48) insures that the set-theoretic intersection of $T_z Z_1$ and $T_z Z_2$ in $T_z M$ equals the zero-section of $T_z M$. It follows that the projection p induces a diffeomorphism $p : T_z Z_1 \xrightarrow{\sim} \overline{T_z Z_1}$. Therefore the map p_* identifies the cycle $[T_z Z_1] \cap [T_z Z_2] \in H_*(T_z M)$ with the cycle $[\overline{T_z Z_1}] \cap [\bar{Z}] \in H_*(E)$. To find the latter, we observe that there is a natural isomorphism

$$E / \overline{T_z Z_1} \simeq T_z M / (T_z Z_1 + T_z Z_2) = T_{1,2} \,.$$

On the other hand, writing $j : \overline{T_z Z_1} \hookrightarrow E$ for the natural embedding, we calculate

$$[\overline{T_z Z_1}] \cap [\bar{Z}] \overset{(2.6.45)}{=} \Big(e(E/\overline{T_z Z_1}) \cdot [E] \Big) \cap [\bar{Z}]$$

$$\overset{(2.6.41)}{=} e(E/\overline{T_z Z_1}) \cdot ([E] \cap [\bar{Z}]) = e(T_{1,2}) \cdot ([E] \cap [\bar{Z}]) = e(T_{1,2}) \cdot [\bar{Z}]. \quad \blacksquare$$

2.7 Convolution in Borel-Moore Homology

In this section we give a general construction of a convolution-type product in Borel-Moore homology. Though looking technically quite involved, the construction is essentially nothing but a "homology-valued" version of the standard definition of the composition of multi-valued maps.

2.7.1. TOY EXAMPLE. We begin with the trivial case of the convolution product. We write $\mathbb{C}(M)$ for the finite dimensional vector space of $\mathbb{C}$-valued functions on a finite set M. Given finite sets M_1, M_2, M_3, define a convolution product

$$\mathbb{C}(M_1 \times M_2) \otimes \mathbb{C}(M_2 \times M_3) \to \mathbb{C}(M_1 \times M_3)$$

by the formula

$$f_{12} * f_{23} : (m_1, m_3) \mapsto \sum_{m_2 \in M_2} f_{12}(m_1, m_2) \cdot f_{23}(m_2, m_3). \tag{2.7.2}$$

Writing d_i for the cardinality of the finite set M_i we may naturally identify $\mathbb{C}(M_i \times M_j)$ with the vector space of $d_i \times d_j$-matrices with complex entries. Then, formula (2.7.2) turns into the standard formula for the matrix multiplication.

As the next step of our toy example, we would like to find a similar convolution construction assuming that M_1, M_2, M_3 are smooth compact manifolds rather than finite sets (note that the compactness condition is a natural generalization of the finiteness condition. The latter was needed in order to make the sum in the RHS of (2.7.2) finite). As one knows from elementary analysis, it is usually the measures and not the functions that can be convolved in a natural way. In differential geometry the role of measures is played by the differential forms. Thus, given a smooth manifold M, we let $\Omega^\bullet(M)$ denote the graded vector space of C^∞-differential forms on M. This is the right substitute for the vector space $\mathbb{C}(M)$ when a finite set is replaced by a manifold.

Let M_1, M_2, M_3 be smooth compact manifolds, and $p_{ij} : M_1 \times M_2 \times M_3 \to M_i \times M_j$ the projection to the (i,j)-factor. Put $d = \dim M_2$. We now define a convolution product

$$\Omega^i(M_1 \times M_2) \otimes \Omega^j(M_2 \times M_3) \to \Omega^{i+j-d}(M_1 \times M_3)$$

by the formula

$$f_{12} * f_{23} = \int_{M_2} p_{12}^* f_{12} \wedge p_{23}^* f_{23}. \tag{2.7.3}$$

Here the expression under the integral is a differential form on $M_1 \times M_2 \times M_3$ of degree $i + j$. In some local coordinates $x_1, \dots, x_c$ on M_1, $y_1, \dots, y_d$ on

M_2, and $z_1, \ldots, z_e$ on M_3, any differential form on $M_1 \times M_2 \times M_3$ has the form

$$f = \sum \omega(x, y, z) dx_{i_1} \wedge \ldots \wedge dx_{i_a} \wedge dy_{j_1} \wedge \ldots \wedge dy_{j_b} \wedge dz_{k_1} \wedge \ldots \wedge dz_{k_c}$$

The integral of such a form f over M_2 is understood to be zero unless the number of the dy's in the expression equals $d = \dim M_2$, in which case it is a well-defined $(i + j - d)$-form on $M_1 \times M_3$ given by the integral over $dy_1 \wedge \ldots \wedge dy_d$ with x's and z's treated as parameters.

The standard properties of differential calculus on manifolds show that convolution (2.7.3) is compatible with the de Rham differential, i.e., we have

$$d(f_{12} * f_{23}) = (df_{12}) * f_{23} + (-1)^j f_{12} * (df_{23}).$$

It follows that the convolution product of differential forms induces a convolution product on the de Rham cohomology

$$(2.7.4) \qquad H^i(M_1 \times M_2) \otimes H^j(M_2 \times M_3) \to H^{i+j-d}(M_1 \times M_3).$$

The latter can be transported, by means of the Poincaré duality, to a similar convolution in homology.

In what follows, we are going to give an alternative "abstract" definition of the convolution product (2.7.4) in terms of algebraic topology. One advantage of such an "abstract" definition is that it works for *any* generalized homology theory, e.g., for K-theory. Such a K-theoretic convolution will be studied in detail in Chapter 5 and applied to representation theory in Chapter 7. Another advantage of the "abstract" definition is that it enables us to make a refined convolution construction "with supports." In the de Rham approach based on Theorem 2.6.4 this would amount to replacing differential forms in formula (2.7.3) by distributions. The problem is however that the $\wedge$-product of distributions is not defined, in general. Thus, the naive attempt of using formula (2.7.3) does not work, and it is essentially this difficulty that makes the "correct" definition below more complicated.

2.7.5. GENERAL CASE. We proceed now to the "abstract" construction of the convolution product. Let M_1, M_2, M_3 be connected, oriented C^∞-manifolds and let

$$Z_{12} \subset M_1 \times M_2, \qquad Z_{23} \subset M_2 \times M_3$$

be closed subsets. Define the set-theoretic composition $Z_{12} \circ Z_{23}$ as

$$(2.7.6) \quad Z_{12} \circ Z_{23} = \{(m_1, m_3) \in M_1 \times M_3 \mid \text{there exists } m_2 \in M_2 \text{ such that } (m_1, m_2) \in Z_{12} \text{ and } (m_2, m_3) \in Z_{23}\}.$$

If we think of Z_{12} (resp. Z_{23}) as a multivalued map from M_1 to M_2 (resp. from M_2 to M_3), then $Z_{12} \circ Z_{23}$ may be viewed as the composition of Z_{12} and Z_{23}.

Example 2.7.7. Let $f : M_1 \to M_2$ and $g : M_2 \to M_3$ be smooth maps. Then $\mathrm{Graph}(f) \circ \mathrm{Graph}(g) = \mathrm{Graph}(g \circ f)$.

We will need another form of definition (2.7.6) in the future. Let $p_{ij} : M_1 \times M_2 \times M_3 \to M_i \times M_j$ be the projection to the (i,j)-factor. From now on, we assume, in addition, that the map

(2.7.8) $\qquad p_{13} : p_{12}^{-1}(Z_{12}) \cap p_{23}^{-1}(Z_{23}) \to M_1 \times M_3$ **is proper.**

We observe that

$$p_{12}^{-1}(Z_{12}) \cap p_{23}^{-1}(Z_{23}) = (Z_{12} \times M_3) \cap (M_1 \times Z_{23}) = Z_{12} \times_{X_2} Z_{23}.$$

Therefore the set $Z_{12} \circ Z_{23}$ defined in (2.7.6) is equal to the image of the map (2.7.8). In particular, this set is a closed subset in $M_1 \times M_3$ for the map in (2.7.6) is proper.

Let $d = \dim_{\mathbb{R}} M_2$. We define a convolution in Borel-Moore homology, cf. [FM]

(2.7.9)
$$H_i(Z_{12}) \times H_j(Z_{23}) \to H_{i+j-d}(Z_{12} \circ Z_{23}), \quad (c_{12}, c_{23}) \mapsto c_{12} * c_{23}$$

by translating the set-theoretic composition into composition of cycles. Specifically put (compare with (2.7.3)):

$$c_{12} * c_{23} = (p_{13})_*\Big((c_{12} \boxtimes [M_3]) \cap ([M_1] \boxtimes c_{23})\Big) \in H_*(Z_{12} \circ Z_{23}),$$

where $c_{12} \boxtimes [M_3] = p_{12}^* c_{12}$, and $[M_1] \boxtimes c_{23} = p_{23}^* c_{23}$ are given by the Künneth formula 2.6.19, and the intersection pairing $\cap$ was defined in 2.6.16. Note that

$$\mathrm{supp}\,((c_{12} \boxtimes [M_3]) \cap ([M_1] \boxtimes c_{23})) \subset (Z_{12} \times M_3) \cap (M_1 \times Z_{23}),$$

so that the direct image is well-defined due to the condition that the map p_{13} in (2.7.8) is proper.

Remarks. (i) The same definition applies in the disconnected case as well, provided $[M_1]$, resp. $[M_3]$, is understood as the sum of the fundamental classes of connected components of M_1, resp. M_3.

(ii) In the special case when M_1, M_2, M_3 are (not necessarily finite) sets with the discrete topology the group $H_*(Z_{ij})$ reduces to the vector space $\mathbb{C}(Z_{ij})$ of $\mathbb{C}$-valued functions on Z_{ij}. The term "proper" in (2.7.8) gets replaced by "has finite fibers," and convolution product (2.7.9) reduces to the one given essentially by formula (2.7.2).

(iii) A similar convolution construction works for any *generalized homology theory* that has pullback morphisms for smooth maps, pushforward morphisms for proper maps and an intersection pairing with supports. This is the case, e.g. for the K-homology theory used in [KL4] and also for the algebraic equivariant K-theory studied in part 5 (though the latter is not a generalized homology theory).

2.7.10. EXAMPLES. **(i)** Let $M_1 = M_2 = M_3 = M$ be smooth, and

$$Z_{12},\ Z_{23} \subset M_\Delta \hookrightarrow M \times M,$$

where $M_\Delta \hookrightarrow M \times M$ is the diagonal embedding. If Z_{12} and Z_{23} are closed then p_{13} is always proper, and moreover

$$Z_{12} \circ Z_{23} = Z_{12} \cap Z_{23} \subset M_\Delta \subset M \times M.$$

In this case we see that the convolution product $*$ reduces to the intersection product $\cap$ defined above in 2.6.15.

(ii) Let M_1 be a point and $f : M_2 \to M_3$ a proper map of connected varieties. Set $Z_{12} = pt \times M_2 = M_2$, and $Z_{23} = \mathrm{Graph}(f)$. Then $Z_{12} \circ Z_{23} = \mathrm{Im}\, f \subset pt \times M_3 = M_3$. Let $c \in H_*(M_2) = H_*(Z_{12})$.

The proof of the following claim is immediate.

Claim 2.7.11. We have $c * [\mathrm{Graph} f] = f_*(c)$.

(iii) Assume now that M_3 is a point, $f : M_1 \to M_2$ is a smooth map of oriented, connected manifolds and $d = \dim M_1 - \dim M_2$ (d may be either positive or negative). Set $Z_{12} = \mathrm{Graph}\ f$, a smooth closed oriented submanifold in $M_1 \times M_2$, and let $Z_{23} = M_2 \times pt = M_2$. Then

$$Z_{12} \circ Z_{23} = M_1 \times pt = M_1.$$

Claim 2.7.12. The assignment given by the formula: $c \mapsto [\mathrm{Graph}\, f] * c$ coincides with the smooth pullback morphism $f^* : H_i(M_2) \to H_{i+d}(M_1)$ introduced in 2.6.26.

2.7.13. Let $Z_{12} \subset M_1 \times M_2$ be a closed subset. Then the embedding of Z_{12} gives, by means of the Künneth formula, a map $H_*(Z_{12}) \to H_*(M_1) \otimes H_*(M_2)$. If in addition the projection $Z_{12} \to M_2$ is proper one can do better. Namely, we will show in Chapter 8, using sheaf theoretic methods, that in this case there is a natural map $H_*(Z_{12}) \to H_*^{ord}(M_1) \otimes H_*(M_2)$. Now, let $Z_{23} \subset M_2 \times M_3$ have a similar property. Then the convolution product $H_*(Z_{12}) \otimes H_*(Z_{23}) \to H_*(Z_{12} \circ Z_{23})$ is a refinement (with supports) of the natural pairing

$$\left(H_*^{ord}(M_1) \otimes H_*(M_2)\right) \otimes \left(H_*^{ord}(M_2) \otimes H_*(M_3)\right) \longrightarrow H_*^{ord}(M_1) \otimes H_*(M_3).$$

The latter is obtained by contracting the middle terms by means of Poincaré duality pairing: $H_*(M_2) \otimes H_*^{ord}(M_2) \to \mathbb{C}$ given by Proposition 2.6.18.

Assume in particular that $M_3 = pt$ and $Z_{23} = M_2 \times pt$. Then, generalizing the setup of example 2.7.10(iii), we obtain a convolution map

$$(2.7.14) \qquad H_i(Z_{12}) \otimes H_j(M_2) \to H_{i+j-d}(M_1) \quad , \quad d = \dim M_2 \, .$$

Observe further that the cap-product (2.6.17) gives rise to a well-defined pairing between Borel-Moore and ordinary homology

$$H_i(Z_{12}) \otimes H_j^{ord}(M_2) \to H^{ord}_{i+j-d}(Z_{12}), \quad (z, c) \mapsto z \cap ([M_1] \times c).$$

Composing it with the pushforward (in ordinary homology) by means of the first projection $Z_{12} \to M_1$ (which is not necessarily proper) we get a counterpart of the convolution map (2.7.14) for ordinary homology

$$(2.7.15) \qquad H_i(Z_{12}) \otimes H_j^{ord}(M_2) \to H^{ord}_{i+j-d}(M_1), \quad d = \dim M_2.$$

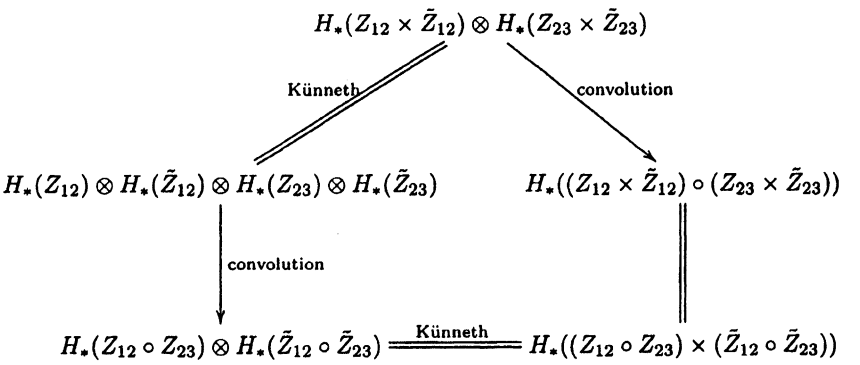
Note that $H_i(Z_{12})$ is still the Borel-Moore homology group.

2.7.16. KÜNNETH FORMULA FOR CONVOLUTION. Let $M_1, M_2, M_3, \tilde{M}_1, \tilde{M}_2, \tilde{M}_3$ be smooth oriented manifolds, and

$$Z_{12} \subset M_1 \times M_2, \quad \tilde{Z}_{12} \subset \tilde{M}_1 \times \tilde{M}_2, \quad Z_{23} \subset M_2 \times M_3, \quad \tilde{Z}_{23} \subset \tilde{M}_2 \times \tilde{M}_3.$$

We view $Z_{12} \times \tilde{Z}_{12}$ as a correspondence in $M_1 \times \tilde{M}_1 \times M_2 \times \tilde{M}_2$ and $Z_{23} \times \tilde{Z}_{23}$ as a correspondence in $M_2 \times \tilde{M}_2 \times M_3 \times \tilde{M}_3$. Then $(Z_{12} \times \tilde{Z}_{12}) \circ (Z_{23} \times \tilde{Z}_{23}) = (Z_{12} \circ Z_{23}) \times (\tilde{Z}_{12} \circ \tilde{Z}_{23})$.

Lemma 2.7.17. *The following diagram formed by convolution maps commutes:*



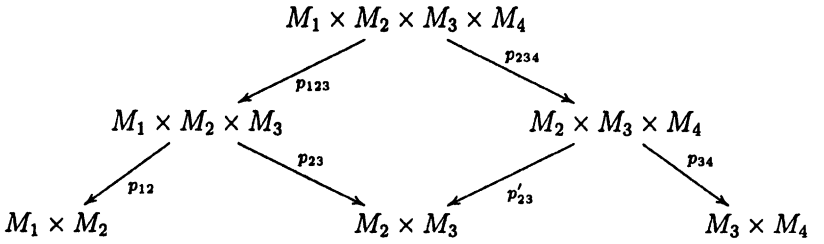
2.7.18. ASSOCIATIVITY OF CONVOLUTION. We maintain the notation of the general convolution setup, as in 2.7.5. Given a fourth oriented manifold,

M_4, and a closed subset $Z_{34} \subset M_3 \times M_4$, the following associativity equation holds in Borel-Moore homology.

$$(c_{12} * c_{23}) * c_{34} = c_{12} * (c_{23} * c_{34}), \tag{2.7.19}$$

where $c_{12} \in H_*(Z_{12})$, $c_{23} \in H_*(Z_{23})$, $c_{34} \in H_*(Z_{34})$.

Proof. We consider the following natural commutative diagram:



Using the diagram and the projection formula 2.6.29, we calculate

$$\begin{aligned}(c_{12} * c_{23}) * c_{34} &= (p_{14})_*(((p_{13})_*(p_{12}^* c_{12} \cap p_{23}^* c_{23}) \boxtimes [M_4]) \cap ([M_1] \boxtimes p_{34}^* c_{34})) \\ &= (p_{14})_*((c_{12} \boxtimes [M_3 \boxtimes M_4]) \cap ([M_1] \boxtimes c_{23} \boxtimes [M_4]) \cap ([M_1 \boxtimes M_2] \boxtimes c_{34})) \\ &= c_{12} * (c_{23} * c_{34}). \quad \blacksquare\end{aligned}$$

2.7.20. BASE CHANGE. Assume given spaces Z, S, $\tilde{S}$, and morphisms $f : Z \to S$ and $\phi : \tilde{S} \to S$. We set $\tilde{Z} := Z \times_S \tilde{S}$, and form a natural cartesian diagram

$$\begin{array}{ccc} \tilde{Z} & \xrightarrow{\tilde{\phi}} & Z \\ {\scriptstyle \tilde{f}}\downarrow & & \downarrow{\scriptstyle f} \\ \tilde{S} & \xrightarrow{\phi} & S \end{array} \tag{2.7.21}$$

We will assume that one of the following two assumptions holds:

(a) The map $\phi : \tilde{S} \to S$ is a locally trivial oriented fibration with smooth fiber, or

(b) S is smooth, the map ϕ is a closed embedding of $\tilde{S}$ as a submanifold of the manifold S and, in addition, there is a smooth variety $M \supset Z$ and a locally trivial fibration $M \to S$ which extends the map $f : Z \to S$ (note that $Z \to S$ is not required to be a fibration).

Under either of these assumptions, we have a well-defined (not necessarily degree preserving) pullback morphism $\tilde{\phi}^* : H_*(Z) \to H_*(\tilde{Z})$ in Borel-Moore homology. In case (a) we use the smooth pullback 2.6.26. In case (b) we

form the fiber product $\tilde{M} := \tilde{S} \times_S M$ and consider the cartesian diagram for $\tilde{M}$, analogous to diagram (2.7.21) for $\tilde{Z}$ but consisting of smooth ambient varieties. In particular, we have a closed embedding $\tilde{\phi} : \tilde{M} \hookrightarrow M$ making $\tilde{M}$ a submanifold of M. We use the restriction with support map $H_*(Z) \to H_*(\tilde{M} \cap Z) = H_*(\tilde{Z})$ as the definition of $\tilde{\phi}^*$.

Proposition 2.7.22. *If either of the two assumptions above hold and, moreover, if the map $f : Z \to S$ is proper, then the following diagram induced by the cartesian square* (5.3.14) *commutes*

$$\begin{array}{ccc} H_*(\tilde{Z}) & \xleftarrow{\tilde{\phi}^*} & H_*(Z) \\ {\scriptstyle \tilde{f}_*}\downarrow & & \downarrow{\scriptstyle f_*} \\ H_*(\tilde{S}) & \xleftarrow{\phi^*} & H_*(S)\,. \end{array}$$

We were unable to find an elementary proof of this result in the literature. A "non-elementary" proof is immediate from the corresponding result for constructible (complexes of) sheaves, see (8.3.14), which is proved in [SGA4], [Iv]. The absence of a completely elementary argument may be not so surprising in view of the absence of a direct elementary construction of the smooth pullback in Borel-Moore homology.

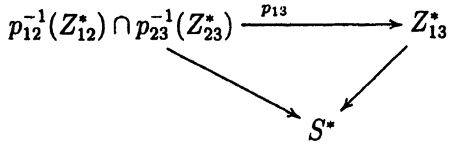
One can show using Proposition 2.7.22 that under the assumption (a) after diagram (2.7.21), the convolution in homology commutes with base change. This is not in general true in case (b). The situation improves however if base change is replaced by the specialization in Borel-Moore homology.

In more detail, let S be a manifold. Let $f_i : M_i \to S$ $(i = 1, 2, 3)$ be three smooth locally trivial fibrations over S. Further let $Z_{12} \subset M_1 \times_S M_2$ and $Z_{23} \subset M_2 \times_S M_3$ be closed subsets. We can then perform all steps of the convolution construction in this "relative framework" over S. To that end introduce the projections $p_{ij} : M_1 \times_S M_2 \times_S M_3 \to M_i \times_S M_j$, and consider the map $p_{13} : p_{12}^{-1}(Z_{12}) \cap p_{23}^{-1}(Z_{23}) \to M_1 \times_S M_3$. Assuming the map is proper, write Z_{13} for its image. Repeating the construction of (2.7.9) one defines a convolution map $* : H_*(Z_{12}) \times H_*(Z_{23}) \to H_*(Z_{13})$.

Next, fix a point $o \in S$, and let $i : \{o\} \hookrightarrow S$ be the embedding (in terms of the base change setup, we are in case (b) with $\tilde{S} = \{o\}$). Put $S^* = S \setminus \{o\}$. Given any map $f : Z \to S$ we write $Z^\circ := f^{-1}(o)$ and $Z^* := f^{-1}(S^*)$. We assume the following:

(1) The natural projection $Z_{ij}^* \to S^*$ is a locally trivial fibration, for any pair (i, j) , $i < j$, $i, j = 1, 2, 3$.

(2) The horizontal map in the diagram



is a morphism of locally trivial fibrations.

The following proposition and its proof are analogous to a result proved in [FM].

Proposition 2.7.23. *Specialization commutes with convolution in Borel-Moore homology, i.e., under the conditions (1)-(2) above the following diagram commutes:*

$$\begin{array}{ccc} H_*(Z_{12}^*) \otimes H_*(Z_{23}^*) & \xrightarrow{\lim_{t\to 0}} & H_*(Z_{12}^\circ) \otimes H_*(Z_{23}^\circ) \\ \downarrow \text{convolution} & & \downarrow \text{convolution} \\ H_*(Z_{13}^*) & \xrightarrow{\lim_{t\to 0}} & H_*(Z_{13}^\circ) \end{array}$$

2.7.24. *Idea of Proof.* Introduce the notation $\mathcal{Z} = p_{12}^{-1}(Z_{12}) \cap p_{23}^{-1}(Z_{23})$. The convolution map $H_*(Z_{12}) \otimes H_*(Z_{23}) \to H_*(Z_{13})$ is by definition the composite of the intersection pairing $\cap : H_*(p_{12}^{-1}(Z_{12})) \otimes H_*(p_{23}^{-1}(Z_{23})) \to H_*(\mathcal{Z})$ and the proper direct image $(p_{13})_* : H_*(\mathcal{Z}) \to H_*(Z_{13})$. Hence, it suffices to prove that both the intersection pairing and the proper direct image commute with the specialization. For the intersection pairing the assertion follows from Proposition 2.6.39. To prove the assertion for the proper direct image, observe that a proper map of pairs: $(X, X') \to (Y, Y')$ gives rise to a morphism of the corresponding long exact sequences of Borel-Moore homology. In particular, the connecting homomorphisms in the long exact sequences commute with the proper direct image. The claim about specialization can be deduced from this fact, since the specialization is constructed by means of a connecting homomorphism in a long exact sequence of Borel-Moore homology. ∎

2.7.25. AN EXPLICIT FORMULA. We perform here a concrete convolution computation in a special case that will be important for us in the future.

Let X_1, X_2, X_3 be complex manifolds and $Y_{12} \subset X_1 \times X_2$, and $Y_{23} \subset X_2 \times X_3$ complex submanifolds. As usual we write $p_{ij} : X_1 \times X_2 \times X_3 \to X_i \times X_j$ for the projection to the (i,j)-factor, and also use the notation pr_{ij} for the projection $T^*(X_1 \times X_2 \times X_3) \to T^*(X_i \times X_j)$, the cotangent bundle companion of p_{ij}. Our goal is to establish a relationship between convolution product for Y_{12} and Y_{23} and the convolution product for their

conormal bundles. Thus we put $Y_{13} := Y_{12} \circ Y_{23}$, and $Z_{ij} = T^*_{Y_{ij}}(X_i \times X_j)$, for any pair (i,j) where $i,j = 1,2,3$, $i < j$.

Theorem 2.7.26. *Assume that Y_{12} and Y_{23} satisfy two conditions:*

(a) *The intersection of $p_{12}^{-1}(Y_{12})$ and $p_{23}^{-1}(Y_{23})$ is transverse;*
(b) *The map $p_{13} : p_{12}^{-1}(Y_{12}) \cap p_{23}^{-1}(Y_{23}) \to Y_{13}$ is a smooth locally trivial oriented fibration with smooth base Y_{13} and smooth and compact fiber F.*

Then the following holds:

(i) We have a set-theoretic equality $Z_{12} \circ Z_{23} = Z_{13}$; moreover,

(ii) The map $\mathrm{pr}_{13} : \mathrm{pr}_{12}^{-1}(Z_{12}) \cap p_{23}^{-1}(Z_{23}) \to Z_{13}$ is a smooth locally trivial oriented fibration with fiber F;

(iii) In $H_*(Z_{13})$ one has an equation: $[Z_{12}] * [Z_{23}] = \chi(F) \cdot [Z_{13}]$, where $\chi(F)$ is the Euler characteristic of F.

2.7.27. Remarks. (i) It is instructive to check the meaning of the theorem in the very special case: $X_1 = pt$, $X_3 = pt$ and $Y_{23} = X_2 \times pt$. Then condition (a) of the theorem holds automatically, and condition (b) says $Y_{12} = pt \times F$ where F is a compact submanifold of X_2. We suggest that the reader check that in this case the theorem boils down to the well-known fact, see e.g. [BtTu], in which the Euler characteristic of the manifold F equals the self-intersection index of the zero-section in T^*F.

(ii) Assume that either of the natural projections $Y_{12} \to X_2 \leftarrow Y_{23}$ has surjective differential, e.g., is a smooth fibration. We will see in the course of the proof of the theorem that in such a case the intersection of $p_{12}^{-1}(Y_{12})$ and $p_{23}^{-1}(Y_{23})$ is necessarily transverse so that condition (a) holds.

(iii) In the special case of Theorem 2.7.26 when the map $p_{12}^{-1}(Y_{12}) \cap p_{23}^{-1}(Y_{23}) \to Y_{13}$ in condition (b) is a bijection, i.e., if $F = pt$, our argument gives more. Namely, in this case the intersection of $\mathrm{pr}_{12}^{-1}(Z_{12})$ and $\mathrm{pr}_{23}^{-1}(Z_{23})$ is transverse (which is not true in general) and, moreover, the projection $\mathrm{pr}_{13} : \mathrm{pr}_{12}^{-1}(Z_{12}) \cap p_{23}^{-1}(Z_{23}) \to Z_{13}$ is an isomorphism.

2.7.28. *Proof of Theorem* 2.7.26. Before entering the proof we would like to emphasize two things. First, we always stick to the following sign convention concerning the isomorphism $T^*(X_i \times X_j) \simeq T^*X_i \times T^*X_j$: our isomorphism is the usual one composed with changing sign in the T^*X_j-factor on the right. Our choice of isomorphism is determined by requiring that, in the case $i = j$, the conormal bundle to the diagonal in $X_i \times X_i$ goes under the isomorphism to the diagonal in $T^*X_i \times T^*X_i$, which is very natural from the point of view of symplectic geometry. Second, the reader should not be mistaken to confuse the variety $\mathrm{pr}_{12}^{-1}(Z_{12}) \simeq Z_{12} \times T^*X_3$ with the conormal bundle to $p_{12}^{-1}(Y_{12})$, which is equal to $Z_{12} \times$ (*zero section* of T^*X_3).

Now, introduce the shorthand notation $X_{12} := X_1 \times X_2$ and $X_{123} := X_1 \times X_2 \times X_3$, etc. We also put $\mathcal{Y} := p_{12}^{-1}(Y_{12}) \cap p_{23}^{-1}(Y_{23})$ and $\mathcal{Z} := \mathrm{pr}_{12}^{-1}(Z_{12}) \cap p_{23}^{-1}(Z_{23})$.

To prove the theorem we must analyze the intersection

$$\mathcal{Z} = \mathrm{pr}_{12}^{-1}(Z_{12}) \cap \mathrm{pr}_{23}^{-1}(Z_{23}) = (Z_{12} \times T^*X_3) \cap (T^*X_1 \times Z_{23}) .$$

This variety clearly projects to $\mathcal{Y}$ by means of the cotangent bundle projection. We may thus study the fibers of each term above over some fixed point $o \in \mathcal{Y}$. The fibers in question will not change if we "linearize" the situation, that is replace each variety by its tangent space at the point o.

Thus, assume for the moment that X_1, X_2, X_3 are vector spaces with o being the origin, and $Y_{12} \subset X_{12}$, $Y_{23} \subset X_{23}$ are vector subspaces, etc. In this linear setup, the second projection $Y_{12} \to X_2$ becomes a linear map, and we write $W_1 \subset X_2$ for its image. To simplify notation we assume first that this map has no kernel, hence gives an isomorphism $Y_{12} \xrightarrow{\sim} W_1$. Inverting this isomorphism, and composing it with the first projection $Y_{12} \to X_1$ we obtain a linear operator $A_1 : W_1 \to X_1$. We see this way that $Y_{12} = \{(A_1(w), w) \in X_1 \oplus X_2 \ , \ w \in W_1\}$. Write $A_1^* : X_1^* \to W_1^*$ for the adjoint operator, and identify W_1^* with $X_2^*/W_1^\perp$. We therefore obtain (keeping our "sign convention" in mind)

$$(2.7.29)\quad (T^*_{Y_{12}} X_{12})|_o = Y_{12}^\perp = \{(\xi_1, \xi_2) \in X_1^* \oplus X_2^* \mid \xi_2 = A_1^*(\xi_1) \bmod W_1^\perp\},$$

where $Y_{12}^\perp \subset (X_1 \oplus X_2)^*$ stands for the annihilator of Y_{12}.

We may replace X_1 by X_3 in the above and repeat the same argument for the projection $Y_{23} \to X_2$. Assuming it has no kernel again, and writing $W_3 \subset X_2$ for its image, in the obvious notation, we then get similarly $Y_{23} = \{(A_3(w), w) \ , \ w \in W_3\}$. We thus obtain

$$(2.7.30)\quad (T^*_{Y_{23}} X_{23})|_o = Y_{23}^\perp = \{(\xi_3, \xi_2) \in X_3^* \oplus X_2^* \mid \xi_2 = A_3^*(\xi_3) \bmod W_3^\perp\}.$$

We keep our linearized setup and combine the information we have about Y_{12} and Y_{23} to derive several consequences. First of all we obtain

$$Y_{12}+Y_{23} = \{(A_1(w_1),\ w_1+w_3,\ A_3(w_3)) \in X_1 \oplus X_2 \oplus X_3 \ , \ w_1 \in W_1 \, , \, w_3 \in W_3\} .$$

It follows that

$$(Y_{12} \oplus X_3) \oplus (X_1 \oplus Y_{23}) = X_1 \oplus (W_1 + W_3) \oplus X_3 .$$

We see that the intersection

$$(2.7.31)\quad \left(p_{12}^{-1}(Y_{12}) \cap p_{23}^{-1}(Y_{23}) \text{ is transverse at } o\right) \Longleftrightarrow W_1 + W_3 = X_2 .$$

(In particular, transversality holds provided either $W_1 = X_2$ or $W_3 = X_2$, cf. Remark 2.7.27(ii).) Furthermore, the linearized version of the LHS above is equal to

$$(2.7.32)\quad (Y_{12} \oplus X_3) \cap (X_1 \oplus Y_{23}) = \{(A_1(w), w, A_3(w_3))\,,\ w \in W_1 \cap W_3\}\,.$$

Note that the RHS may be conveniently expressed in terms of the vector space $W = W_1 \cap W_3$ and the operator $A = A_1 \oplus A_3 : W \to X_1 \oplus X_3$, $w \mapsto A_1(w) + A_3(w)$.

We next turn to the dual spaces and observe that (2.7.29) and (2.7.30) yield ("sign convention" again!):

$$(Y_{12}^{\perp} \oplus X_3^*) \cap (X_1^* \oplus Y_{23}^{\perp}) = \{(\xi_1, \xi_2, \xi_3) \in X_1^* \oplus X_2^* \oplus X_3^* \mid A_1^*(\xi_1) = \xi_2 \mathrm{mod}\ W_1^{\perp}\ \&\ \xi_2 = A_3^*(\xi_3) \mathrm{mod}\ W_3^{\perp}\,.$$

Note that LHS here is the fiber of $\mathcal{Z} = \mathrm{pr}_{12}^{-1}(Z_{12}) \cap \mathrm{pr}_{23}^{-1}(Z_{23})$ at point o. The RHS may be expressed in terms of the operator $A^* : X_1^* \oplus X_3^* \to W^*$, dual to A. Because of our sign convention for the identification of $(X_1 \oplus X_3)^*$ with $X_1^* \oplus X_3^*$, we have $A^* : (\xi_1, \xi_3) \mapsto A_1^*(\xi_1) - A_3^*(\xi_3)$. (To see that signs are right check the special case where $X_1 = X_2 = X_3$ and both Y_{12} and Y_{23} are diagonals). We thus find

(2.7.33)
$$\mathcal{Z}|_o = \{(\xi_1, \xi_3, \xi_2) \mid (\xi_1, \xi_3) \in \mathrm{Ker}(A^*)\ \&\ \xi_2 = A_1^*(\xi_1) \bmod (W_1^{\perp} \cap W_3^{\perp})\},$$

where the apparently non-symmetric equation $\xi_2 = A_1^*(\xi_1) \bmod (W_1^{\perp} \cap W_3^{\perp})$ is equivalent to $\xi_2 = A_2^*(\xi_2) \bmod (W_1^{\perp} \cap W_3^{\perp})$, which recovers the symmetry.

Further, recall the projection $p_{13} : X_{123} \to X_{13}$ and set $x = p_{13}(o)$. The induced tangent map $T_o X_{123} \to T_x X_{13}$ becomes, in our linearized setup, the natural projection $(p_{13})_* : X_1 \oplus X_2 \oplus X_3 \to X_1 \oplus X_3$. From formula (2.7.32) we find

$$(2.7.34)\qquad (p_{13})_*\Big((Y_{12} \oplus X_3) \cap (X_1 \oplus Y_{23})\Big) = \mathrm{Im}\ A\,.$$

We also have the linear projection (along T^*X_2-factor) $\mathrm{pr}_{13} : T_o^* X_{123} \to T_x^* X_{13}$ of the corresponding cotangent spaces. We see from (2.7.33) that the image and the kernel of the restriction of this linear projection to $\mathcal{Z}|_o$ are given by

$$(2.7.35)\quad \mathrm{pr}_{13}(\mathcal{Z}|_o) = \mathrm{Ker}\,(A^*),\quad \mathrm{Ker}\,(\mathrm{pr}_{13}|_{(\mathcal{Z}|_o)}) = W_1^{\perp} \cap W_3^{\perp}$$

This completes our analysis of the "linearized" situation under the assumptions that the projections $Y_{12} \to X_2 \leftarrow Y_{23}$ are both injective. In the general case write $K_1 \subset X_1$ and $K_3 \subset X_3$ for the respective kernels. Let $X_1' \subset X_1$ and $X_3' \subset X_3$ be arbitrary complementary vector

spaces to the kernels so that we have $X_i = K_i \oplus X_i'$, $i = 1,3$. Also put $Y_{12}' = Y_{12} \cap (X_1' \oplus X_2)$, and for $Y_{23}' = Y_{23} \cap (X_2 \oplus X_3')$. It is then easy to see that all the above arguments go through once X_i, $i = 1,3$ is replaced by X_i' and the projections $Y_{12} \to X_2 \leftarrow Y_{23}$ are replaced by $Y_{12}' \to X_2 \leftarrow Y_{23}'$. The direct summands K_1 and K_3 effectively split off and play no essential role.

We are now in a position to prove claims (i) and (ii) of the theorem. Assumption (a) of the theorem saying that $p_{12}^{-1}(Y_{12})$ and $p_{23}^{-1}(Y_{23})$ intersect transversely implies that $T_o\mathcal{Y}$ equals the intersection of the tangent spaces to $p_{12}^{-1}(Y_{12})$ and $p_{23}^{-1}(Y_{23})$, respectively. Hence, we have $T_o\mathcal{Y} = (Y_{12} \oplus X_3) \cap (X_1 \oplus Y_{23})$, and formula (2.7.34) yields $(p_{13})_*(T_o\mathcal{Y}) = \operatorname{Im} A$. Further, let $p_{\mathcal{Y}} : \mathcal{Y} \to Y_{13}$ denote the restriction of the projection p_{13}. Since assumption (b) of the theorem implies that the map $p_{\mathcal{Y}}$ has surjective differential, the equation above yields $T_x Y_{13} = (p_{\mathcal{Y}})_*(T_o\mathcal{Y}) = \operatorname{Im} A$. Therefore, for the cotangent space we obtain

$$T_x^* Y_{13} = (\operatorname{Im} A)^\perp = \operatorname{Ker}(A^*).$$

But then the first equation in (2.7.35) yields

$$\operatorname{pr}_{13}(\mathcal{Z}|_o) = T_x^* Y_{13}. \tag{2.7.36}$$

This proves part (i) of the theorem.

Next, we combine assumption (a) of the theorem with equation (2.7.31) to conclude that $W_1 + W_3 = X_2$. Hence, $W_1^\perp \cap W_3^\perp = (W_1 + W_3)^\perp = 0$, and the second equation in (2.7.35) implies that the map

$$\operatorname{pr}_{13} : \mathcal{Z}|_o \to T_x^* X_{13} \quad \text{is injective}. \tag{2.7.37}$$

To prove part (ii) of the theorem, we consider the subbundle $p_{\mathcal{Y}}^* T_{13}^* \subset T^* X_{123}|_{\mathcal{Y}}$, the pullback of the cotangent bundle on X_{123} by means of $p_{\mathcal{Y}}$. We can factorize the natural projection $\operatorname{pr}_{13} : \mathcal{Z} \to T^* X_{13}$ as the composition in the top row of the following diagram

$$\begin{array}{ccccc} \mathcal{Z} & \xrightarrow{\varepsilon} & p_{\mathcal{Y}}^* T_{13}^* & \longrightarrow & T^* X_{13} \\ & & \downarrow & & \downarrow \\ & & \mathcal{Y} & \longrightarrow & X_{13} \end{array}$$

Note that the square in the diagram is cartesian, due to the natural isomorphism $p_{\mathcal{Y}}^* T_{13}^* \simeq \mathcal{Y} \times_{X_{13}} T^* X_{13}$. Note further that, by property (2.7.37), the map $\varepsilon : \mathcal{Z} \to p_{\mathcal{Y}}^* T_{13}^*$ in the top row is a closed embedding. Moreover, formula (2.7.36) shows that the image of this embedding equals $\mathcal{Y} \times_{X_{13}} Z_{13}$. It follows that the cartesian square restricts, by means of the embedding ε,

to the following cartesian square

$$\begin{array}{ccc} \mathcal{Z} & \xrightarrow{\pi} & \mathcal{Y} \\ \downarrow{\scriptstyle \mathrm{pr}_{13}} & & \downarrow{\scriptstyle p_{\mathcal{Y}}} \\ Z_{13} & \xrightarrow{\overline{\pi}} & Y_{13} \end{array} \tag{2.7.38}$$

The vertical map $p_{\mathcal{Y}}$ on the right of this diagram is, by assumption (b) of the theorem, a fibration with fiber F. Since the diagram is cartesian the vertical map on the left is a fibration with the same fiber, and part (ii) of the theorem follows.

We can now prove the convolution formula of part (iii). First, it is clear that whenever two submanifolds intersect transversely their conormal bundles have "clean" intersection, see 2.6.47, in the cotangent bundle. Thus the intersection $\mathrm{pr}_{12}^{-1}(Z_{12}) \cap \mathrm{pr}_{23}^{-1}(Z_{23})$ is clean, and Proposition 2.6.47 applies. The proposition says

$$[\mathrm{pr}_{12}^{-1}(Z_{12})] \cap [\mathrm{pr}_{23}^{-1}(Z_{23})] = e(\mathbf{T}/(\mathbf{T}_1 + \mathbf{T}_2)) \cdot [\mathcal{Z}], \tag{2.7.39}$$

where $\mathbf{T}_1$, $\mathbf{T}_2$, and $\mathbf{T}$ stand for the normal bundles at $\mathcal{Z}$ to $\mathrm{pr}_{12}^{-1}(Z_{12})$, $\mathrm{pr}_{23}^{-1}(Z_{23})$, and T^*X_{123}, respectively.

Let $T_{\mathcal{Y}/Y_{13}}$ denote the relative tangent bundle on $\mathcal{Y}$, the tangent bundle to the fibers of the projection $p_{\mathcal{Y}} : \mathcal{Y} \to Y_{13}$. We have a natural isomorphism of vector bundles on $\mathcal{Z}$:

$$\mathbf{T}/(\mathbf{T}_1 + \mathbf{T}_2) \simeq \pi^* T_{\mathcal{Y}/Y_{13}} \quad \text{where} \quad \pi : \mathcal{Z} \to \mathcal{Y},$$

is the cotangent bundle projection. To check this, one first linearizes the setup at a point $o \in \mathcal{Y}$. In the linear case the result follows easily from formulas (2.7.29) - (2.7.35). We leave the details to the reader.

Using formula (2.7.39) and the definition of convolution, we see that to complete the proof of part (iii) we must compute the direct image of the class $e(\pi^* T_{\mathcal{Y}/Y_{13}}) \cdot [\mathcal{Z}]$ under the projection $\mathrm{pr}_{13} : \mathcal{Z} \to Z_{13}$. To that end we use the base change Theorem 2.7.22 for the cartesian square (2.7.38). We find

$$\begin{aligned} &(\mathrm{pr}_{13})_* \Big(e(\pi^* T_{\mathcal{Y}/Y_{13}}) \cdot [\mathcal{Z}] \Big) \\ &\quad = (\mathrm{pr}_{13})_* \pi^* \Big(e(T_{\mathcal{Y}/Y_{13}}) \cdot [\mathcal{Y}] \Big) = \overline{\pi}^* (p_{\mathcal{Y}})_* \Big(e(T_{\mathcal{Y}/Y_{13}}) \cdot [\mathcal{Y}] \Big) . \end{aligned}$$

But $p_{\mathcal{Y}} : \mathcal{Y} \to Y_{13}$ being a locally trivial fibration with fiber F, the direct image of $e(T_{\mathcal{Y}/Y_{13}}) \cdot [\mathcal{Y}]$ under $p_{\mathcal{Y}}$ is known, see e.g. [BtTu], [Sp], to be equal

to $\chi(F)[Y_{13}]$. Thus we obtain

$$\begin{aligned}
[Z_{12}] * [Z_{23}] &= (\mathrm{pr}_{13})_*\Big([\mathrm{pr}_{12}^{-1}(Z_{12})] \cap [\mathrm{pr}_{23}^{-1}(Z_{23})]\Big) \\
&= (\mathrm{pr}_{13})_*\Big(e(\mathbf{T}/(\mathbf{T}_1 + \mathbf{T}_2)) \cdot [\mathcal{Z}]\Big) \\
&= (\mathrm{pr}_{13})_*\Big(e(\pi^* T_{\mathcal{Y}/Y_{13}}) \cdot [\mathcal{Z}]\Big) \\
&= \overline{\pi}^*(\chi(F)[Y_{13}]) = \chi(F) \cdot \overline{\pi}^*[Y_{13}] \\
&= \chi(F) \cdot [Z_{13}]. \quad \blacksquare
\end{aligned}$$

2.7.40. THE CONVOLUTION ALGEBRA [Gi4]. Let M be a smooth complex manifold, let N be a (possibly singular) variety, and let $\pi : M \to N$ be a proper map. Put $M_1 = M_2 = M_3 = M$ and $Z = Z_{12} = Z_{23} = M \times_N M$ in the general convolution setup 2.7.5. Explicitly we have

$$Z = \{(m_1, m_2) \in M \times M \mid \pi(m_1) = \pi(m_2)\}.$$

It is obvious that $Z \circ Z = Z$. Therefore we have the convolution map, cf. (2.7.9)

$$H_*(Z) \times H_*(Z) \to H_*(Z).$$

The following corollary is an immediate consequence of (2.7.19).

Corollary 2.7.41. *$H_*(Z)$ has a natural structure of an associative algebra with unit. The unit is given by the fundamental class of $M_\Delta \subset Z$.*

Choose $x \in N$ and set $M_x = \pi^{-1}(x)$. Apply the convolution construction for $M_1 = M_2 = M$ and M_3 a point. Let $Z = Z_{12} = M \times_N M$ and $Z_{23} = M_x \subset M \times \{pt\}$. We see immediately that $Z \circ M_x = M_x$.

Corollary 2.7.42. *$H_*(M_x)$ has a natural structure of a left $H_*(Z)$-module under the convolution map.*

We give some examples.

Example 2.7.43. Let N be a point and M a compact manifold. Then according to the above constructions we have $Z = M \times M$. Hence we have

$$H_\bullet(Z) \simeq H_\bullet(M) \otimes H_\bullet(M) \simeq H_\bullet(M) \otimes H_\bullet(M)^* \simeq \mathrm{End}\, H_\bullet(M),$$

where the first isomorphism is the Künneth theorem and the second one arises from Poincaré duality. On the other hand, the convolution action $H_\bullet(Z) \times H_\bullet(M) \to H_\bullet(M)$, see (2.7.14), clearly gives an algebra homomorphism $H_\bullet(Z) \to \mathrm{End}\, H_\bullet(M)$. One can check that this homomorphism is equal to the composition of the chain of isomorphisms above.

Thus, the convolution algebra $H_\bullet(Z)$ is isomorphic to a matrix algebra, and the convolution action makes $H_\bullet(M)$ a simple $H_\bullet(Z)$-module.

Example 2.7.44. Let Y be smooth and compact, N smooth and connected. We put $M = Y \times N$ and let $\pi : M \to N$ be the second projection, a proper map. Then $Z = Y \times Y \times N$, and we have isomorphisms similar to those in Example 2.7.43:

$$H_\bullet(Z) \simeq H_\bullet(Y) \otimes H_\bullet(Y) \otimes H_\bullet(N) \simeq$$
$$\simeq H_\bullet(Y) \otimes H_\bullet(Y)^* \otimes H_\bullet(N)^* \simeq (\mathrm{End}\, H_\bullet(Y)) \otimes H^\bullet(N).$$

Furthermore, the cohomology $H^\bullet(N)$ has a natural commutative algebra structure, and one can check that the composition of the chain of isomorphisms above gives an isomorphism of the convolution algebra, $H_\bullet(Z)$, with a graded algebra tensor product $(\mathrm{End}\, H_\bullet(Y)) \otimes H^\bullet(N)$.

Further, for any $x \in N$ we have $M_x = \pi^{-1}(x) \simeq Y$. Therefore, the convolution action makes $H_\bullet(Y)$ an $H_\bullet(Z)$-module, hence yields an algebra homomorphism $H_\bullet(Z) \to \mathrm{End}\, H_\bullet(Y)$. One can check that this homomorphism is equal to the following composition

$$H_\bullet(Z) \xrightarrow{\sim} (\mathrm{End}\, H_\bullet(Y)) \otimes H^\bullet(N) \xrightarrow{Id \otimes \varepsilon} (\mathrm{End}\, H_\bullet(Y)) \otimes \mathbb{C} = \mathrm{End}\, H_\bullet(Y)$$

where the first map is the algebra isomorphism above, and the second map is induced by the augmentation homomorphism $\varepsilon : H^\bullet(N) \to H^0(N) \simeq \mathbb{C}$.

2.7.45. BASE LOCALITY FOR CONVOLUTION. We now study the behavior of convolution under restriction to appropriate open subsets.

In the setup of 2.7.40 let U be an open subset of N. Write M_U for $\pi^{-1}(U)$ and let

$$Z_U = M_U \times_U M_U = Z \cap (M_U \times M_U).$$

Lemma 2.7.46. *(a) The natural restriction map in homology $H_*(Z) \to H_*(Z_U)$ is an algebra homomorphism.*

(b) If $x \in U$ then the $H_*(Z_U)$-module structure on $H_*(M_x)$ is the one coming from the $H_*(Z)$-action by means of the restriction: $H_*(Z) \to H_*(Z_U)$.

Proof. This follows from the excision axiom for relative cohomology (see [Sp]). A sheaf theoretic "explanation" of the lemma will be given later in Section 8.6. ∎

2.7.47. THE DIMENSION PROPERTY. [Gi4] Let M_1, M_2, M_3 be smooth varieties of real dimensions m_1, m_2, m_3 respectively. Let $Z_{12} \subset M_1 \times M_2$ and $Z_{23} \subset M_2 \times M_3$ and let

$$p = \frac{m_1 + m_2}{2}, \quad q = \frac{m_2 + m_3}{2}, \quad r = \frac{m_1 + m_3}{2}.$$

Then it is obvious from (2.7.9) that convolution induces a map (assuming that p, q and r are integers)

$$H_p(Z_{12}) \times H_q(Z_{23}) \to H_r(Z_{12} \circ Z_{23}).$$

We say that this is the property that "the middle dimension part is always preserved."

We investigate the special case where $M_1 = M_2 = M_3 = M$, and $\dim_{\mathbb{R}} M = m$. Set $Z = M \times_N M$ and $H(Z) = H_m(Z)$. We have

Corollary 2.7.48. *$H(Z)$ is a subalgebra of $H_*(Z)$.*

Proof. This is immediate from the dimension property. ∎

The last result is especially concrete in view of the following

Lemma 2.7.49. *Let $\{Z_w\}_{w \in W}$ be the irreducible components of Z indexed by a finite index set W. If all the components have the same dimension then the fundamental classes $[Z_w]$ form a basis for the convolution algebra $H(Z)$.*

Proof. This follows from Proposition 2.6.14. ∎

In a similar way, one derives from formula (2.7.14)

Corollary 2.7.50. *The convolution action of the subalgebra $H(Z) \subset H_*(Z)$ on $H_*(M_x)$ is degree preserving, that is, for any $i \geq 0$ we have*

$$H(Z) * H_j(M_x) \subset H_j(M_x).$$

The fact that the middle dimensional homology behaves nicely under convolution is similar in spirit to the following result of symplectic geometry. Assume in the above setup that (M_i, ω_i), $i = 1, 2, 3$ are symplectic algebraic manifolds. Equip $M_i \times M_j$ with the symplectic form $\omega_{ij} = p_i^* \omega_i - p_j^* \omega_j$. Then we have

Proposition 2.7.51. *If $Z_{12} \subset M_1 \times M_2$ and $Z_{23} \subset M_2 \times M_3$ are both isotropic algebraic subvarieties then $Z_{12} \circ Z_{23}$ is again an isotropic subvariety (in $M_1 \times M_3$).*

Proof. We must prove that the tangent space at the generic point of any irreducible component of the variety $Z_{13} := Z_{12} \circ Z_{23}$ is an isotropic vector subspace. To that end set $Z_{12,23} = p_{12}^{-1}(Z_{12}) \cap p_{23}^{-1}(Z_{23})$. Then, by definition, Z_{13} is the image of $Z_{12,23}$ under the projection p_{13}. We may assume without loss of generality that Z_{13} is irreducible. Hence, there exists U, a non-empty Zariski open subset of the non-singular locus of $Z_{12,23}$ such that $p_{13}(U)$ is a Zariski dense subset in the non-singular locus of Z_{13}. Moreover, we may choose U so that the restriction map $p_{13} : U \to Z_{13}$ has surjective differential at any point of U.

It suffices to show that the 2-form ω_{13} vanishes on $p_{13}(U)$. By surjectivity of the differential of $p_{13} : U \to Z_{13}$ this is equivalent to the vanishing of the 2-form $p_{13}^*\omega_{13}$ on U. To prove the latter we use the trivially verified identity

$$p_{13}^*\omega_{13} = p_{12}^*\omega_{12} - p_{23}^*\omega_{23}\,.$$

We will show that both terms on the RHS of the identity vanish on U.

To that end, consider U as a smooth locally closed subvariety of $p_{12}^{-1}(Z_{12})$. Since Z_{12} is isotropic with respect to the form ω_{12}, a slight modification of Proposition 1.3.30 (one needs to replace Z_{12} by $p_{12}^{-1}(Z_{12})$ and ω_{12} by $p_{12}^*\omega_{12}$) implies that $p_{12}^*\omega_{12}$ vanishes on U. Similarly, we deduce that $p_{23}^*\omega_{23}$ vanishes on U. Hence, by identity, the form $p_{13}^*\omega_{13}$ vanishes on U, and the proposition follows. ∎

Warning. There exists an example of lagrangian subvarieties $Z_{12} \subset M_1 \times M_2$ and $Z_{23} \subset M_2 \times M_3$ such that $Z_{12} \circ Z_{23}$ is *not* lagrangian (it has components of dimension $< 1/2 \dim (M_1 \times M_3)$.)